

Applications of Extreme Value Theory in Financial Risk Modelling

DISSERTATION
der Universität St. Gallen,
Hochschule für Wirtschafts-,
Rechts- und Sozialwissenschaften (HSG)
zur Erlangung der Würde eines
Doktors der Wirtschaftswissenschaften

vorgelegt von
Christian Thomas Reich
von
Sennwald (St. Gallen)

Genehmigt auf Antrag der Herren

Prof. Dr. Heinz Zimmermann
und
Prof. Dr. Joachim Grammig

Dissertation Nr. 2893

Difo-Druck GmbH, Bamberg 2004

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Die Universität St. Gallen, Hochschule für Wirtschafts-, Rechts- und Sozialwissenschaften (HSG), gestattet hiermit die Drucklegung der vorliegenden Dissertation, ohne damit zu den darin ausgesprochenen Anschauungen Stellung zu nehmen.

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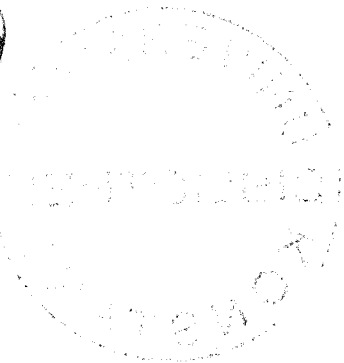
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Der Rektor:

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To my brother

Acknowledgements

The present thesis would not have been possible without my academic teacher and thesis advisor Professor Heinz Zimmermann. It is to him I am most indebted. He introduced me to finance as an undergraduate student, awakened and continuously deepened my interest in the area and encouraged me to become his project assistant and to write a doctoral thesis. Furthermore, he provided me with an excellent research and working environment both at the Swiss Institute of Banking and Finance (s/bf) at the University of St. Gallen as well as at the Wirtschaftswissenschaftliches Zentrum (WWZ) at the University of Basel, respectively. I am very grateful for his advice and guidance.

Many thanks also go to Professor Joachim Grammig. He spontaneously agreed to become my co-advisor and provided many helpful suggestions and critical comments.

Moreover, I would like to thank Professor John Bennett for examining my writing and composing the language referee report.

Special thanks go to my colleagues at the Swiss Institute of Banking and Finance (s/bf) and at the Wirtschaftswissenschaftliches Zentrum (WWZ), respectively, both for their friendship as well as for creating an encouraging and inspiring research environment. In particular, I thank Christian Buhl and Patrick Wegmann for the numerous academic discussions on the subject of quantitative risk management, giving me many new insights and ideas and helping me significantly improve the quality of my work. I would also like to thank Stefan Beiner, David Rey and Markus Schmid for their careful proofreading of parts of the manuscript.

I thankfully acknowledge the financial funding of the sponsors of CARISMA supporting my work at the Swiss Institute of Banking and Finance (s/bf) and at the Wirtschaftswissenschaftliches Zentrum (WWZ), respectively.

The main parts of the present book have been presented at research seminars of the Wirtschaftswissenschaftliches Zentrum (WWZ), Adelboden, 2002, and at

different CARISMA Workshops on financial risk management, Zurich, 2001-2004. I thank the respective participants for their helpful comments and discussions.

My doctoral education was deeply influenced by my stay at the London School of Economics (LSE). I thankfully acknowledge the partial funding of the Swiss National Science Foundation (SNF). Also, I would like to thank all the teachers and especially Jon Danielsson, Professor Paul Embrechts and Andrew Patton for their helpful comments and suggestions. In addition, I thank all my student colleagues for giving me an unforgettable time in London.

My warmest thanks go to my friend Claudia Saupper for her professional support as well as for sharing many of my ups and downs in the last years. Last but not least, I thank my parents for their support and for offering me a safe haven in good and difficult times. I will always be thankful to them.

I dedicate this book to my brother. To him I owe my deepest gratitude for his inspiration from an early age and his ongoing influence on my personal development.

Igis, January 2004

Christian Th. Reich

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Introduction

0.1 Modern Risk Management

Risk is the fundamental element that influences financial behaviour. In its absence, the financial system necessary for efficient allocation of resources would be vastly simplified. In that world, only a few institutions and financial instruments would be needed, and the practice of finance would require relatively elementary analytical tools. However, in the real world, risk is ubiquitous. Much of the structure of the financial system we see serves the function of the efficient distribution of risk. Much of the financial decision making by households, business firms, governments, and especially financial institutions is focused on the management of risk. Measuring the influence of risk, and analyzing ways of controlling and allocating it, require a wide range of sophisticated mathematical and computational tools. Indeed, mathematical models of modern finance practice contain some of the most complex applications of probability, optimization, and estimation theories. Those applications challenge the most powerful of computational technologies.

A central issue in modern risk management is the measurement or modelling of risk, respectively.¹ The need to quantify risk arises in many different contexts. For instance, a regulator has to quantify the risk exposure of a financial institution in order to determine the amount of capital the institution has to hold as a buffer against unexpected losses. Similarly, the clearing house of an exchange needs to set margin requirements for investors trading on the exchange.

In order to measure, i.e. forecast risk, it is necessary to assume a model which in turn is estimated with market data. This requires a number of as-

¹There exist some comprehensive textbooks about integrated risk management, see i.e. Crouhy, Galai and Mark (2001) or Jorion (2000). Culp (2001) explicitly focuses on the risk management process.

assumptions regarding both model design and the statistical properties of the data. Consequently, it is not possible to create a perfect risk model, and a risk forecaster needs to weigh pros and cons of various models and data choices to create what inevitably can only be an imperfect model.

Since risk modelling is at the heart of the present thesis, we start with the key issues (stylized facts and risk model performance evaluation issues) which should be addressed when assessing the *properties* of risk measurement models in practice:²

- *Goodness of risk forecasts* (i.e. backtesting results, violation ratios). A good model should provide accurate forecasts across different assets, time horizons, and risk levels within the same asset class.
- *Risk volatility* (i.e. the standard deviation of the VaR forecasts over the sample period, the minimum and maximum VaR forecast over the sample period). If the forecast is too volatile, the costs of portfolio rebalancing would be (too) large, if VaR limits were strictly adhered to.
- *Non-normality of returns and Volatility clustering* (the first stylized fact of fat tails is well recognized and indeed addressed by the use of Extreme Value Theory (EVT), the second stylized fact implies that returns go through periods when volatility is high and periods when it is low). Regarding the volatility clustering, we will see in Chapter 3 that GARCH based models may reduce (eliminate) dependence in residuals, making the assumption that returns are identically and independent distributed (*iid*) more realistic.³ However, there is a trade-off, since these models provide necessarily more volatile risk forecasts.
- *Extreme clustering*. While it is a stylized fact that returns exhibit dependence, it is less established whether extremes are clustered. Danielsson (2000) suggests the following definition: Data is said to have extreme clusters if the arrival time between events is not *iid*.
- *Violation clustering* (i.e. when the time between VaR violations is not *iid*). It should be noted though that violation clustering depends both on the VaR forecast method and on the data. Danielsson and Morimoto (2000) point out that it is impossible to create a VaR model that does

²This is based on and extends Danielsson (2002).

³*iid* means essentially that the volatility is identical across all time periods and the volatility is independent over time.

not result in some type of violation clustering, either globally, or at least locally.

- *Asymmetry of the return distribution* (i.e. a feature of conditional volatility models, such as GARCH, the implicit assumption of symmetry of the return distribution). For this issue, the EGARCH model is sometimes used for volatility forecasting.
- *Measuring horizon*; Danielsson (2002) shows that shorter estimation horizons do not appear to contribute to more accurate forecasting, but longer estimation horizons do lead to lower risk volatility. However, longer horizons imply that the estimation of the volatility does less reflect the current volatility.
- *Holding period*; there are essentially two methods of doing 10 day VaR calculations in practice: use non-overlapping 10 day returns, or use a scaling law to convert one day VaRs to 10 day VaRs. With the first method we have a data problem, the second one depends on strong distribution assumptions, i.e. that returns are *normally* and *iid*.
- *Non-linear dependence*; to properly measure the dependence structure, one needs to use the related concepts of bivariate EVT and copulas. We address the respective issues in Chapter 2 and 4.

Different risk measures have been proposed, the most popular being the Value at Risk (VaR).⁴ The main drawback of the VaR concept is that it is not a coherent risk measure, in the sense of Artzner et al. (2002). Moreover, VaR related models are often based on the normal distribution assumption. In reality, however, returns often are not normally distributed but have fat tails. In risk management, where we are interested in the stress case, we are mainly concerned about the behaviour of the tails of distributions. Hence, an adequate modelling of the tails is crucial. Methods based on the normal distribution assumption are likely to underestimate tail risk.⁵

⁴Note that VaR is also the concept proposed by the Basel Committee for Banking Supervision. See Danielsson, Embrechts, Goodhart, Keating, Münnich, Renault and Shin (2001) for a critical assessment.

⁵However, while the normal distribution has its shortcomings, it should be noted that its practical advantages make it a good choice if *fast* and accurate estimation of various risk statistics for a portfolio driven by a *large number of risk factors* is considered.

0.2 Extreme Value Theory (EVT)

Extreme event risk is present in all areas of risk management. Whether we are concerned with market, credit, operational or insurance risk, one of the greatest challenges to the risk manager is to implement risk management models which allow for rare but damaging events, and permit the measurement of their consequences. In market risk, we might be concerned with the day to day determination of the VaR for the losses we incur on a trading book due to adverse market movements. In credit or operational risk management our goal might be the determination of the risk capital we require as a cushion against irregular losses from credit downgradings and defaults or unforeseen operational problems.

Extreme Value Theory (EVT), widely used in the actuarial field, is currently experiencing a boom in the field of finance, especially in risk management. EVT is mainly used to address the estimation issues concerning large losses in leptokurtic and potentially asymmetric distributions. Basically, EVT uses limiting distributions for extreme returns, regardless of the actual distribution. Consequently, we do not have to make (any) assumptions about the underlying distribution function of the portfolio returns in order to calculate extreme quantiles and event probabilities. EVT is in this sense unique as a statistical discipline in that it develops techniques and models for describing the unusual rather than the usual.

However, EVT is not a recent innovation indeed. For many years, it has been successfully applied in the area of environmental design. The estimators of extreme behaviour of sea levels, rainfall, pollution concentrations, temperatures etc., belong to the most important applications.⁶ Other fields where EVT plays an important role are modelling of insurance losses⁷ and teletraffic data⁸ as well as survival analysis.⁹ The theoretical foundation behind the EVT was derived in the late 1920's by Fisher and Tippett (1928) and later extended by Gnedenko (1943). The importance of their theorem for modelling sample extrema is to be compared with the role of the central limit theorem for modelling averages.

On a univariate level, state of the art models reflect two stylized facts exhibited by most financial return series, namely the stochastic volatility and the fat-tailedness of conditional return distributions over short time horizons. The goal of the Multivariate Extreme Value Theory (MEVT) is to construct a

⁶ See for example Smith (1989), Coles (1999a).

⁷ See McNeil (1996) and McNeil and Saladin (1997).

⁸ See Resnick (1997).

⁹ See Coles (1999a).

methodology to calculate risk measures, i.e. the VaR and the expected shortfall, which directly come from the extremal dependence structure between portfolio components. This is achieved by considering their multivariate extreme value probability distributions. However, the extension of univariate to multivariate modelling is far from being obvious. One way of understanding MEVT is as the study of *copulas* which arise in the limiting multivariate distribution of componentwise block maxima. Basically, the copula is a function that links univariate margins to the full multivariate distribution. Hence, the risk is splitted into two parts: the individual risks (i.e. the marginal distributions) and the dependence structure between them (i.e. the copula function).

The aim of the present thesis is to contribute to improve risk measurement as part of an integrated risk management framework. The (main) focus is on measuring market risk. The main starting question is how to improve risk modelling using EVT. In particular, we address the following main research questions:

- What are the main assumptions underlying the theory of extreme values? What are the main models and tools EVT has to offer? (Chapter 1)
- Do different financial markets crash jointly or is the fall of one the gain to another market? How do markets behave in crisis periods? (Chapter 2)
- How can we improve the modelling of heteroskedastic financial time series using EVT? (Chapter 3)
- How can we build bivariate risk models using EVT and related (tail) dependence concepts? (Chapter 4)
- What are the limits of EVT as a tool to model financial risk? (Chapter 5)

0.3 Organization of this Book

Chapter 1 is an introduction. Many fields in modern science and engineering have to deal with events which are rare but have significant consequences. EVT is considered to provide the basis for the statistical modelling of such extremes. The potential of EVT applied to financial problems has been recognized in the past few years. The aim of the first chapter is to introduce the fundamentals of EVT, to show its practical aspects for estimating, to show its basic applications in Financial Risk Measurement and to critically discuss its underlying assumptions. The (parametric) block maxima and Peaks-over-threshold (POT)

methods are introduced, and estimating issues as well as diagnostic tools are discussed and implemented for the example of the Dow Jones Index series. Furthermore, the calculation of VaR and S-VaR via EVT is introduced and basic VaR models are backtested.

Chapter 2 presents an exploratory analysis about extremal linkages in crisis periods. We characterize asset return linkages during crisis periods (or spillovers) by an extremal dependence measure. We apply a non-parametrical approach to measure bivariate exceedance probabilities and the respective dependence function (or copula). Contrary to correlation analysis, this non-parametric measure is not predisposed towards the normal distribution and can account for non-linear relationships.

In our extensive survey across different asset classes we measure extremal dependence between single-title risk and market risk, extremal dependence between international stock market returns, spillovers across foreign exchange rate regimes and extremal dependence between credit spreads and market returns.

We calculate non-parametric estimates for the expected number of crashes, given that there is at least one crash. We find conditional crash expectation estimates ranging from moderate 6.5% up to 29%. Furthermore, we test non-parametrically whether the observed number of bivariate threshold exceedances suggests tail dependence. As our hypothesis tests for independence are rejected for confidence levels of 5% and 1% in almost all cases, we conclude that extreme dependence is existing in the empirical data and may be relevant for firm-wide risk management.

In Chapter 3 we apply conditional and unconditional EVT-based risk measurement models and look for some evidence from the Swiss market. The main objective of the chapter is to survey and evaluate the performance of different univariate risk measurement methodologies. We calculate the daily Value at Risk (VaR) and Expected Shortfall (S-VaR) using unconditional and conditional models based on Extreme Value Theory (EVT) and compare it with standard models. We chose single asset returns of stocks contained in the Swiss Market Index rather than (more diversified) stock indices as in previous studies.

Besides the Historical simulation model, Variance-covariance model and an unconditional EVT-based VaR model, we apply standard GARCH models and the approach proposed by Frey and McNeil (2000), which combines pseudo-maximum-likelihood fitting of GARCH models to estimate the current volatility and EVT for estimating the tail of the innovation distribution of the GARCH model. The approach is used to estimate conditional quantiles (e.g. the VaR) and conditional expected shortfall (the expected size of a return exceeding the VaR), the latter being an alternative measure of tail risk with better theoretical

properties than VaR.

Using backtesting of historical daily return series we improve our VaR and S-VaR estimates with the conditional EVT-based model, compared to models which ignore the heavy tails of the innovations (e.g. non-EVT-based models) and/or the stochastic nature of the volatility (unconditional models).

In Chapter 4, we combine the theory of extreme values and dependence modelling in a bivariate risk measurement framework. We calculate the portfolio Value-at-Risk (VaR) and Expected Shortfall (S-VaR) using an unconditional bivariate model. The proposed model combines Extreme Value Theory (EVT) and copula estimation in a simulation framework. The following main steps are implemented within the model:

- Estimating the parameters for a given copula from historical data (i.e. fitting the data to the normal, kimeldorf sampson and other copulas, respectively)
- Random generation from the estimated copulas
- Calculation of portfolio returns
- Estimating VaR and S-VaR, respectively, from P/L distribution of the portfolio.

Different copulas are assumed and we backtest the proposed model both for VaR and S-VaR, using a data set consisting of historical daily return series of 15 titles of the Swiss Market Index (SMI). We compare the model with the Historical simulation and the Variance-covariance models. The results for the given data sample show that for quantiles higher than 95%, our model performs better than the traditional ones. Furthermore, we get different results when assuming different dependence structures; e.g. assuming a copula with strong lower tail dependence behaviour (e.g. the kimeldorf sampson or the BB3 copula) gives better results than assuming a normal copula. We conclude that the former copulas better describe the dependence structure for the given portfolios and sample. Finally, we find that for the S-VaR the assumption of normality seems useless.

We conclude that the choice of the dependence structure is crucial in financial risk measurement and for making decisions in Financial Risk Management.

In the last chapter advantages of EVT in finance are shown and, more importantly, limits of EVT as a tool to model financial risk are critically assessed.

Recent years have witnessed an enormous growth in Financial Risk Modelling both for regulatory and internal risk management purposes. There are many

	<i>Univariate EVT</i>	<i>Multivariate EVT</i>
<i>Parametric Estimation</i> <i>(ML estimation)</i>	<i>Unconditional (Chapter 1, Chapter 4):</i> Classical EVT (Block maxima model) Peak over threshold (POT)	Bivariate Block Max. Models Bivariate POT Models Extreme Value Copulas Stable Tail Dependence Function (STDF)
	<i>Conditional (Chapter 3):</i> Dynamic EVT	Tail dependence measures
<i>Non-parametric Estimation</i>	Hill estimator (Chapter 2) Extensions and Alternatives to the Hill estimator	Nonparametric estimation of Copulas (Chapter 4) Nonparametric estimation of the STDF and Spillover probabilities (Chapter 2)

Figure 1: Organization of the book: an overview of the methodologies used.

reasons for this; stronger perception of the importance of risk management, deregulation enabling more risk taking, and technical advances encouraging both risk taking and facilitating the estimation and forecasting of risk. However, there are some limits inherent to risk modelling in general, beyond the use of EVT. We conclude with a discussion of the main important limits.

Figure 1 gives an overview of the methodologies used in the present book.¹⁰ We distinguish between uni- and multivariate EVT as well as between parametric and non-parametric estimation techniques. Parametric estimation consists of estimating parameters of the asymptotic distribution by assuming that realized extremes are drawn exactly from an extreme value distribution. The commonly used parametric technique is Maximum Likelihood (ML). Non-parametric estimation is based on the direct tail estimation of the parent variable (i.e. the return) and does not assume that extremes are drawn *exactly* from the asymptotic distribution.

¹⁰Note that the overview (Figure 1) is not meant to be comprehensive but to give an illustration of the contents of the present book.

Chapter 1

EVT for Tail-Related Risk Measures

1.1 Introduction

Broadly speaking, there are two principal kinds of models for extreme values. The classical group of models are the block maxima models; these are models for the largest observations collected from large samples of identically distributed observations. A more modern group of models, however, are the Peak-over-threshold models (POT); these are models for all large observations which *exceed a high threshold*. The POT models are generally considered to be the more useful for practical applications, due to their more efficient use of the (often limited) data on extreme values.

There exists a wide range of textbooks in the area of EVT. Treatments of the mathematical foundations can be found in Leadbetter, Lindgren and Rootzén (1983) or in Resnick (1987). On the statistical side, Reiss and Thomas (2001), Kotz and Nadarajah (2000), Finkenstädt and Rootzén (2003) and Coles (2001) give thorough overviews of statistical techniques for extreme value modelling. Finally, a detailed treatment of the role of EVT for insurance and financial applications is given by Embrechts, Klüppelberg and Mikosch (2001).

Besides the mentioned textbooks, there are several papers introducing EVT to *Financial Risk Management*. Longin (2000), McNeil (1999b), Embrechts, Resnick and Samorodnitsky (1998a, 1998b) and Danielsson and De Vries (1997b) were among the first to use EVT for risk measurement purposes. Longin (2000)

uses a parametric model and presents three main advantages of EVT over classical VaR models. McNeil (1999b) uses the block maxima method for the calculation of quantile risk measures for financial return series. Resnick and Samorodnitsky (1998a, 1998b) introduce EVT as a risk management tool. Danielsson and De Vries (1997b) propose a semi-parametric method (based on the Hill estimator as introduced in Section 1.3.3) for VaR evaluation. They model the largest risk parametrically, while the smaller risks are captured by the non-parametric empirical distribution function.

More recently, Smith (2003) gives a survey of how EVT answers a variety of questions of interest to an applied scientist in finance. He reviews parts of univariate EVT and discusses estimation, diagnostics, multivariate extremes and max-stable processes. Kellezi and Gilli (2000) introduce the fundamentals of EVT as well as practical aspects for estimating and assessing statistical models for tail-related risk measures. McNeil (1999a) provides an overview of the role of EVT in risk management, concentrating on the POT model. Bensalah (2000) applies univariate EVT techniques to a series of daily exchange rates of Canadian/US dollars over a 5-year period. Klüppelberg (2003) reviews aspects of VaR and its estimation based on EVT. She presents results of a comprehensive investigation of the extremal behaviour of some of the most important continuous and discrete time series models that are of current interest in finance. Finally, Gavin (2000) argues that EVT is a practical tool to estimate tail risk and emphasizes on the application of an EVT model rather than on its theoretical properties, using graphical tools to visualize the results.

The present chapter is an introduction to EVT. First, the block maxima model and the POT model are introduced. Next, the issues of threshold selection, parameter estimation and model diagnostics are discussed. The following two sections introduce EVT in financial risk measurement and give an application for the Dow Jones Index return series, i.e. VaR and S-VaR are calculated and backtested. Finally, underlying assumptions and potential applications are shown in the last section.

1.2 The Theory of Extreme Values

1.2.1 Classical Extreme Value Theory

Similar to the central limit theorem we are interested in a sequence of normalizing coefficients a_n, b_n in order to obtain a class of limit distributions for M_n as $n \rightarrow \infty$, whereas $M_n = \max \{X_1, \dots, X_n\}$ and $X_1, \dots, X_n$ is a sequence of in-

dependent, identically distributed random variables with common distribution function F . In financial applications we usually calculate

$$X_t = (\log S_t - \log S_{t-1}), \quad (1.1)$$

where S_t is the value of the asset at the end of day t and X_t the respective log return. If sequences of constants $a_n > 0$ and b_n exist, such that the sequence of normalized maxima $(M_n - b_n/a_n)$ satisfies (or converges in distribution to) a limit law H

$$\Pr \left\{ \frac{M_n - b_n}{a_n} \leq x \right\} = F^n(a_n x + b_n) \rightarrow H(x) \text{ as } n \rightarrow \infty, \quad (1.2)$$

where H is a non-degenerate distribution function,¹ then there are only three fundamental types of extreme value limit laws (three extremal types):

$$\text{Type I : } \Lambda(x) = \exp \left\{ -\exp \left[-\frac{(x-b)}{a} \right] \right\}, \quad -\infty < b < \infty, \quad (1.3)$$

$$\text{Type II : } \Phi_\alpha(x) = \begin{cases} 0, & x \leq b \\ \exp \left\{ -\left(\frac{(x-b)}{a} \right)^{-\alpha} \right\}, & x > b \end{cases} \quad (1.4)$$

$$\text{Type III : } \Psi_\alpha(x) = \begin{cases} \exp \left\{ -\left[-\left(\frac{(x-b)}{a} \right)^\alpha \right] \right\}, & x < b \\ 1, & x \geq b, \end{cases} \quad (1.5)$$

with parameters $a > 0$ and b and, in the case of the distributions of type II and III, $\alpha > 0$.² Collectively, the three classes of distributions are termed extreme value distributions, with type I, II and III known as the Gumbel, Fréchet and Weibull families, respectively. In other words, the limiting distribution of sample maxima follows one of the three distributions introduced above, whatever the parent distribution function F . If condition 1.2 holds we say that the unknown distribution F is in the maximum domain of attraction of H ,

¹For a thorough treatment of the limiting behaviour of sample extrema, see Embrechts, Klüppelberg and Mikosch (2001), Chapter 3.

²The theorem that the limit distribution must be an extremal type is called Fisher-Tippett theorem (1928). For a sketch of the respective proof see Embrechts, Klüppelberg and Mikosch (2001), p 122.

$F \in MDA(H)$.³ For the sake of convenience, these three types may be combined into the following single Generalized Extreme Value (GEV) distribution $H_{\xi;\mu,\sigma}$:

$$H_{\xi;\mu,\sigma}(x) = \begin{cases} \exp \left\{ - \left[1 + \xi \left(\frac{x - \mu}{\sigma} \right) \right]^{-1/\xi} \right\} \\ \exp \left[- \exp \left\{ - \left(\frac{x - \mu}{\sigma} \right) \right\} \right] \end{cases} \quad \text{if } \begin{cases} \xi \neq 0 \\ \xi = 0 \end{cases}, \quad (1.6)$$

with scale parameter σ and location parameter μ . The parameter ξ is called tail index or shape parameter and parametrizes the heaviness of the tails. The bigger the ξ , the heavier the tail. The case $\xi > 0$ corresponds to the Fréchet distribution with shape parameter $\alpha = 1/\xi$, the case $\xi < 0$ corresponds to the Weibull distribution with shape parameter $\alpha = -1/\xi$ and $\xi = 0$ corresponds to the Gumbel distribution. Since the Fréchet distribution represents the only limit law that governs heavy-tailed behaviour, it is, naturally, of special interest in financial applications. The latter case, where $\xi > 0$, arises in situations where the tail of the distribution F decays like a power function, which means that the distribution is in the domain of attraction of the Fréchet distribution. Distributions falling under this class include the Pareto, Burr, loggamma, Cauchy and the t-distribution. The class of distributions governed by the Gumbel distribution characterizes an exponentially decreasing tail and includes the normal, exponential, gamma and lognormal distribution. Distributions with finite upper bound, like the beta and uniform distribution, are found in the maximum domain of attraction of the Weibull distribution.⁴

The aforementioned theory for the modelling of extreme events is called the classical approach or block maxima method. Within this method, one takes the maximum value observed in a certain block or subperiod, obtains a series of such block maxima and fits them to one of the extreme value limit distributions. In practice, the choice of block size can be critical. The choice amounts to a trade-off between bias and variance: blocks that are too small mean that

³Note that all commonly encountered continuous distributions are in the maximum domain of attraction of an extreme value distribution, see Embrechts, Klüppelberg and Mikosch (2001), p. 153-157. Furthermore, all the common continuous distributions used in finance and insurance are in $MDA(H)$.

⁴For a detailed characterization of the three types see Embrechts, Klüppelberg and Mikosch (2001); the Fréchet case on p. 130, the Gumbel case on p. 142 and the Weibull case on p. 134, respectively.

approximation by the limit model in Equation 1.6 is likely to be poor, leading to bias in estimation and extrapolation; large blocks on the other hand generate few block maxima, leading to large estimation variance.

Overall, the block maxima method may be a wasteful of data if one block happens to contain more extreme events than another. There are two well-known *general* characterizations. One is based on the behavior of the r largest order statistics within a block, for small values of r .⁵ If an entire time series of, say, daily observations is available, however, then better use is made of the data by avoiding altogether the procedure of blocking. Such an alternative is presented in the next section.

1.2.2 The Peak-over-threshold (POT) model

As mentioned before, rather than dividing the data set into subperiods and selecting the maximum value in each sub-sample (block maxima models), the POT method offers a more efficient use of the data set, where all values exceeding a high pre-specified threshold are considered.

Within the POT class of models we further distinguish two styles of analysis. There are non-parametric models built around the Hill estimator (see for example Danielsson (2002) and Danielsson and De Vries (1997)) and the fully parametric models based on the generalized Pareto distribution (GPD). There is little to pick and choose between these approaches, as both are theoretically justified and empirically useful when used correctly. In our analysis, we will use the fully parametric POT method, referring to it as GPD or POT model,⁶ respectively, one reason being that the respective model is fully implemented in S-Plus for Windows.

Conditional on the event that the random variable X is larger than the threshold u , and denoting these exceedances by y (in other words the probability that a loss exceeds the threshold u by more than an amount y), we are interested in estimating the following conditional probability⁷

$$\Pr(X > u + y \mid X > u) = \frac{1 - F(u + y)}{1 - F(u)}, \quad 0 < y. \quad (1.7)$$

If the parent distribution function F were known, the distribution of threshold exceedances in 1.7 would also be known. Since, in practical applications, this is

⁵See Coles (2001), p. 66, or Leadbetter, Lindgren and Rootzén (1982), Chapter 2.

⁶See McNeil (1999a) and Frey and McNeil (2000) for a more detailed description of the GPD model.

⁷See Coles (2001), p. 74.

not the case, approximations that are broadly applicable for high values of the threshold are sought.⁸ Pickands (1975) has shown that if the parent distribution F is in the MDA of the GEV, then the exceedances over a high threshold follow a Poisson process. As a consequence, the Generalized Pareto distribution (GPD) is the limiting distribution for the distribution of the excesses $(X - u)$. Hence, for $u \rightarrow \infty$, the distribution function of $(X - u)$, conditional on $X > u$, is approximately⁹

$$G_{\xi, \sigma}(x) = \begin{cases} 1 - \left(1 + \frac{\xi x}{\sigma}\right)^{-1/\xi} \\ 1 - \exp\left(-\frac{x}{\sigma}\right) \end{cases} \quad \text{if } \begin{cases} \xi \neq 0 \\ \xi = 0 \end{cases} \quad (1.8)$$

This theorem shows that for a sufficiently large u the conditional excess distribution function is well approximated by a GPD. The choice of threshold must be high enough for the limit theorem to be valid but not too high in order to have efficient estimation (i.e. bias variance trade-off). If block maxima have approximating distribution H , then threshold excesses have a corresponding approximate distribution within the generalized Pareto family. Moreover, the parameters of the generalized Pareto distribution of threshold excesses are uniquely determined by those of the associated GEV distribution of block maxima (also known as duality between the GEV and the GPD). In particular, the parameter ξ is again the dominant shape parameter, as it was for the GEV distribution.

The issue of threshold choice is analogous to the choice of block size in the block maxima approach, implying a balance between bias and variance. In this case, too low a threshold u is likely to violate the asymptotic basis of the model, leading to bias; too high a threshold will generate few excesses (or, in other words, too few highest order statistics) with which the model can be estimated, leading to high variance. The standard practice is to adopt as low a threshold as possible, subject to the limit model providing a reasonable approximation. Two parametric methods are available for this purpose: one is an exploratory technique carried out prior to model estimation (e.g. the mean residual life plot); the other is an assessment of the stability of parameter estimates, based on the fitting of models across a range of different thresholds. Both methods are described in Section 1.3.1.

⁸Note that this parallels the use of the called generalized extreme value distribution (GEV) as an approximation for the distribution of maxima of long sequences when the parent population is unknown.

⁹See Coles (2001), p. 74-75 and Embrechts, Klüppelberg and Mikosch (2001), p. 158.

Coles (2001) gives a unifying characterization of the extremes of processes that encompasses each of the two models derived above. This characterization is based on the theory of point processes. There are two good reasons for considering the latter. First, it provides an interpretation of extreme value behaviour that unifies all the models introduced so far. Second, the model provides a more flexible framework for modelling extremes of non-stationary processes.¹⁰

1.3 Threshold Selection, Estimation Issues and Model Diagnostics

1.3.1 Threshold selection

The so-called mean residual life plot (also called mean excess function) is based on the mean of the GPD. Suppose the GPD is valid as a model for the excesses of a threshold u_0 , generated by a series $X_1, \dots, X_n$, of which an arbitrary term is denoted by X . The corresponding mean is given by¹¹

$$E(X - u_0 | X > u) = \frac{\sigma_{u_0}}{1 - \xi}, \quad (1.9)$$

provided that $\xi < 1$. But if the GPD is valid for excesses of the threshold u_0 , it should equally be valid for all thresholds $u > u_0$, subject to the appropriate change of scale parameter to σ_u . Hence, for $u > u_0$,

$$\begin{aligned} E(X - u | X > u) &= \frac{\sigma_u}{1 - \xi} \\ &= \frac{\sigma_{u_0} + \xi \mu}{1 - \xi}. \end{aligned} \quad (1.10)$$

Hence, for $u > u_0$, $E(X - u | X > u)$ is a linear function of u . Furthermore, $E(X - u | X > u)$ is simply the mean of excesses of the threshold u , for which the sample mean of the threshold excesses of u provides an empirical estimate. According to 1.10, these estimates for the mean excesses are expected to change linearly with u , at levels for u for which the GPD model is appropriate. Therefore, the following locus of points

¹⁰Extremes of non-stationary sequences are (briefly) discussed in Section 1.6.

¹¹See Coles (2001), p. 78-79 and Embrechts, Klüppelberg and Mikosch (2001), p. 161 and p. 295.

$$\left\{ \left(u, \frac{1}{n_u} \sum_{i=1}^{n_u} (x_{(i)} - u) \right) : u < x_{\max} \right\}, \quad (1.11)$$

where $x_{(1)}, \dots, x_{(n_u)}$ consists of the n observations that exceed u , and $x_{\max}$ is the largest of the X_i , is called the mean residual life plot (Mres-plot).¹² Above a threshold u at which the GPD provides a valid approximation to the excess distribution, the Mres-plot should be approximately linear in u . Consequently, we chose the threshold u in the region where the Mres-plot is stable.^{13 14}

As indicated above, an additional (or alternative) method for the choice of the threshold is to *estimate* (e.g. with maximum likelihood) the model at a range of thresholds, and to look for stability of parameter estimates. Above a level u_0 at which the asymptotic motivation for the GPD is valid, estimates of the shape parameter ξ should be approximately constant, while estimates of σ_u should be linear in u , due to 1.10.¹⁵

McNeil and Saladin (1997) review the POT method and test what kind of thresholds are required and what sample sizes are necessary to give accurate estimates of quantiles.

The choice of the threshold (i.e. the choice of k) is one of the very key issue in extreme value analysis. In what follows, we present two important contributions which can supplement the usual graphical analysis (as suggested above).

Danielsson and De Vries (2002) propose, based on the Hill estimator, a bootstrap procedure for estimating the optimal tail threshold in extreme value analysis. They use a two step subsample bootstrap procedure for estimating the threshold such that the bias and variance of the estimated tail index decline at the same rate and achieve convergence in probability to normal distribution function of the estimated tail index.

Beirlant, Vynckier and Teugels (1996) show that, based on classical ideas on regression diagnostics, algorithms can be constructed searching for an appropriate set of extreme values to use in the estimation procedure.

¹² As mentioned, the mean residual life plot is also called mean excess function, see McNeil (1999a).

¹³ Figure 1.2 shows the Mres-plot for the Dow Jones Index series. Confidence intervals are added, based on the approximate normality of sample means.

¹⁴ For the choice of the threshold in the backtesting procedure it is (apparently) not feasible to examine the fitted model carefully every day and to choose a new threshold for the tail estimator each time. We might fix the threshold at a certain quantile. Alternatively, Danielsson and De Vries (2000) propose a double subsample bootstrap procedure and assess its accuracy with Monte Carlo experiments.

¹⁵ See Coles (2001), p. 83. Figure 1.3 shows parameter estimates ($\hat{\xi}$ and $\hat{\sigma}$) against threshold for the Dow Jones data.

1.3.2 Parameter estimation

Having determined a threshold, the parameters of the GPD can be estimated. Many techniques have been proposed for parameter estimation in extreme value models. These include graphical techniques based on versions of probability plots; moment-based techniques in which functions of model moments are equated with their empirical equivalents; procedures in which the parameters are estimated as specified functions of order statistics (i.e. the Hill estimator); and maximum likelihood estimation. The latter is probably the most general method in statistical applications. Suppose that the values $y_1, \dots, y_k$ are the k excesses of a threshold u . For $\xi \neq 0$ the log likelihood can be derived as¹⁶

$$\ell(\xi, \sigma) = -k \ln \sigma - \left(\frac{1}{\xi} + 1 \right) \sum_{i=1}^k \ln \left(1 + \frac{\xi}{\sigma} y_i \right), \quad (1.12)$$

provided that $(1 + \sigma^{-1} \xi y_i) > 0$ for $i = 1, \dots, k$; otherwise, $\ell(\xi, \sigma) = -\infty$. In practice, this method works well for the range $-1/2 < \xi < 1/2$, which is often the case. The asymptotic theory of maximum likelihood estimation for EVT is given in Smith (1985). Analytical maximization of the log-likelihood is not possible, so numerical techniques are required, taking care to avoid numerical instabilities when $\xi \approx 0$ in 1.12, and ensuring that the algorithm does not fail due to evaluation outside of the allowable parameter space.

In order to make inferences (i.e. to calculate confidence intervals) on parameters, a method called profile likelihood is often applied.¹⁷ The log-likelihood for θ can be formally written as $\ell(\theta_i, \theta_{-i})$, where θ_{-i} denotes all components of θ excluding θ_i . The profile log-likelihood is defined as

$$\ell_p(\theta_i) = \max_{\theta_{-i}} \ell(\theta_i, \theta_{-i}). \quad (1.13)$$

That is, for each value of θ_{-i} , the profile log-likelihood is the maximized log-likelihood with respect to all other components of θ . Numerical evaluation of the profile likelihood for any of the individual parameters ξ and σ is straightforward. For example, to obtain the profile log-likelihood for ξ , we fix $\xi = \xi_0$, and maximize the log-likelihood with respect to the remaining parameters, i.e. σ . This is repeated for a range of values of ξ_0 . The corresponding maximized values of the log-likelihood constitute the profile log-likelihood for ξ .

¹⁶ See Coles (2001), p. 80.

¹⁷ See Coles (2001), p. 34.

1.3.3 Non-parametric approaches

A non-parametric alternative to select the threshold and an alternative to the (parametric) maximum likelihood estimation is a family of estimators based on functions of (highest) order statistics, the most popular being Hill's estimator $\hat{\alpha}$:¹⁸

$$\hat{\xi}_{k,n}^{(H)} = \hat{\alpha}^{-1} = \frac{1}{k} \sum_{j=1}^k \ln \frac{X_{(j)}}{X_{(k+1)}}, \quad (1.14)$$

where k equals the number of highest order statistics $X_{(1)} \geq \dots \geq X_{(n)}$, and where $\hat{\alpha}$ is the corresponding tail index estimate.¹⁹ The lower the α , the fatter the tails of the distribution. Furthermore, for a t -distribution, the tail index α equals the number of degrees of freedom. To determine the choice of k , the number of highest order statistics, Huang (1992) shows that one can pick k such that it is in the range which minimizes the respective asymptotic mean squared error. Consequently, minimizing the sample mean squared error is an appropriate selection criterion.²⁰ Alternatively, one can simply compute $\hat{\alpha}$ for different k and select the threshold in the region over which $\hat{\alpha}$ is more or less constant.²¹ A detailed description of Hill's asymptotic properties as well as its various derivations are to be found in Embrechts, Klüppelberg and Mikosch (2001).²²

An extension of Hill's estimator that covers the whole range of ξ is the moment estimator, proposed by Dekkers (1991). Because Hill's estimator is inconsistent if $\xi \leq 0$, it will notoriously estimate $\xi > 0$ even in a thin tailed case, and thus it is useful to apply an estimator that takes into account thin tailed distributions, too. The moment estimator is:²³

$$\hat{\xi}_{k,n}^{(M)} = 1 + H_{k,n}^{(1)} + \frac{1}{2} \left(\frac{(H_{k,n}^{(2)})^2}{H_{k,n}^{(2)}} - 1 \right)^{-1}, \quad (1.15)$$

¹⁸ See for example Hill (1975) or Embrechts, Klüppelberg and Mikosch (2001).

¹⁹ Note that $\hat{\alpha}$ is the reciprocal of the shape parameter ξ . The former has a value of around 3 for fat tailed distributions.

²⁰ See for example Straetmans (1998), p. 73.

²¹ Figure 1.4 shows the Hillplot for the Dow Jones data.

²² For properties of the Hill estimator see Embrechts, Klüppelberg and Mikosch (2001), p. 330. Many results have been shown for processes with various kinds of dependence, see Embrechts, Klüppelberg and Mikosch (2001), p. 336.

²³ See Embrechts, Klüppelberg and Mikosch (2001), p. 339.

where $H_{k,n}^{(1)}$ corresponds to $\hat{\alpha}^{-1}$ and $H_{k,n}^{(2)}$ corresponds to

$$\hat{\xi}_{k,n} = \frac{1}{k} \sum_{j=1}^k \left(\log \frac{X_{(j)}}{X_{(k+1)}} \right)^2. \quad (1.16)$$

Another estimator consistent for all ξ values based on order statistics is the Pickands estimator:²⁴

$$\hat{\xi}_{k,n}^{(P)} = \frac{1}{\ln 2} \ln \frac{X_{(k)} - X_{(2k)}}{X_{(2k)} - X_{(4k)}}, \quad 1 \leq k \leq n/4. \quad (1.17)$$

Although the consistency holds for any ξ and all intermediate sequences $k \rightarrow \infty$ and $k/n \rightarrow 0$, it is very sensitive to the choice of the intermediate order statistics which are used for estimation. Because this estimator is based only on few order statistics, even a small alteration of k can yield a significant change in the estimate. As a solution to this problem Drees (1993, 1995) proposed a mixture of Pickands estimators

$$\hat{\xi}_{n,v}^{(D)} = \sum_{k=1}^m c_{nk} \hat{\xi}_{k,n}^{(P)}, \quad (1.18)$$

where m is an intermediate sequence and the scores c_{nk} , $1 \leq k \leq m$, are generated by a probability measure v on $B(0, 1]$, that is $c_{nk} = v((k-1)/m, k/m]$.²⁵ The advantage of this estimator, in the literature often described as the refined Pickands estimator or Drees-Pickands estimator, is its robustness against an unsuitable choice of k for all values of ξ .

Although the Hill estimator was not originally motivated by a regression framework, the basic idea has an immediate regression interpretation. Diebold, Schuermann and Strouhair (2000) suggest to estimate α by a linear regression of the log tail of the empirical survival function on an intercept and the logs of the k most extreme values. The regression approach may have the advantage that the extensive kit of powerful tools developed for regression can potentially be brought to bear on the tail estimation problem.²⁶

²⁴ For derivation and asymptotic properties see Embrechts, Klüppelberg and Mikosch (2001), p. 327.

²⁵ See Drees (1995).

²⁶ See also Longin (1996) and Tsay (2002).

1.3.4 Model diagnostics

The reason for fitting a statistical model to data is to make conclusions about some aspect of the population from which the data were drawn. Such conclusions can be sensitive to the accuracy of the fitted model, therefore, it is necessary to check that the model fits well. Three graphical techniques, in particular, are commonly used. The first is the *quantile plot*. Given an ordered sample of independent observations

$$x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}, \quad (1.19)$$

from a population with estimated distribution function $\hat{F}$, a quantile plot consists of the points²⁷

$$\left\{ \left(\hat{F}^{-1} \left(\frac{i}{n+1} \right), x_{(i)} \right) : i = 1, \dots, n \right\}. \quad (1.20)$$

The name quantile plot derives from the fact that the quantities $x_{(i)}$ and $\hat{F}^{-1}(i/(n+1))$ each provide estimates of the $i/(n+1)$ -quantile of the distribution F . If $\hat{F}$ is a reasonable estimate of F , then the quantile plot should consist of points close to the unit diagonal.

A slightly different tool, which is used in order to have a guide to heavy-tailedness is the *QQ-plot*, see for example Frey and McNeil (2000). It examines the hypothesis that the losses come from an exponential distribution, i.e. from a distribution with a medium-sized tail. The quantiles of the empirical distribution function on the x-axis are plotted against the quantiles of the exponential distribution function on the y-axis. The plot is

$$\left\{ \left(X_{i,n}, G_{0,1}^{-1} \left(\frac{n-i+1}{n+1} \right) \right) : i = 1, \dots, n \right\},$$

where $X_{i,n}$ denotes the i -th order statistic, and $G_{0,1}^{-1}$ is the inverse of the distribution function of the exponential distribution (i.e. a special case of the GPD). The points should lie approximately along a straight line if the data are an *i.i.d.* sample from an exponential distribution.

Another graphical tool is the so-called *probability plot*, which contains basically the same information as the quantile plot, expressed on a different scale. However, the perception gained on different scales can be important, hence, what

²⁷See Coles (2001), p. 37.

looks like a reasonable fit on one scale, may look poor on the other. Again, given the ordered sample of independent observations

$$x_{(1)} \leq x_{(2)} \leq \dots \leq x_{(n)}, \quad (1.21)$$

from a population with estimated distribution function $\hat{F}$, the probability plot consists of the points²⁸

$$\left\{ \left(\hat{F}(x_{(i)}), \frac{i}{n+1} \right) : i = 1, \dots, n \right\}. \quad (1.22)$$

If $\hat{F}$ is a reasonable model for the population distribution F , the points of the probability plot should lie close to the unit diagonal. Substantial departures from linearity provide evidence of a failure in $\hat{F}$ as a model for the data.

Next, we want to estimate the distribution of the excesses over a high threshold u , which is defined to be (recall Equation 1.7)

$$F_u(y) = \Pr \{ X - u \leq y \mid X > u \}. \quad (1.23)$$

In words, the *excess distribution* represents the probability that a loss exceeds the threshold u by at most an amount y , given the information that it exceeds the threshold.²⁹ Given our EVT results from the last sections, the GPD is the natural model for the unknown excess distribution above a sufficiently high threshold u . Hence, as shown in Figure 1.8 (upper figure) as a smooth curve, we can estimate the GPD model for the excess distribution and compare it with the empirical distribution, as shown by points.

Finally, we want to assess the tail of the underlying distribution. A tail estimator is based on the GPD approximation to the excess distribution above the threshold u . With $x = u + y$, we write

$$F(x) = (1 - F(u)) H_{\xi, \sigma}(x - u) + F(u), \quad (1.24)$$

for $x > u$.³⁰ The aim is to use 1.24 to construct a tail estimator, therefore we need an estimate of $F(u)$. For this purpose we take the obvious empirical estimator $(n - N_u)/n$, where n is the total of data points and N_u is the number of data points (out of n) exceeding the threshold u . Putting this estimator for

²⁸See Coles (2001), p. 37.

²⁹Compare equation 1.7, which shows the conditional probability that a loss exceeds the threshold u by more than an amount y .

³⁰See McNeil (1999a), p.6.

$F(u)$ and our maximum likelihood estimates of the parameters of the GPD (i.e. $\hat{\xi}$ and $\hat{\sigma}$) together, we have the following empirical tail estimator

$$\hat{F}(x) = 1 - \frac{N_u}{n} \left(1 + \hat{\xi} \frac{x - u}{\hat{\sigma}} \right)^{-1/\hat{\xi}}, \quad (1.25)$$

which is only valid for $x > u$. For practical use we again have a trade-off between bias and variance. A high u reduces bias in estimating excess function. A low u reduces variance in estimating excess function and $F(u)$.

1.4 Quantiles and Risk Measures

Risk management does not usually take into account the whole distribution of the value of a portfolio but only makes predictions about the loss potential in stress situations. As a consequence, it does make sense to concentrate on the tails of the respective distributions which mainly determine the stress loss potential both for single risks as well as for a combination of risks. The application of EVT in finance is particularly important for Value-at-Risk (VaR) calculation where the estimation of extreme quantiles of portfolio returns is of interest.

It is often more convenient and practical to interpret extreme value models in terms of quantiles, rather than individual parameter values. Suppose that a GPD with parameters σ and ξ is a suitable model for exceedances of a threshold u by a variable X . That is, the quantile $x > u$,

$$\Pr(X > x | X > u) = \left[1 + \xi \left(\frac{x - u}{\sigma} \right) \right]^{-1/\xi}, \quad (1.26)$$

It follows that

$$\Pr(X > x) = \zeta_u \left[1 + \xi \left(\frac{x - u}{\sigma} \right) \right]^{-1/\xi}. \quad (1.27)$$

where $\zeta_u = \Pr(X > u)$. Hence the level x_m which is exceeded on average once every m observations (i.e. a general version of VaR) is the solution of

$$\zeta_u \left[1 + \xi \left(\frac{x_m - u}{\sigma} \right) \right]^{-1/\xi} = \frac{1}{m}. \quad (1.28)$$

Rearranging,

$$x_m = u + \frac{\sigma}{\xi} [(m \zeta_u)^\xi - 1], \quad (1.29)$$

provided that m is sufficiently large to ensure that $x_m > u$. x_m is called the m -observation return level. In practice, we estimate

$$\hat{x}_m = u + \frac{\hat{\sigma}}{\hat{\xi}} [(m \hat{\zeta}_u)^{\hat{\xi}} - 1]. \quad (1.30)$$

Consequently, the plot of x_m (i.e. the VaR) against m (i.e. the quantile) on a logarithmic scale produces the so-called return level plot. Therefore, the generalized Pareto threshold model provides a direct method for the estimation of VaR. The return level plot is simply a graph of VaR against risk, on a convenient scale.³¹

Alternatively, we define the measure of extreme risk in terms of the (loss) distribution function F and let $1 > q \geq 0.95$. VaR^{32} is then defined as the q th quantile of the distribution F

$$\text{VaR}_q = F^{-1}(q), \quad (1.31)$$

where F^{-1} is the inverse of the distribution function F . Like F itself, the VaR_q is a theoretical quantity which we do not know. The goal is to estimate VaR_q . For a given probability $q > F(u)$, the VaR estimate is calculated by inverting the tail estimator in Equation 1.25. Hence, we have the following VaR estimate³³

$$\widehat{\text{VaR}}_p = \hat{x}_q = u - \frac{\hat{\sigma}}{\hat{\xi}} \left(\left(\frac{n}{N_u} (1 - q) \right)^{-\hat{\xi}} - 1 \right). \quad (1.32)$$

In standard statistical language, this is a quantile estimate, where the quantile is an unknown parameter of the underlying distribution. Furthermore, it is possible to give a confidence interval for $\widehat{\text{VaR}}_p$, using the method of profile likelihood. This yields an asymptotic interval in which we have confidence that the VaR lies. The resulting asymmetric confidence interval reflects a fundamental

³¹The logarithmic scale is chosen in order to emphasize the effect of extrapolation, see Coles (2001). Compare Figure 1.5 (lower left figure).

³²Note that here we quantify market risk with the estimation of the market risk VaR, or related measures of tail risk. Some ideas, however carry over to operational and credit risk settings.

³³See McNeil (1999a), p. 7. Note the similarity of 1.32 and 1.30, respectively. The former is the more general case of the latter.

asymmetry in the problem of estimating a high quantile for heavy-tailed data; it is easier to bound the interval below than to bound it above.

The expected shortfall (S-VaR) is the expected loss size, given that VaR is exceeded (i.e. a mean of an excess distribution), or

$$S - VaR_q = E[X | X > VaR_q]. \quad (1.33)$$

S-VaR is coherent in the sense of Artzner, Delbaen, Eber and Heath (2002). S-VaR is related to VaR by

$$S - VaR_q = VaR_q + E[X - VaR_q | X > VaR_q], \quad (1.34)$$

where the second term on the right is the mean of the excess distribution $F_{VaR_q}(y)$ over the threshold VaR_q . McNeil (1999a) shows that VaR and S-VaR are related through the following equation

$$\frac{S - VaR_q}{VaR_q} = \frac{1}{1 - \xi} + \frac{\beta - \xi u}{(1 - \xi)VaR_q}, \quad (1.35)$$

which can be estimated by substituting data-based estimates for everything which is unknown to obtain for the S-VaR

$$\widehat{S - VaR_p} = \widehat{VaR_p} \left(\frac{1}{1 - \hat{\xi}} + \frac{\hat{\beta} - \hat{\xi}u}{(1 - \hat{\xi})} \right). \quad (1.36)$$

Note that the first term in the explanation on the right hand side is a constant, the second term is almost zero. Therefore, the estimated S-VaR is equal to the estimated VaR multiplied by a constant. Finally, confidence intervals for the estimator $\widehat{S - VaR_p}$ can be added, using the profile likelihood technique.

1.5 An Application for the Dow Jones Index Return Series

1.5.1 The POT method at work

In the following section, the estimation procedures are implemented for the Dow Jones Index series. The data consists of 19'427 data points, from October 1928 to December 2001. Figure 1.1 (upper figure) shows the plot of the respective time series. Evidently, the level of the process changes dramatically over the observed period, hence, issues about extremes of daily behavior are swamped by

long-term time variation in the series. However, an approximation to stationarity can be obtained by taking logarithms of ratios of successive observations - the log-daily returns. Indeed, Figure 1.1 (lower figure) suggests a reasonably successful transformation to stationarity.

For the sake of convenience, the log-daily returns are first re-scaled, i.e. multiplied by a factor of 100, $\tilde{X} \rightarrow 100\tilde{X}$. Next, in order to make a choice about the threshold u , the mean residual life plot is estimated as depicted in Figure 1.2. The plot is initially linear, but shows substantial curvature in the range $-2 \leq u \leq 2$. For $u > 5$ the plot is reasonably linear when judged relative to confidence intervals, suggesting that we set $u = 5$ (i.e. a threshold of a 5% log-daily return). This threshold results in a total number of 49 exceedances, as the following estimation output for the Dow Jones Index series suggests.³⁴

\$threshold (the chosen threshold after re-scaling the data):

[1] 5

\$nexc (the number of exceedances of the chosen threshold):

[1] 49

\$conv (true or false indicator of whether the likelihood has been satisfactorily maximized or not):

[1] T

\$nllh (negative log-likelihood at the maximum likelihood estimate):

[1] 77.83434

\$mle (maximum likelihood estimates of σ and ξ , respectively):

[1] 1.1865877 0.4173744

\$rate (proportion of points exceeding the threshold):

[1] 0.002522263

\$se (approximate standard errors of σ and ξ , respectively):

[1] 0.3606846 0.2748518

\$

The maximum likelihood estimates of the GPD scale and shape parameters are $(\hat{\sigma}, \hat{\xi}) = (1.186, 0.417)$, with standard errors of 0.360 and 0.274, respec-

³⁴ All the computations are performed in the statistical language S-Plus, using the software package S-Plus 2000 for Windows. See Krause and Olson (2000) for a good introduction to S-Plus. Most of the functions are written by Prof. Stuart Coles, "http://www.maths.bris.ac.uk/~masgc/ismev/summary.html", some by Prof. Alexander McNeil, "http://www.math.ethz.ch/~mcneil/software.html". Gençay, Selçuk and Ulugülyağci (2001b) provide a step-by-step guideline for extreme value analysis in the MATLAB environment.

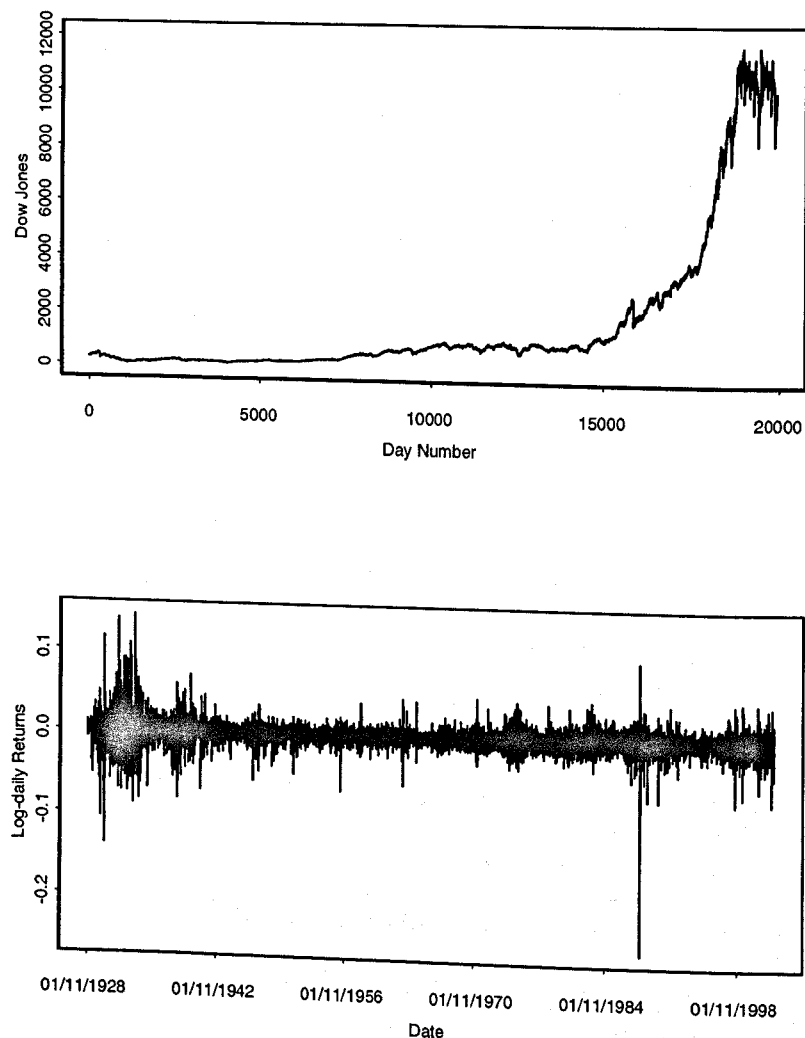


Figure 1.1: Dow Jones Index series (upper figure) and Log-daily returns of Dow Jones Index series (lower figure).

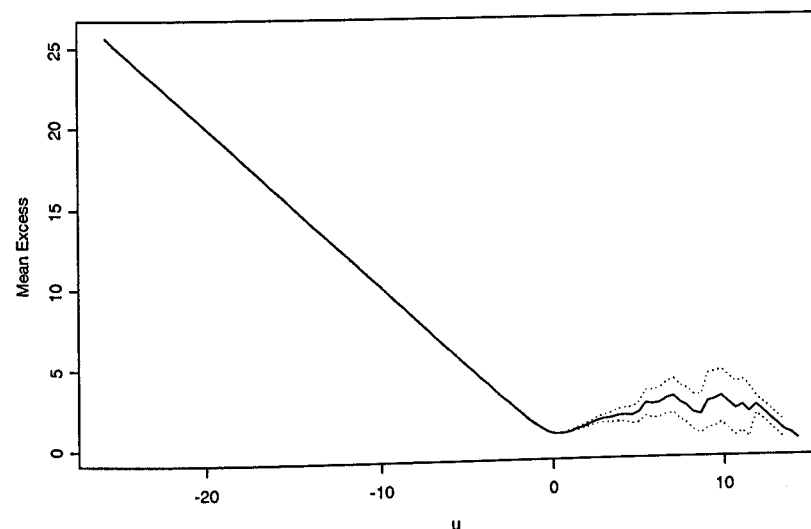


Figure 1.2: Mean residual life plot of Dow Jones log-daily returns. Note that the data has been re-scaled. The plot shows reasonable linearity for $u > 5$, if judged relative to confidence intervals.

tively.³⁵ The validity of the chosen threshold of 5 can be further assessed by checking stability with respect to u of the maximum likelihood estimates for the reparameterized model. Estimates are carried out over the range $u = 0$ to $u = 7$, equally spaced in order to give 100 estimates in total. The output is shown in Figure 1.3, supporting the use of threshold $u = 5$.

Alternatively, we can estimate the shape parameter ξ non-parametrically by means of the Hillplot. Figure 1.4 shows the Hill estimates for different numbers of highest order statistics for the Dow Jones data. The relation between the shape parameter ξ and the Hill-estimator $\hat{\alpha}$ (tail index) is given by

$$\hat{\xi} = 1/\hat{\alpha}. \quad (1.37)$$

³⁵The respective tail index estimate is 2.39, indicating fat tails in the return distribution.

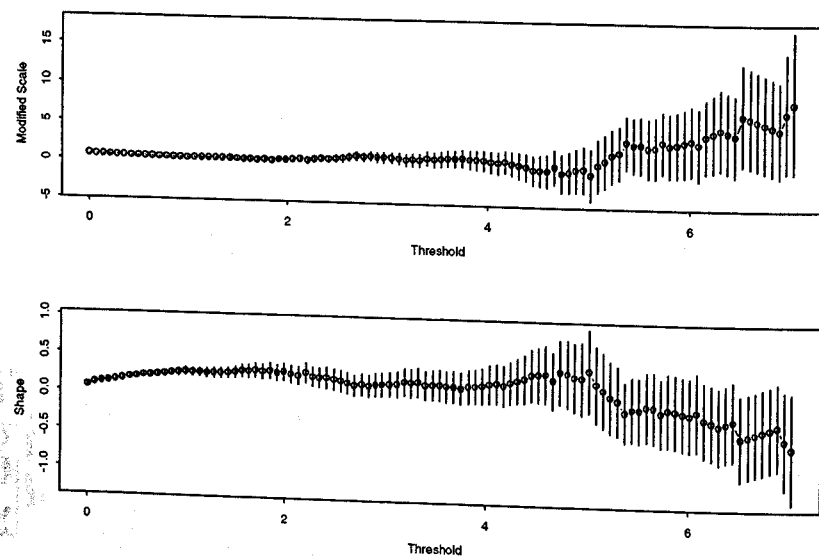


Figure 1.3: Maximum likelihood estimates of the reparameterized generalized Pareto model (i.e. of $\hat{\sigma}$ and $\hat{\xi}$) as a function of the threshold, for Dow Jones log-daily returns, with 95% confidence intervals.

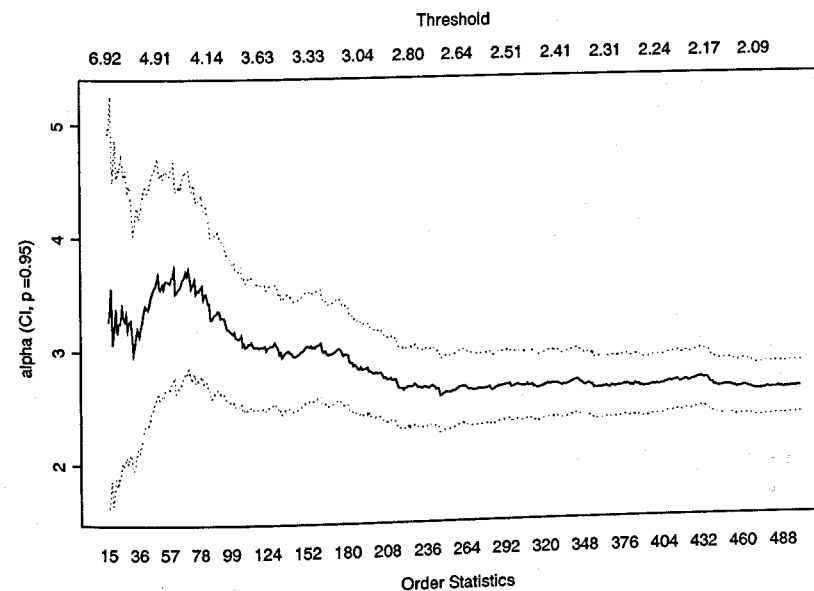


Figure 1.4: Hillplot for Dow Jones data; the plot shows the value of the tail index (using 1.14) on the vertical axis and the threshold and number of highest order statistics, respectively, on the horizontal one. Number of highest order statistics = 500.

Figure 1.5 shows different diagnostics of the fitted object. The goodness-of-fit in the quantile plot as well as in the probability plot, respectively, seem a bit unconvincing, as shown by departures from linearity. However, the confidence intervals on the return level plot suggest that the model departures are not large after allowance for sampling. The return level plot illustrates the very large uncertainties that accrue once the model is extrapolated to higher levels (quantiles). Furthermore, the concave departure from the ideal shape in the QQ-plot (Figure 1.6) indicates that the tails are heavier tailed than those from an exponential distribution. Convexity, on the other hand, would indicate a thinner tailed distribution. Overall, the diagnostic analysis does lend support to the fitted GPD model. As a consequence, we further assess the tails of the distribution for risk management purposes.

1.5.2 Financial Risk Measurement

In risk management, we are interested in the losses of our profit/loss distribution function. In order to calculate the exceedances of a certain loss level, we change the sign of the returns, because the POT method defines exceedances *above* a threshold u . As we are interested in risk measures (i.e. the VaR and the S-VaR), we start by calculating summary statistics about the Dow Jones series. E-views gives the output as shown in Figure 1.7. Clearly, the hypothesis of a normal distribution can be rejected.³⁶ Furthermore, the kurtosis is very high (30.97), confirming the suggestion of fat tails. The maximum positive and negative daily returns over the sample period are 14.34% and 25.63%, respectively (with a mean of some 0.01%). The skewness of 0.6608 indicates an asymmetric distribution. Again we use the POT method to estimate the parameters

\$threshold:

[1] 5

\$nexc:

[1] 60

\$conv:

[1] T

\$nllh:

[1] 90.1165

\$mle:

³⁶E-views applies a Jarque-Béra test with the null-hypothesis of normally distributed returns.

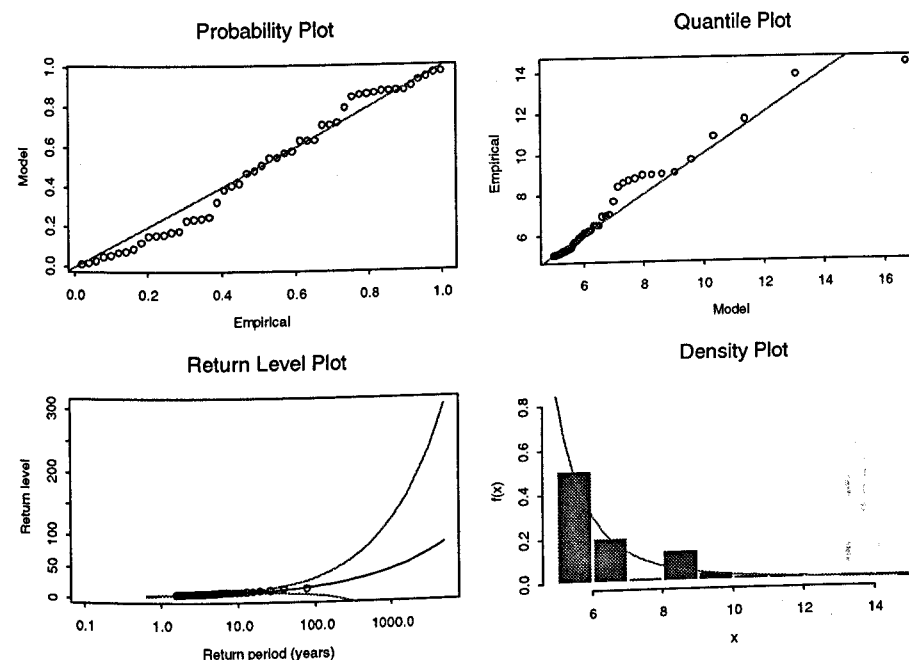


Figure 1.5: Diagnostic plots for threshold excess analysis of Dow Jones log-daily returns; probability plot, quantile plot, return level plot and density plot are shown, respectively.

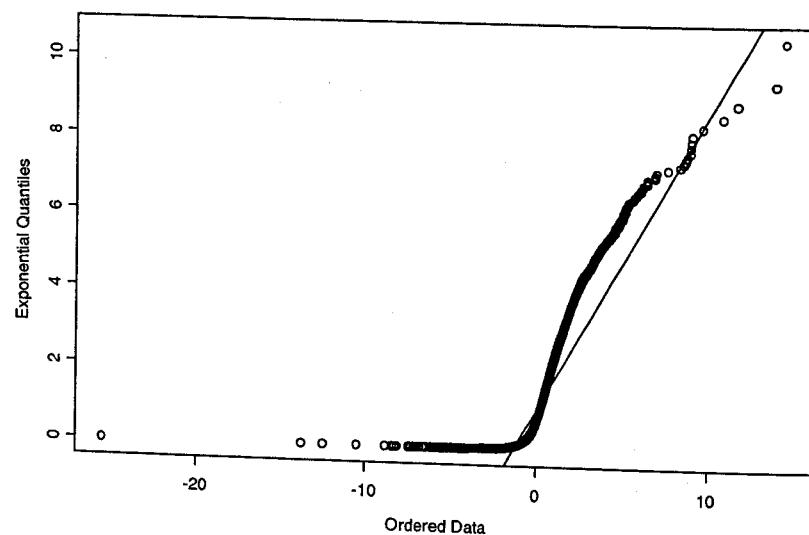


Figure 1.6: QQ-plot for the Dow Jones Index data against the exponential distribution. The concave departure from the ideal shape indicates that the tails are heavier tailed than those from an exponential distribution.

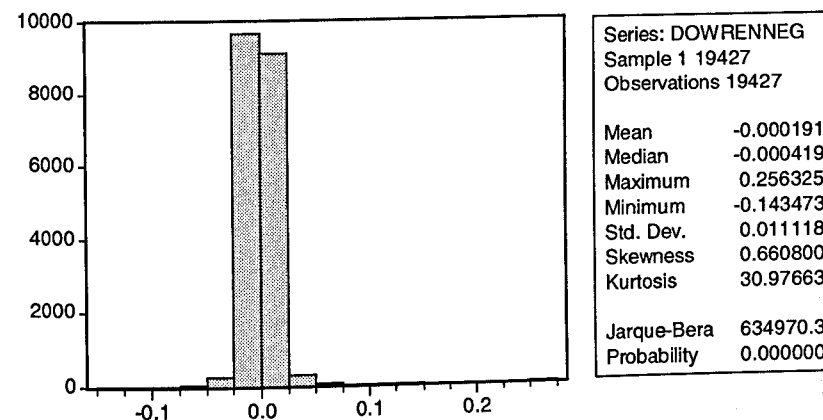


Figure 1.7: Summary statistics for Dow Jones Index series.

[1] 1.2091141 0.3120524

\$rate:

[1] 0.003088485

\$se:

[1] 0.2364285 0.1522880

\$

The excess (loss) distribution is shown in Figure 1.8 (upper figure). The estimated GPD model for the excess distribution, is shown as a smooth curve. The empirical distribution of the 49 extreme values is shown by points in the same figure. It is evident that the GPD model fits these excesses reasonably well. The tail of the underlying distribution is depicted in Figure 1.8 (lower figure). In the latter figure, the y-axis indicates the tail probabilities $1 - F(x)$. The x-axis shows the different thresholds. The points represent the 60 exceedances and the solid curve shows how the tail estimation formula allows extrapolation into the area where the data become a sparse and unreliable guide to their unknown parent distribution.

In risk management we are interested in the VaR and S-VaR for different extreme quantiles. Table 1.1 shows the (static) results for the Dow Jones series. The Historical VaR, Normal VaR and the Extreme VaR (Extreme S-VaR) are

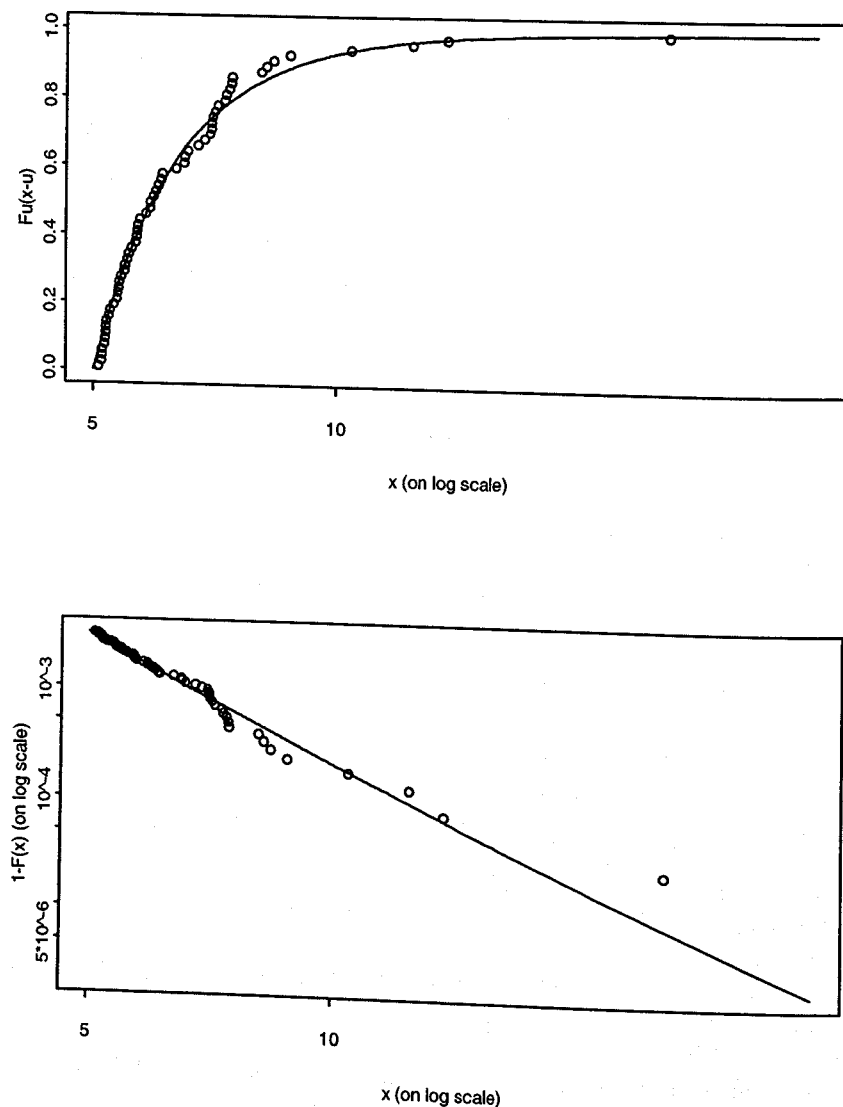


Figure 1.8: Excess (loss) distribution for Dow Jones data (upper figure, using 1.23) and Tail of underlying distribution (lower figure, using 1.24).

	Historical VaR	Normal VaR	Extreme VaR	Extreme S-VaR
95%	1.55698	1.828978	2.750450	3.487626
99%	3.217397	2.587253	3.810712	5.028822
99.5%	4.28637	2.862989	4.459170	5.971421
99.9%	6.89149	3.435587	6.634202	9.133045
99.99%	12.447267	4.024862	12.426339	17.552491

Table 1.1: Daily relative VaR's for different quantiles; Historical VaR, Normal VaR and Extreme VaR, respectively.

shown for different extreme quantiles (i.e. the 95%, 99%, 99.5%, 99.9% and the 99.99% quantiles).³⁷ Clearly, both the VaR based on the normal distribution assumption and the Historical VaR underestimate the risk for high quantiles. The 99.9% VaR estimated using extreme value theory is 6.63%. In Figure 1.9 (upper figure) the 99.9% VaR is shown on the horizontal x-axis. The vertical dotted line intersects with the tail estimate at the point (6.63, 0.001) and allows the VaR estimate to be read off the x-axis. The dotted curve is a tool to enable the calculation of the confidence interval for the VaR. The first y-axis on the left shows the respective quantile, i.e. the quantile scale. The second y-axis on the right of the graph is a confidence scale. The horizontal dotted line corresponds to 95% confidence: the x-coordinates of the two points where the dotted curve intersects the horizontal line are the boundaries of the 95% confidence interval (6.19, 7.26). Note that we can observe a wider 99% confidence interval by dropping the horizontal line down to the value 99 on the (second y) confidence axis. Furthermore, the confidence interval is asymmetric as expected. The asymmetric confidence intervals for the estimates (i.e. the risk measures) are obtained using the profile likelihood method. The estimate for the S-VaR is 9.13%, which is marked with a second vertical line in Figure 1.9 (lower figure). Again using the profile likelihood method a 95% coincidence interval for the S-VaR estimate is added (7.90, 14.35; again the x-coordinates of the two points where the dotted curve intersects the horizontal line). Note that the uncertainty about the value of this (coherent) risk measure is large, however, this was to be expected with such heavy-tailed data.

³⁷ Note that the VaRs are calculated using the whole range of data. Normal VaR is calculated with the respective standard deviations over the range, for Historical VaR the data is ordered, and the respective quantiles are read from the order statistics.

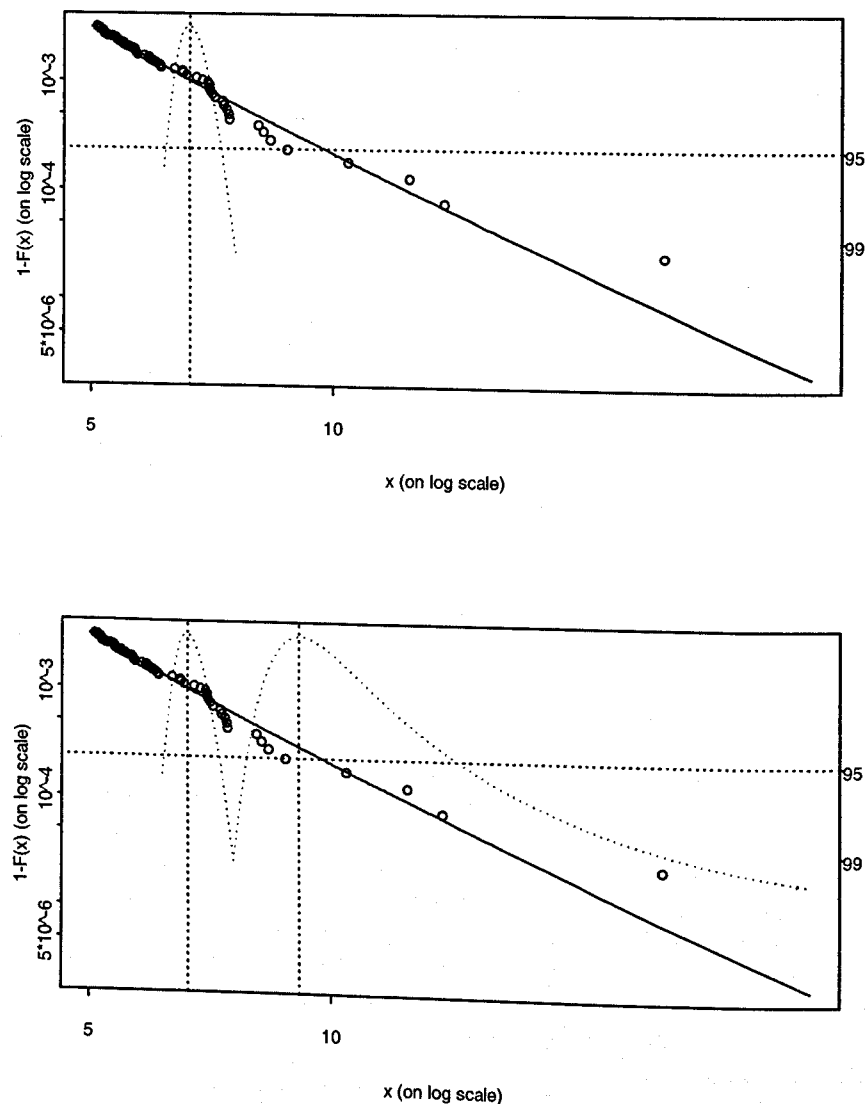


Figure 1.9: 99.9% Extreme VaR for Dow Jones series, with 95% confidence intervals (upper figure), 99.9% Extreme VaR and Extreme S-VaR for Dow Jones series, with confidence intervals (lower figure)

Finally, the VaR calculation methods are backtested. To backtest the VaR models on a historical series, we calculate the daily VaRs for different days, using a time window of 1000 days each time.³⁸ Hence, we use about four years of data for each VaR calculation. Every time, the VaR estimate (i.e. the predicted maximum value which is not exceeded in $(1-q)\%$ of all cases) is compared with the realization of the next day. In a long backtest like this it is less feasible to examine the fitted model carefully every day (as done above) and to choose a different threshold each time. Instead, the threshold (or the number of exceedances, respectively) is fixed. As a consequence, the algorithm to estimate the parameters may fail from time to time. In such cases, we take the VaR estimation of the preceding day. Table 1.2 shows the expected number of exceedances and the actual number of exceedances for the Historical VaR, the Normal VaR and the Extreme VaR calculation methods. Clearly, for the 95% quantile, the EVT calculation does not help to improve the VaR calculation. A simple historical VaR does the best job. However, for quantiles of 99% or higher, the Extreme VaR gives most accurate results, i.e. the least difference from the expected number of violations. The results are well in line with Frey and McNeil (2000).³⁹ Figure A.1 in the appendix shows the backtesting results for the Historical VaR and the Normal VaR, respectively.⁴⁰ The figures on the left show the log-daily returns and, superimposed, the estimated VaRs.⁴¹ The figures on the right show the differences between the realized VaRs and the respective predicted VaRs. Hence, values above the zero y-axis show VaR estimates which were exceeded. The figure is aimed to give hints about the magnitude of the VaR violations.

1.6 Underlying Assumptions and Applications

1.6.1 The *iid* assumption

The models discussed in Sections 1.2.1 and 1.2.2 assume an underlying series of independent and identically (*iid*) distributed random variables. This makes

³⁸Note that with 19427 datapoints, we calculate 18427 daily VaRs. I gratefully thank Claudia Saupper for her professional support with programming in S-Plus.

³⁹Note that Frey and McNeil (2000) applied a binomial test in order to test for too few or too many violations. For further details, see Chapter 3.

⁴⁰In addition, the Appendix shows the backtesting results for Historical VaR (Figure A.2) and Normal VaR (Figure A.3), respectively, for the 95%, 99%, 99.5% and 99.9% quantiles.

⁴¹Note the step-like pattern for the Historical VaR result. The "next" step is taken, if another extreme value is added to range.

	95%	99%	99.5%	99.9%
Expected	0921	184	092	18
Normal VaR	0826	319	231	61
Historical VaR	0954	226	130	48
Extreme VaR	1205	198	114	36

Table 1.2: Backtesting results; expected and realized number of observations, length of test: 18427 observations.

the arguments simpler, but it is widely agreed that high-frequency financial asset returns are conditionally heteroscedastic, and hence not *iid*. In particular, extreme conditions often persist over several consecutive observations, bringing into question the appropriateness of the discussed models.

Coles (2001) deals with stationary sequences - series that may be dependent through time, but that have a homogeneity of stochastic behaviour. Dependence in stationary series can take many different forms, and it is impossible to develop a general characterization of the behaviour of extremes unless some constraints are imposed. It is usual to assume a condition that limits the extent of long-range dependence at extreme levels; extreme events are close to independent at times that are far enough apart. Coles (2001) eliminates long-range dependence at extreme levels in this way, focusing attention on the effect of short-range dependence. Moreover, he argues that such dependence in data can be ignored, so far as the modelling of block maxima is concerned. Diebold, Schuermann and Stroughair (2000) agree, however, name the drawbacks; the aggregation (e.g. the use of weekly, monthly or annual maxima) reduces efficiency, and the problem of the appropriate amount of aggregation is introduced.

Regarding the POT approach, various suggestions have been made for dealing with the problem of dependent exceedances in the threshold exceedance model. The most widely-adopted method is declustering, which corresponds to a filtering of the dependent observations to obtain a set of threshold excesses that are approximately independent. Coles (2001) describes the following steps:

1. Using an empirical rule to define *clusters of exceedances*
2. Identifying the maximum excess within each cluster
3. Assuming cluster maxima to be independent, with conditional excess distribution given by the GPD

4. Fitting the GPD to the cluster maxima

The method is simple, but has its limitations. In particular, results can be sensitive to the arbitrary choices made in cluster determination and there is arguably a wastage of information in discarding all data except the cluster maxima. Leadbetter, Lindgren and Rootzén (1983) give a detailed account of the whole development of the general characterization of extremes of stationary processes. Overall, subject to certain limitations, the usual extreme value limit models are still applicable in the presence of temporal dependence.

Another alternative to deal with dependence is to estimate the tail of the conditional, rather than the unconditional, distribution. The two coincide in the *iid* case only. The idea is to fit the tail of the conditional distribution by first fitting a conditional volatility model (such as a GARCH model), standardizing the data by the estimated conditional standard deviation, and then estimating the tail index of the standardized data. The idea was proposed by Frey and McNeil (2000) and, independently, by Diebold, Schuermann and Stroughair (2000). If the adopted volatility model is a good approximation to the true volatility dynamics, then the standardized residuals will be approximately *iid*, legitimating the use of the standard EVT models.⁴²

Another issue is to deal with non-stationary processes - processes whose stochastic behaviour changes through time, perhaps due to trends or seasonality. In such cases, little can be derived in terms of formal probability results, and it is usual to adapt a pragmatic approach of using the standard extreme value models as basic templates and enhance them by statistical modelling. In a linear trend model, for example, variations through time in the observed process can be modeled as a linear trend in the location parameter of the appropriate extreme value model, such as the GEV distribution. Using the notation $H(\xi; \mu, \sigma)$ to denote the GEV distribution with parameters ξ, μ and σ , it follows that a suitable model for Z_t , the annual maximum sea-level for example, might be

$$Z_t \sim H(\xi; \mu(t), \sigma), \quad (1.38)$$

where

$$\mu(t) = \beta_0 + \beta_1 t, \quad (1.39)$$

for parameters β_0 and β_1 . The time-homogenous model forms a special case of the time-dependent model, with $\beta_1 = 0$, in which case $\mu = \beta_0$.⁴³ For some

⁴²The estimation of the tail of the conditional distribution is shown in the application in Chapter 3.

⁴³For more complex models, see Coles (2001), Chapter 6.

simple studies of the theoretical properties of particular classes of non-stationary processes, see Leadbetter, Lindgren and Rootzén (1983).

1.6.2 The power law assumption

Most of the tail estimation literature assumes that returns are in the maximum domain of attraction of a Fréchet distribution, so that the tail of the survival function follows a power law. The crucial question are how to check that assumption, and what the evidence is. We will come back to the latter questions in the last chapter of the present thesis.

1.6.3 Applications of EVT in finance and insurance

There are several potential applications of EVT in finance and insurance. Some examples are:

- Improving risk management by better estimating the tails of return distributions.
- The measurement of diversification effects under heavy tails, which implies that standard portfolio theory which is based on risk-return profiles in a mean variance framework is not strictly correct.
- Derivatives pricing, especially for exotic derivatives (e.g. basket options), is strongly affected by the presence of heavy tails. Some examples are options where the payout is a function of the largest value of an underlying security over a given period of time, options triggered whenever a well specified extreme move takes place or Alternative Risk Transfer (ART) products as there are catastrophe bonds. For some products one may be able to derive analytic formulas, for others careful simulation is required.
- Margin setting in futures markets, since the latter is well known to be sensitive to the occurrence of large price changes. EVT could be used to derive the margin level for a given probability of margin violation desired by margin committees or brokers.

1.7 Conclusion

The aim of Chapter 1 has been to introduce the fundamentals of EVT, to show its practical aspects for estimating, to show its basic applications in Financial

Risk Measurement and to critically discuss its underlying assumptions.

The main virtue of EVT is that it gives the user a critical view on and a methodological toolkit for issues like skewness, fat tails, rare events and stress scenarios. More specifically, the first analysis in the present chapter showed the potential of EVT when calculating tail-related risk measures for extreme quantiles. The concept of VaR and S-VaR will be introduced in the context of conditional model in Chapter 3. The use of a conditional EVT approach will show how to further improve the measurement of risk.

Chapter 2

Extremal Linkages in Crisis Periods

2.1 Introduction

Dependencies between different kinds of risks are an important issue to be considered in firm-wide risk management. It is possible, for example, that strong adverse market movements are accompanied by insufficient market liquidity and an increase in credit risk premia. The accurate modelling of dependencies is a challenge on the way to an integrated risk management framework including market, liquidity, credit and possibly other risks.

Risk management does not usually take into account the whole distribution of the value of a portfolio but only makes predictions about the loss potential in stress situations. As a consequence, it does make sense to concentrate on the tails of the respective distributions which mainly determine the stress loss potential both for single risks and for a combination of risks. The classical technique to model the tails of distributions is EVT, which has often been applied in risk management. In addition to the modelling of extreme events of single time series, EVT also permits to model and determine multivariate extremal dependencies between different kinds of risk.

The pitfalls of using *correlations* in the analysis of tail dependence between risks are described in Embrechts, McNeil and Straumann (2000). Linear correlation is a natural dependence measure for multivariate normally and, more generally, elliptically distributed risks. For other distributions, however, lin-

ear correlations can be misleading. Several empirical studies show that the multivariate normal distribution underestimates the frequency of joint extreme market exceedances. In addition, linear correlation coefficients can be zero even if there is a high spill-over probability. Mainly for this reason we do not want to rely on the statistical concept of linear correlation as a measure of the interdependence between markets during times of stress.

In order to avoid the potential problems of the linear correlation measure, multivariate EVT has been employed. Basically, there are parametric and non-parametric approaches to estimate the parameters of the respective distributions. While the number of univariate contributions increases steadily, multivariate financial applications are only beginning to appear. Longin and Solnik (2000) and Starica (1999, 2000) apply parametric models for tail dependence. Longin and Solnik (2000) use the logistic function¹ as proposed by Gumbel (1960) to model the form of the dependence function. They explore the use of multivariate extreme value methods for stock market returns and reject the null hypothesis of multivariate normality for the left tail. Starica (1999, 2000) models time series in a constant conditional correlation framework and estimates an empirical measure for tail dependence. He uses the latter to calculate probabilities of large movements in an exchange rate conditional on the occurrence of extreme returns in another exchange rate and finds a high level of dependence between the extreme movements of most of the currencies in the EU. Straetmans (1998, 2000a, 2000b), Hartmann, Straetmans and De Vries (2000) apply fully non-parametric approaches. The analysis builds on the so-called Stable Tail Dependence Function (STDF)² which measures the amount of interdependence between the tail probabilities of multiple random variables. With these estimates they calculate conditional spillover probabilities.³ Hartmann, Straetmans and De Vries (2000) find co-crashes between stock and bond markets and some evidence of extreme cross-border linkages. Bac, Karolyi and Stulz (2001) go another line of research and apply a (nonlinear) multinomial logistic regression model to evaluate contagion in financial markets. They find that contagion, when measured by the coincidence within and across regions of extreme shocks, is predictable and depends on regional interest rates, exchange rate changes and conditional stock return volatility. Their approach is, however, similar to the multivariate extreme value approach.

¹Note that the logistic function is also known as the Gumbel copula, an extreme value copula, see Chapter 4. Longin and Solnik (2000) estimate the copula by means of Maximum Likelihood.

²See also Embrechts, De Haan and Huang (2000) for the theoretical foundations.

³Spillover means that a crash in one market has external effects on another one.

In the present chapter, we focus on bivariate EVT and specifically on *extremal* dependence between different kinds of risk. We calculate a *linkage measure* which indicates whether markets move together in turbulent periods or not. We apply the model proposed by Straetmans (1998) and Hartmann, Straetmans and De Vries (2000), using a non-parametric approach. Our starting questions for the present chapter are:

- How to measure extremal linkages?
- Do financial markets crash jointly or is the fall of one the gain to another market?
- How is extremal dependence, for different asset classes, different from dependence as assumed by the (bivariate) normal distribution?
- Is tail dependence a prevalent phenomenon across different asset classes?

The results can provide useful insights for risk and economic capital calculations, e.g. VaR forecasting. We will use the results of this chapter further in Chapter 4 to set up a model for the calculation of portfolio VaR. It should be noted at this point that the focus in the present chapter is on measures of tail dependence and estimating spillover probabilities, whereas in Chapter 4 the focus is on general dependence concepts and their applications in portfolio risk management.

In our extensive survey across different asset classes, we measure extremal dependence between single assets and market indices, extremal dependence between international stock market returns, spillovers across foreign exchange rate regimes and extremal dependence between credit spreads and market returns. We directly measure and report the expected number of exceedances of a certain threshold, conditional on the event that there is at least one exceedance. Because of the use of non-parametric estimators for the limiting tail dependence function, the probability law of the joint process can be left unspecified, this in contrast to correlation-based approaches which measure market linkages.

In a related work, Buhl, Reich and Wegmann (2002) study the extremal dependence of market and liquidity risk, the former being measured through the market return and the latter being measured through the relative bid-ask spread. They apply a non-parametrical approach to measure bivariate exceedance probabilities and the respective dependence function. Their analysis for the Swiss Market indicates moderate tail dependence in that roughly 10% of the exceedances of the 99% quantile are co-exceedances.

The chapter is structured as follows.⁴ In the following section, we show how to measure extremal dependence by means of the Stable Tail Dependence Function (STDF) and we discuss its asymptotic and in-sample properties. Estimation and testing procedures are discussed in the third section. In Section 4, we implement the model for different data sets. We draw the conclusions and give an outlook in the last section.

2.2 The Measurement of Extremal Linkages

In statistics, the concept of asymptotic dependence can be formally defined in terms of the conditional exceedance probability. Let (X, Y) represent a pair of asset returns, and let y and x be the excess levels above which we speak of an extreme event, i.e. of a market crash.⁵ Furthermore, we again adopt the convention to take the negative of a return, in order to study all extreme events in the first quadrant. It follows that (X, Y) are asymptotically or tail independent if the conditional exceedance probability (or conditional crash probability) defined on the joint probability distribution converges to zero, i.e.

$$\lim_{y, x \rightarrow +\infty} \Pr\{Y > y \mid X > x\} = 0. \quad (2.1)$$

The ranking of asset returns according to the criterion in 2.1 about asymptotic dependency may be reversed by changing the conditioning time series. In order to have a numéraire invariant version for asymptotic independence, we estimate the following:

$$\lim_{y, x \rightarrow +\infty} \frac{\Pr\{X > x, Y > y\}}{\Pr\{X > x \text{ or } Y > y\}} = 0. \quad (2.2)$$

In words, asymptotic independence means that the probability of a simultaneous "crash" (or extreme event), given that there is at least one crash, is equal to zero. This definition does not allow, however, to make any inference about the causality of the crashes.

In practice, we may be interested in the likelihood of extremal but bounded spillovers. Generally, let κ be the number of crashes that simultaneously occur in a bivariate system. In a finite-sample framework, we have the following

⁴The results of this chapter have been presented at the CaRisMa workshop I / 2002 in Zurich. I thank the participants of the respective workshop, especially Dr. Patrick Wegmann and Christian Buhl, for helpful comments.

⁵More generally, we speak of a crisis, e.g. of a currency crisis, a stock market crisis, etc.

expression for the likelihood of simultaneous declines in κ different financial markets (i.e. the conditional crash probability)

$$\Pr\{\kappa = 2 \mid \kappa \geq 1\} = \frac{\Pr\{X > x, Y > y\}}{\Pr\{X > x \text{ or } Y > y\}}. \quad (2.3)$$

An alternative expression for the asymptotic dependence is $E\{\kappa \mid \kappa \geq 1\}$, i.e. the expected number of crashes or attacks that may simultaneously occur, given that there is a crash somewhere (i.e. the conditional crash expectation). From elementary theory of probability, we have

$$E\{\kappa \mid \kappa \geq 1\} = \Pr\{\kappa = 2 \mid \kappa \geq 1\} + 1 = \frac{\Pr\{X > x\} + \Pr\{Y > y\}}{1 - \Pr\{X \leq x, Y \leq y\}}, \quad (2.4)$$

or written in a compact way

$$E\{\kappa \mid \kappa \geq 1\} = \frac{p_1 + p_2}{p_{12}}. \quad (2.5)$$

In order to estimate the exceedance probabilities p_1, p_2 and p_{12} , it is crucial to choose the right distribution function, which should correctly capture the empirical regularities of the respective asset returns. We can proceed either parametrically or non-parametrically. The disadvantage of the former is the nonnestedness of alternative parametric models in the parameter space. As a consequence, the estimates are dependent on the maintained hypothesis. In order to circumvent this problem, Straetmans (1998) suggests a robust non-parametric approach. The tail is estimated without any knowledge about the underlying distribution function.

In the following sections we give the theoretical basis as well as estimators for the two linkage measures 2.3 and 2.5, respectively.⁶

2.2.1 Uni- and Multivariate Extreme Value Theory Results

The conditional crash probability 2.3 and the conditional crash expectation 2.5 are estimated by applying (multivariate) EVT. Following Straetmans (1998), this results in a two-step approach. In the first, *univariate* step, the stylized fact of tail fatness is exploited in order to estimate the univariate exceedance probabilities. For this purpose, univariate EVT provides the respective limit law

⁶The following sections are heavily based on Straetmans (1998).

(i.e. the three extremal types of distributions introduced in the first chapter) for the maxima of a return series in which all fat tailed distributions are nested with respect to their tail index. Conditional upon the estimates of the tail index, inverse quantile estimators are calculated in order to estimate the univariate probabilities. Straetmans (1998) shows that the univariate tail probabilities can be expressed in closed form when n (i.e. the sample size) becomes large.

In the second, *bivariate* step, we will identify the so-called Stable Tail Dependence Function (STDF), which links the multivariate exceedance probability to the univariate exceedance probabilities. Consistently, the tail index as well as the tail dependence function are estimated non-parametrically.

The probability that the pair $(X_{n,n}, Y_{n,n})$ of the first n random variables is below a certain level x and y , respectively, is given by the following distribution function:⁷

$$\Pr \{X_{n,n} \leq x, Y_{n,n} \leq y\} = F^n(x, y). \quad (2.6)$$

Bivariate extreme value theory studies the limiting distribution of the pair of order statistics $(X_{n,n}, Y_{n,n})$, appropriately scaled. One is interested in suitable normalizing constants $a_n > 0, c_n > 0$ and d_n such that:⁸

$$\lim_{n \rightarrow +\infty} \Pr \left\{ \frac{X_{n,n} - b_n}{a_n} \leq x, \frac{Y_{n,n} - d_n}{c_n} \leq y \right\} = H(x, y), \quad (2.7)$$

where $H(x, y)$ is the *bivariate extreme value distribution* (a limiting distribution) and F is said to be in the domain of attraction of H . The class of limit distribution functions in the above equation is the class of max-stable distributions.⁹ If the definition for max-stability holds for $H(x, y)$, then it should also hold for the univariate counterparts H_1 and H_2 , for the same set of scaling constants.¹⁰ Furthermore, max-stable marginals are one of the three classical extreme value distributions studied by Gnedenko (1943), as introduced in the first chapter of the present thesis.

There is, unfortunately, no closed form solution for the bivariate exceedance probability, i.e. the dependence function. In the next section we will identify the extremal dependence structure by means of the STDF, which allows us to specify $H(x, y)$ or, in words, the bivariate tail non-parametrically.

⁷Note that $X_{i,n}$ represents the i -th ascending order statistic of X with $X_{1,n} \leq \dots \leq X_{n,n}$.

⁸Recall the Block Maxima approach or Classical Extreme Value theory from the first chapter. This is essentially its bivariate extension.

⁹For a definition of max-stable distributions, see Embrechts, Klüppelberg and Mikosch (2001), p. 121.

¹⁰See Straetmans (1998), p. 62.

2.2.2 The Stable Tail Dependence Function: Asymptotic and In-Sample

Suppose that $F(x, y)$ is a bivariate probability distribution function with continuous and strictly increasing marginal distributions F_1 and F_2 . The dependence function C_F (also and more common called copula) of F is defined by

$$C_F(u, v) = F(F_1^{-1}(u), F_2^{-1}(v)), \quad 0 \leq u \leq 1, 0 \leq v \leq 1, \quad (2.8)$$

where

$$F_i^{-1}(x) = \inf \{y \mid F_i(y) \geq x\}, \quad i = 1, 2, \quad (2.9)$$

are the general inverse functions of F_1 and F_2 . Note that through the transformation in 2.8, the marginal distributions of F are uniformly distributed. Copulas represent a way to extract the dependence structure from the joint distribution and to distinguish between the marginal behaviour and the dependence.

In order to make inferences about extremal dependencies, we need a tail version of the dependence function C_F , the STDF l_F as defined in Huang (1992):¹¹

$$l_F(u, v) := \lim_{t \rightarrow +0} t^{-1} [1 - D_F(1 - tu, 1 - tv)] \quad (2.10)$$

$$= \lim_{t \rightarrow +0} t^{-1} [1 - F(Q_1(tu), Q_2(tv))] \quad (2.11)$$

$$= -\ln G(u^{-1/\alpha_1}, v^{-1/\alpha_2}), \quad (2.12)$$

where F is the distribution function of (X, Y) , and $Q_i := (1 - F_i)^{-1}$ represents the corresponding marginal quantile functions for some positive u, v and t . Multivariate Extreme Value Theory (MEVT) deals with existence conditions,¹² properties and estimators for this function as can be seen in Huang (1992)¹³ and De Haan and De Ronde (1998). Applied to our conditional crash probability $\Pr(\kappa = 2 \mid \kappa \geq 1)$, we have the following asymptotic version¹⁴

¹¹A tail version of a copula, or in other words an extreme value copula which is estimated non-parametrically in what follows.

¹²The limit function is guaranteed to exist if the joint distribution of X and Y is in the domain of attraction of a multivariate extreme value distribution.

¹³For some general results on the estimation of the STDF, see Huang (1992), Chapter 2. For the specific estimation and testing procedures of the STDF in a bivariate tail, see Huang (1992), Chapter 3.

¹⁴See Hartmann, Straetmans and De Vries (2000), p. 7.

$$\Pr\{\kappa = 2 \mid \kappa \geq 1\} \rightarrow \lim_{t \rightarrow +0} \frac{\Pr\{X > Q_1(tu), Y > Q_2(tv)\}}{\Pr\{X > Q_1(tu) \text{ or } Y > Q_2(tv)\}} = \frac{u+v}{l_F(u,v)} - 1, \quad (2.13)$$

and the respective asymptotic analogue for the conditional expectation $E(\kappa \mid \kappa \geq 1)$,

$$E\{\kappa \mid \kappa \geq 1\} \rightarrow \lim_{t \rightarrow +0} \frac{t^{-1} \Pr\{X > Q_1(tu)\} + t^{-1} \Pr\{Y > Q_2(tv)\}}{t^{-1} (1 - \Pr\{X \leq Q_1(tu), Y \leq Q_2(tv)\})} = \frac{u+v}{l_F(u,v)}. \quad (2.14)$$

Exploiting the homogeneity property of the STDF,¹⁵ one can show that the bivariate excess probability and the marginal probabilities are related via the STDF,¹⁶ i.e. the following approximation identity (for large n) results in a finite-sample framework:

$$p_{12,n} \approx l_F(p_{1,n}; p_{2,n}), \quad (2.15)$$

with $p_{1,n} = 1 - F_1(a_n, y)$, $p_{2,n} = 1 - F_2(c_n, y)$, and $p_{12,n} = 1 - F(a_n x, c_n y)$, where $p_{1,n}$ and $p_{2,n}$ represent the univariate sample exceedance probabilities. Consequently, the joint probability $p_{12,n}$ only depends on the marginal probabilities $p_{1,n}$ and $p_{2,n}$, and once l_F is known, and using 2.3 and 2.5, respectively, we get the following expressions for the conditional crash probability and the conditional crash expectation, respectively

$$\Pr\{\kappa = 2 \mid \kappa \geq 1\} := \frac{p_1 + p_2}{l_F(p_1, p_2)} - 1, \quad (2.16)$$

$$E\{\kappa \mid \kappa \geq 1\} := \frac{p_1 + p_2}{l_F(p_1, p_2)}. \quad (2.17)$$

In the next section we derive estimators for the univariate probabilities, the STDF and the resulting estimators for the linkage measures.

2.2.3 Estimating the Univariate Probabilities, the STDF and the Linkage Measure

In order to estimate the (possible) out-of-sample probability $\Pr\{X > x \text{ or } Y > y\}$, we concentrate on the following estimator:

¹⁵The property states that the STDF is asymptotically homogenous of degree one, e.g. $l(au, av) = al(u, v)$, $\forall a > 0$.

¹⁶See Hartmann, Straetmans and de Vries (2000).

$$\widehat{p}_{12,n} = \widehat{l}_F(\widehat{p}_{1,n}; \widehat{p}_{2,n}). \quad (2.18)$$

The estimation is done in two steps. In the first step, the univariate exceedance probabilities for fat tailed marginals are estimated as follows:¹⁷

$$\widehat{p}_{i,n} = \frac{k_i}{n} \left(\frac{X_{n-k_i,n}}{s} \right)^{\widehat{\alpha}_i}, \quad i = 1, 2, \quad (2.19)$$

where $X_{n-k_i,n}$ is the $(n - k)$ -th ascending order statistic from a sample of size n , such that $\lim(1/k(n)) = 0$, but $k = o(n)$, and where the extreme probability-quantile combination $(\widehat{p}_{i,n}, s)$ has to be such that $s > X_{n-k_i,n}$. The idea behind the univariate probability estimate is that it extends the empirical distribution function outside the domain of the sample by means of its asymptotic Pareto tail. However, in risk management we are interested not only in the exceedance probability, but also in the VaR of our position. Therefore, we apply the following estimator for the value of the quantile¹⁸

$$\widehat{x}_{i,p} = \left(\frac{n}{k_i} (1 - p) \right)^{-1/\widehat{\alpha}_i} X_{i,n}, \quad i = 1, 2, \quad (2.20)$$

where p is the respective quantile (e.g. the 0.99 quantile). The quantile estimators are still conditional upon knowledge of the tail indexes α_1 and α_2 , which can be estimated by use of the Hill estimator $\widehat{\xi}_i$.¹⁹

$$\widehat{\xi}_i = \frac{1}{\widehat{\alpha}_i} = \frac{1}{k_i} \sum_{j=0}^{k_i-1} \ln \frac{X_{n-j,n}}{X_{n-k_i,n}}, \quad i = 1, 2, \quad (2.21)$$

where k equals the number of highest order statistics used in the estimation of the univariate exceedance probability, and where $\widehat{\alpha}_i$ is the corresponding tail index estimate. Note that the lower the alpha, the fatter the tails of the distribution. Furthermore, for a t-distribution, the tail index α_i equals the number of degrees of freedom. Figure 2.1 shows an example of a Hill plot.

In the second step, we estimate the bivariate tail dependence function, the STDF. This can be done by either adopting a specific functional form for the

¹⁷See Dekkers (1991), De Haan, Jansen, Koedijk and De Vries (1994) or Embrechts, Klüppelberg and Mikosch (2001), p. 330.

¹⁸See Embrechts, Klüppelberg and Mikosch (2001), p. 330.

¹⁹See Hill (1975), Jansen and De Vries (1991) or Embrechts, Klüppelberg and Mikosch (2001). Recall Section 1.3.3 in the introductory chapter.

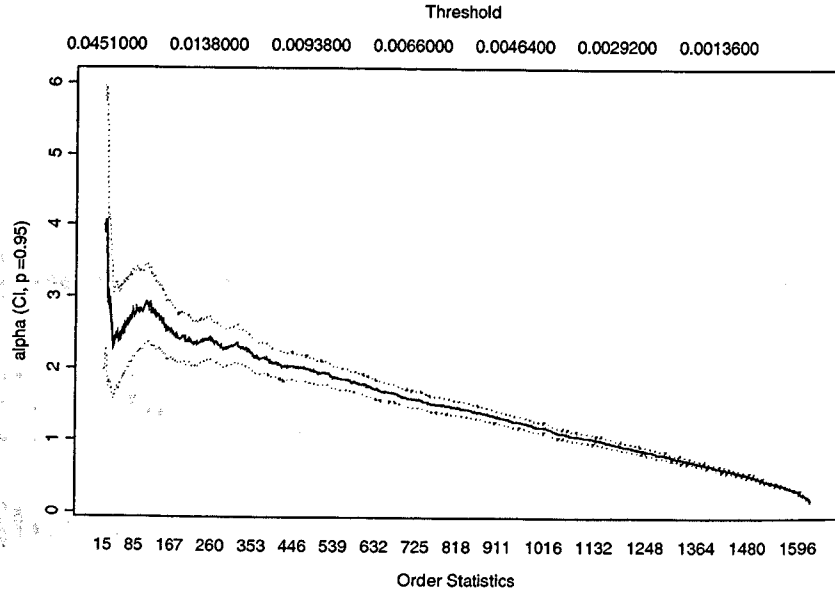


Figure 2.1: Example of a Hill plot, for the SMI index. The plot shows the value of the tail index (using 2.21) on the vertical axis and the threshold and number of highest order statistics, respectively, on the horizontal one.

STDF²⁰ or by proceeding non-parametrically.²¹ Recall definition 2.15 from the last section, here in terms of $(\hat{p}_{1,n}; \hat{p}_{2,n})$:

$$\hat{l}_F(\hat{p}_{1,n}; \hat{p}_{2,n}) = \frac{n}{m} \cdot \frac{1}{n} \sum_{i=1}^n \mathbf{I} \left\{ X > Q_1 \left(\frac{m \hat{p}_{1,n}}{n} \right) \text{ or } Y > Q_2 \left(\frac{m \hat{p}_{2,n}}{n} \right) \right\}, \quad (2.22)$$

where $\mathbf{I}$ stands for the indicator function, m is the number of highest order statistics and the summation represents the number of points in the area

$$\left\{ (X_i, Y_i)_{i=1, \dots, n} \mid X_i > Q_1 \left(\frac{m \hat{p}_{1,n}}{n} \right) \text{ or } Y_i > Q_2 \left(\frac{m \hat{p}_{2,n}}{n} \right) \right\}. \quad (2.23)$$

Q_1 and Q_2 are so large that there may be no observations left in the area defined by Equation 2.23. However, the homogeneity property allows us to scale up the arguments of the quantile function in order to decrease the quantile values and increase the number of excesses. Naturally, the estimator should then be premultiplied by the inverse of this 'scaling up' factor in order to leave $\hat{l}_F$ invariant. A popular candidate for this operation is the following polar transformation²²

$$\hat{p}_{1,n} = \hat{\rho}_n \cos \hat{\theta}_n \text{ and } \hat{p}_{2,n} = \hat{\rho}_n \sin \hat{\theta}_n. \quad (2.24)$$

The approximate homogeneity of $\hat{l}_F$ results in

$$\hat{l}_F(\hat{p}_{1,n}; \hat{p}_{2,n}) \approx \hat{\rho}_n \hat{l}_F(\cos \hat{\theta}_n, \sin \hat{\theta}_n). \quad (2.25)$$

Furthermore, conditional upon knowledge of the inverse quantile estimates, the angle θ and the corresponding radius ρ can now be consistently estimated by

$$\hat{\theta}_n = \arctan(\hat{p}_{2,n}/\hat{p}_{1,n}) \text{ and } \hat{\rho}_n = \sqrt{\hat{p}_{1,n}^2 + \hat{p}_{2,n}^2}. \quad (2.26)$$

Again, $\hat{l}_F(\cos \hat{\theta}_n, \sin \hat{\theta}_n)$ can be estimated non-parametrically by the number of points in the area

$$\left\{ (X_i, Y_i)_{i=1, \dots, n} \mid X_i > Q_1 \left(\frac{m}{n} \cos \hat{\theta}_n \right) \text{ or } Y_i > Q_2 \left(\frac{m}{n} \sin \hat{\theta}_n \right) \right\}, \quad (2.27)$$

²⁰See for example Longin and Solnik (2000). They apply a logistic function.

²¹See Straetmans (1998), p.71.

²²See Huang (1992).

divided by n . Next, we replace Q_1 and Q_2 by their empirical counterparts, and it follows the estimator of the dependence function which was first introduced by Huang (1992)²³

$$\hat{l}_F(\hat{p}_{1,n}; \hat{p}_{2,n}) = m^{-1} \hat{p}_n \sum_{i=1}^n \mathbf{I} \left\{ X_i > X_{[n-m \cos \hat{\theta}_n], n} \text{ or } Y_i > Y_{[n-m \sin \hat{\theta}_n], n} \right\}. \quad (2.28)$$

Finally, we estimate our two linkage measures, i.e. the following estimator for 2.16

$$\widehat{\Pr} \{ \kappa = 2 \mid \kappa \geq 1 \} := \frac{\hat{p}_1 + \hat{p}_2}{\hat{l}_F(\hat{p}_1, \hat{p}_2)} - 1, \quad (2.29)$$

as well as for 2.17

$$E \{ \kappa \mid \kappa \geq 1 \} := \frac{\hat{p}_1 + \hat{p}_2}{\hat{l}_F(\hat{p}_1, \hat{p}_2)}. \quad (2.30)$$

Note that if both returns are completely dependent in the tails, $E \{ \kappa \mid \kappa \geq 1 \} \approx 2$, and the markets co-crash with certainty. On the other side, without extreme co-movements in the two markets $E \{ \kappa \mid \kappa \geq 1 \} \approx 1$. Note that in the following empirical work we report the conditional crash expectation $E \{ \kappa \mid \kappa \geq 1 \}$ as linkage measure.

The entire estimation procedure depends essentially on three easily to calculate estimators 2.19, 2.21 and 2.28. Conditional on the proper choice of the threshold parameters m and k , respectively, the three estimators are asymptotically normally distributed. In order to determine m and k , Huang (1992) suggests to pick them such that they are in the range which minimizes the respective asymptotic mean-squared errors. In small samples, however, it is best practice to determine m and k by plotting the estimators as a function of the thresholds and to select m and k in the region over which the estimators are more or less constant.²⁴

²³See Huang (1992), p. 42.

²⁴See Section 1.3.1 in the introductory chapter for a discussion about how to choose the threshold.

2.3 Data and Hypothesis Testing

Starting from observing extreme crashes in stock markets and currency crises, we want to apply the extremal linkage measure as proposed in the last sections in order to detect possible spillover effects. Financial market linkages are studied for different asset classes. First, in order to have extremal dependence results for the Swiss market, the dependence between 14 SMI-titles²⁵ and the SMI itself is estimated. The dataset contains daily returns from 12.03.1993 to 28.12.2001, a total of 2295 observations. Furthermore, in order to compare liquid with less liquid stocks, we include 5 middle and small capitalized stocks,²⁶ respectively, in our bivariate analysis.

Second, the same analysis is carried out for the extremal dependence between the major international stock markets, i.e. between the G-7 (Japan, France, Germany, USA, Canada, UK and Italy) stock index returns. We use both local currency denominated returns and US\$ denominated stock returns. Stock price indices are on a daily basis and are extracted from datastream, the time series starting on 30.12.1987 and ending on 11.04.2002, a total of 3581 observations.

Third, we calculate extremal dependence between exchange rates under a free float regime and under a fixed exchange rate regime, respectively. Previous studies as in Straetmans (1998) found that fixed exchange rate regimes have fatter tails. We use daily data from 10.04.1987 to 12.04.2002, a total of 3915 observations. We use bilateral nominal spot exchange rates for seven countries (Belgium, France, Netherlands, Italy, UK, Ireland and Denmark) relative to the £ Sterling. Cross-currency CHF rates and Dmark rates are calculated by exploiting the no triangular arbitrage restriction. Straetmans (1998) carried out a similar analysis for a different time period and a different quantile. Furthermore, he compared US\$ currency returns (floating regime) with Dmark currency returns (fixed regime), respectively.

Finally, the extremal dependence between the credit spread and the market return is calculated. The data has been provided by Moody's. We use the credit spread, having daily data for industrial and public-utility long-term bonds with ratings from Aaa to Baa (dating from 08.04.1997 to 18.04.2002, 1170 observations) and weakly data for industrial and public-utility intermediate-term bonds with ratings from Aaa to Baa (dating from 06.08.1997 to 17.04.2002, 243

²⁵All the data throughout the chapter is extracted from Datastream. The titles included are ABB, Adecco, Baloise, Credit Suisse, EMS, Julius Bär, Nestlé, Novartis, Richemont, Roche, Sulzer, SwissRe, UBS and Zurich.

²⁶The mid cap titles are Agie Charmilles, Huber & Suhner, Conzetta, Oz Holding and Jelmoli, the small cap titles Batigroup, Kardex, Nextrom, Swiss Steel and Eichhof.

observations).

For testing purposes, Peng (1999) proposes a hypothesis test for asymptotic independence under the null hypothesis of asymptotic dependence. As the power of this test is very low, one may simply examine the number of simultaneous extreme observations in order to test for asymptotic dependence. The null hypothesis H_0 states that the asset return distributions are independent. Under the null hypothesis, the number of observations of asset returns simultaneously exceeding the p -quantile is binomially distributed $B(n, p^2)$. Furthermore, the expected number of values for an independent sample consisting of n datapoints in the bivariate exceedance region is $n \times (1 - p)^2$. Hence, for our given data sample (extremal dependence between single stocks and the SMI) consisting of 2295 datapoints and the 99% quantile, we expect 0.23 observations to be in the bivariate exceedance region. In order to test for extremal dependence, we apply a binomial test, where we test for significant departure of the realized number of observations from the expected number of observations. More specifically, we perform a one-sided binomial test of the null hypothesis of independence against the alternative of extremal dependence. A p -value less than 0.05 (0.01) will be interpreted as evidence against the null hypothesis.

2.4 An Exploratory Analysis

2.4.1 Extremal dependence between single stocks and the SMI

Univariate Analysis

Table 2.1 shows the minimum and maximum values of the continuously compounded returns of the considered SMI-titles. For example, the daily maximum (minimum) return in the given period was 16.30% (22.57%). Furthermore, columns 6 and 7 show the tail-index estimators $\hat{\alpha}_i$ for the left hand side (i.e. losses) and for the right hand side (i.e. gains) of the respective empirical distributions. As it is common practice, the number of highest order statistics used in the estimation, k , is selected in the region over which the Hill-plot is more or less constant. We separate between the left and right tails in order to examine the different tail characteristics of positive and negative returns. Note that the tail-index estimates for the left tail, i.e. the loss side estimates, are in all cases smaller than their right tail counterparts, indicating fatter tails on the left hand

Title	Min	Max	k left	k right	Tail left	Tail right
ABB	-0.2037	0.1262	35	27	2.5940	2.7367
ADECCO	-0.2234	0.1630	18	25	3.4037	4.2517
BALOISE	-0.0922	0.1219	29	60	2.6702	3.3613
CS GROUP	-0.1406	0.1281	33	34	2.8902	3.2362
EMS	-0.0867	0.0870	20	36	3.2154	3.1857
JULIUS BAER	-0.1001	0.0948	30	25	3.5932	5.2576
NESTLE	-0.0604	0.0593	24	47	4.3011	3.7258
NOVARTIS	-0.0748	0.1823	11	20	4.4014	3.3987
RICHEMONT	-0.1358	0.1136	40	35	3.2289	3.4176
ROCHE	-0.0725	0.0725	23	34	4.0469	4.5825
SULZER	-0.1839	0.1586	23	40	3.5778	2.7996
SWISS RE	-0.1902	0.1243	40	39	2.6490	2.9700
UBS	-0.1542	0.1054	22	52	2.9386	3.0309
UNAXIS	-0.1384	0.1062	30	53	3.4388	3.4676
ZURICH	-0.2257	0.1408	28	20	2.5880	2.7760

Table 2.1: Univariate analysis for market returns. Min. and Max. returns, the number of highest order statistics and Hill estimates are given in columns 2/3, 4, 5 and 6/7, respectively.

side.²⁷ This is consistent (in almost every case) with the observed asymmetry between minimal and maximal stock returns, as shown in the left part of the table.

Table 2.2, on the other hand, shows the estimates for different quantiles, applying formula 2.20. We calculate the 95%, 99% and the 99.9% quantile values. These estimates are particularly interesting for risk management, as they can be regarded as the (daily) VaR measures for the respective SMI-titles. The values are well in line with the empirical quantiles. Furthermore, we calculate the VaR based on the normal distribution assumption. The respective results are shown in Table 2.3. As can be seen, the risk, as measured through the VaR based on the normal distribution assumption, is underestimated for high quantiles, such as the 99% and the 99.9% quantiles, while it is adequately measured for the 95% quantile.

Bivariate Analysis

Before showing the results, we calculate the linkage measure, assuming a bivariate normal distribution. Figure 2.2 shows, under the assumption of bivariate normally distributed random variables, the expected number of crashes

²⁷In order to calculate the tail index estimate for a normally distributed return series, we simulated a lognormal price process (with normally distributed returns), and found a tail index of about 12.

Title	VaR 95% left tail	VaR 99% left tail	VaR 99.9% left tail	VaR 95% right tail	VaR 99% right tail	VaR 99.9% right tail
ABB	0.0326	0.0607	0.1475	0.0279	0.0503	0.1166
ADECCO	0.0405	0.0649	0.1277	0.0502	0.0733	0.1260
BALOISE	0.0250	0.0457	0.1082	0.0299	0.0482	0.0957
CS GROUP	0.0320	0.0559	0.1239	0.0337	0.0554	0.1129
EMS	0.0176	0.0291	0.0595	0.0175	0.0290	0.0597
JUL. BAER	0.0317	0.0495	0.0940	0.0402	0.0545	0.0845
NESTLE	0.0257	0.0373	0.0638	0.0221	0.0341	0.0632
NOVARTIS	0.0275	0.0397	0.0670	0.0235	0.0378	0.0744
RICHEMONT	0.0345	0.0568	0.1159	0.0358	0.0573	0.1123
ROCHE	0.0232	0.0345	0.0609	0.0243	0.0345	0.0571
SULZER	0.0402	0.0631	0.1200	0.0285	0.0507	0.1153
SWISS RE	0.0210	0.0386	0.0920	0.0253	0.0435	0.0945
UBS	0.0292	0.0505	0.1105	0.0276	0.0470	0.1004
UNAXIS	0.0389	0.0622	0.1214	0.0390	0.0620	0.1205
ZURICH	0.0304	0.0566	0.1379	0.0272	0.0485	0.1112

Table 2.2: VaR estimation results using the non-parametric Hill estimator approach. Estimations are shown for the left and right tail, respectively, and for different quantiles.

Title	Normal VaR 95%	Normal VaR 99%	Normal VaR 99.9%
ABB	0.0335	0.0473	0.0630
ADECCO	0.0420	0.0594	0.0790
BALOISE	0.0273	0.0386	0.0513
CS GROUP	0.0319	0.0451	0.0599
EMS	0.0172	0.0242	0.0323
JULIUS BAER	0.0294	0.0415	0.0552
NESTLE	0.0212	0.0299	0.0398
NOVARTIS	0.0221	0.0313	0.0416
RICHEMONT	0.0342	0.0482	0.0642
ROCHE	0.0210	0.0296	0.0394
SULZER	0.0323	0.0457	0.0608
SWISS RE	0.0257	0.0363	0.0483
UBS	0.0289	0.0408	0.0543
UNAXIS	0.0372	0.0526	0.0699
ZURICH	0.0305	0.0431	0.0574

Table 2.3: VaR estimations based on the normal distribution assumption, shown for different quantiles.

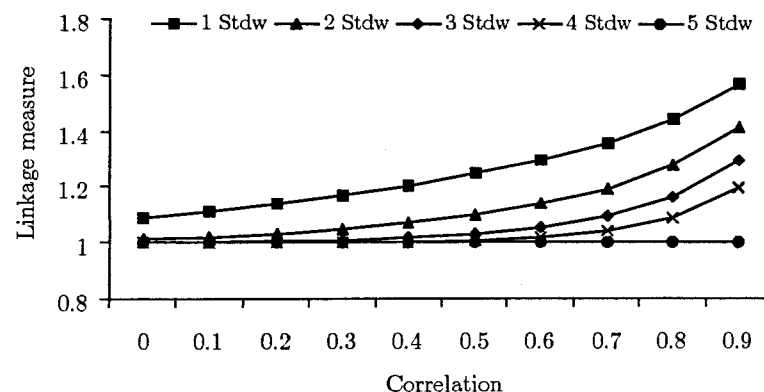


Figure 2.2: Expected number of crashes that may simultaneously occur, given that there is at least one crash (vertical axis), for different values of the correlation coefficient. The figure shows the results for bivariate normally distributed random variables. The five lines correspond to different excess values x and y of 1, 2, 3, 4 and 5 standard deviations.

according to Equation 2.30, for different values of the correlation coefficient and different values of excess levels x and y . At high excess levels of five standard deviations, the conventional correlation is clearly misleading, resulting in $E[\kappa | \kappa \geq 1] \approx 1$, meaning extremal independence, for all values of the correlation coefficient.²⁸

Table 2.4 contains the estimation results for the SMI-titles and SPI-titles, respectively.²⁹ The correlations are given in column 2, the number of highest order statistics k in column 3 and the conditional crash expectation as in 2.30 in column 4. Note that the correlation measures reported might easily read as suggesting higher linkage values than the results in column 4. However, as the correlation calculation assumes a normal distribution, we would find values of the linkage measure that are much less than the calculated ones.

It is important that the conditional crash probabilities throughout this chapter are calculated using univariate probabilities of 1%. This means that we do

²⁸Note that the conditional expectation for jointly normally distributed returns approaches 1.01 in the limiting case.

²⁹All the computations in the present chapter are done using MATLAB and Excel, respectively. The bivariate part is a joint work with Christian Buhl and Patrick Wegmann.

not calculate the exceedance probabilities over given levels of x and y as crash levels, but we rather take the exceedance probability as given (one percent or the 99% VaR).³⁰ Consequently, the linkage measure has the following interpretation: given that there is at least one crash, meaning that the one percent quantile is exceeded, what is the probability that there is a second crash? Hence, crashes are defined as exceedances of the 99% VaR values.

We infer from Table 2.4 that for the SMI-titles, on average, every third crash is a co-crash. Furthermore, the table shows some results for firms with an average market capitalization of 100 Mio CHF (i.e. small capitalized firms) and an average market capitalization of 500 Mio CHF (i.e. middle capitalized firms), respectively. Clearly, for firms with a lower market capitalization, the extremal dependence is, as expected, much lower. For the low cap firms, the average conditional probability is (only) 6.5%.³¹

A value for the conditional expectation of 1.10 means that the conditional bivariate probability for a simultaneous crash is 10%. In order to better understand what this implies, we derive the unconditional bivariate crash probability, using elementary probability theory. A value of 1.10 means that a simultaneous crash is 18 times more likely than for the case of independence. Assuming independence, the simultaneous exceedance probability of the 1% quantile is 0.01%, which happens once every forty years. However, for the value of 1.10 a simultaneous exceedance happens once every second year. After all, a seemingly small probability of 10% means that we are far from independence.

The calculation of the conditional crash expectation is justified by the results of the binomial test for independence. Column 5 of Table 2.4 shows the realized number of joint extremes, column 6 the p-values for the binomial test. The null hypothesis of independence in the tails is rejected for all SMI-title-returns, and the same is true for all but three SPI-title-returns, on a 5% significance level.³²

Figure 2.3 shows the example of a highly correlated SMI-title, namely Novartis. Values of the STDF (stable tail dependence function) and the conditional crash expectation are plotted against different values of the threshold k . Figure 2.4 shows the same result for the example of Eichhof, a low capitalized stock. It can be inferred that there is a trade-off between the validity of the applied measure for extremal dependence (i.e. small values for the number of highest

³⁰ This is in contrast to Hartmann, Straetmans and De Vries (2000) who set extreme returns and calculate the respective univariate probabilities.

³¹ Note that the conditional crash probability is calculated by simply subtracting one from the conditional expectation, according to 2.29.

³² Note that for the titles where the null is not rejected, we observe only one or zero joint extremes.

Title	Corr.	k	Ce	# obs.	p-value	Market Value
ABB	0.5887	80	1.2352	10	0.000	21103
ADECCO	0.4169	75	1.1529	06	0.000	8849
BALOISE	0.5442	60	1.3054	11	0.000	5149
CREDIT SUISSE	0.7661	70	1.3943	13	0.000	45933
EMS	0.4582	70	1.2531	08	0.000	2520
JULIUS BAER	0.5134	80	1.2712	11	0.000	3043
NESTLE	0.7263	65	1.4142	10	0.000	84641
NOVARTIS	0.7725	100	1.3469	11	0.000	106139
RICHEMONT	0.3708	50	1.1592	05	0.000	12186
ROCHE	0.7291	70	1.3199	09	0.000	82644
SULZER	0.467	60	1.2122	06	0.000	2692
SWISS RE	0.6798	62	1.4142	15	0.000	30220
UBS	0.7225	52	1.3875	13	0.000	44538
UNAXIS	0.3929	90	1.1264	04	0.000	2412
ZURICH	0.7193	65	1.3720	13	0.000	27757
AVERAGE	0.5911	70	1.2910	9.6	0.000	31989
AGIE CHARMIL.	0.1914	52	1.1310	06	0.000	450
HUBER+SUHNER	0.1918	55	1.1110	04	0.000	461
CONZETTA	0.1065	51	1.0160	00	1.000	511
OZ HOLDING	0.4495	43	1.3820	11	0.000	557
JELMOLI	0.2573	80	1.2430	04	0.000	584
AVERAGE	0.2393	56	1.1770	05	0.200	512
BATIGROUP	0.1933	45	1.0970	04	0.000	103
KARDEX I	0.1085	30	1.0100	00	1.000	110
NEXTROM	0.1647	47	1.1080	02	0.022	135
SWISS STEEL	0.1495	45	1.0610	02	0.022	139
EICHHOF	0.0855	55	1.0510	01	0.205	145
AVERAGE	0.1403	44	1.065	1.8	0.249	127

Table 2.4: Bivariate analysis for selected titles. Correlations, number of highest order statistics, conditional crash expectation and p-values for the binomial test are given in columns 2,3,4 and 6, respectively. Insignificant p-values are shown in italics.

order statistics k) and the stability of the linkage measure. Finally, Figures 2.5 and 2.6 show the realized number of joint extremes for Novartis and Eichhof, respectively, in the scatterplots.

2.4.2 Spillovers between international stock markets

Tables 2.5 (in US\$ denominated returns) and 2.6 (in local denominated returns) contain the estimation results for the pairs of G-7 stock index returns. Column 2 shows the correlations between index returns.³³ Furthermore, column 3 shows

³³ Note that the average correlation is 0.352 for the US\$ denominated returns and 0.358 for the local denominated returns.

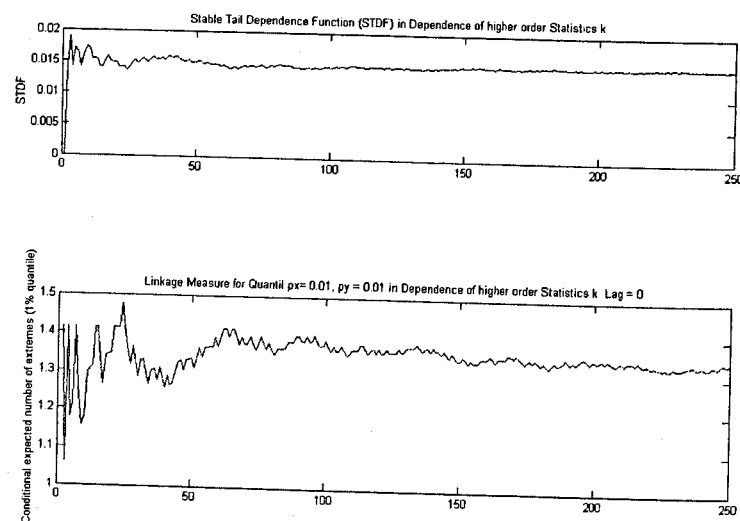


Figure 2.3: Example of high extremal dependence; Novartis. Stable tail dependence function (STDF) (upper figure, using 2.28) and Conditional crash expectation for different values of k (lower figure, using 2.30), with marginal probabilities of $(p_1, p_2) = (1\%, 1\%)$.

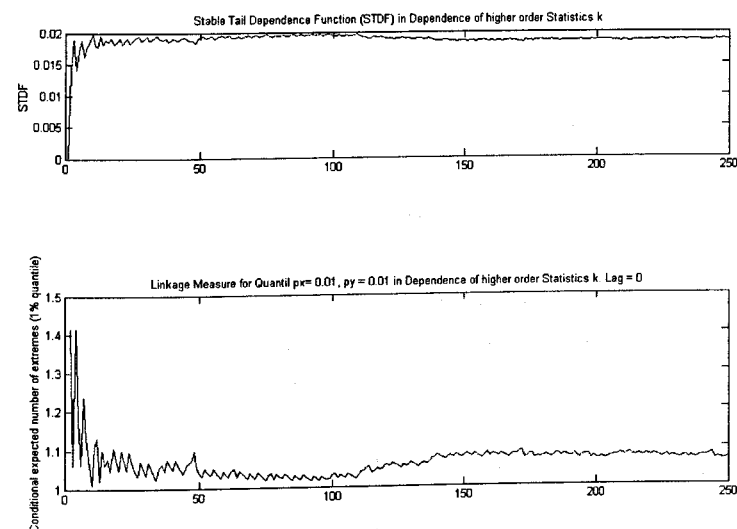


Figure 2.4: Example of low extremal dependence; Eichhof. Stable tail dependence function (STDF) (upper figure, using 2.28) and Conditional crash expectation for different values of k (lower figure, using 2.30), with marginal probabilities of $(p_1, p_2) = (1\%, 1\%)$.

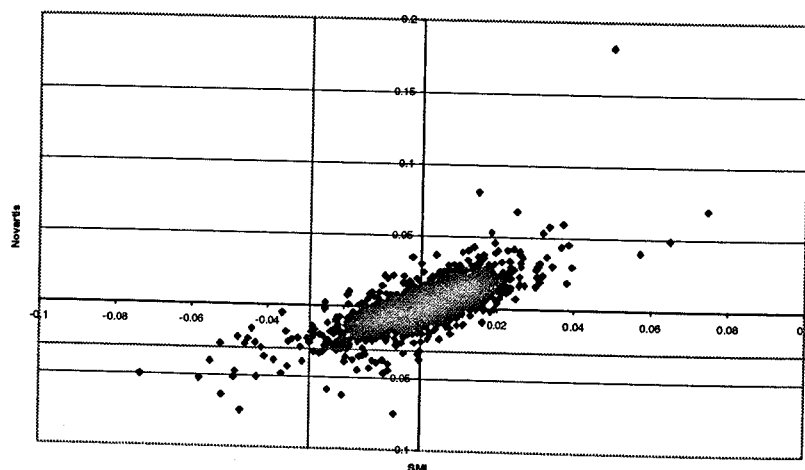


Figure 2.5: Scatterplot of the returns of Novartis vs. SMI: an example of high extremal dependence (11 observed joint extreme values).

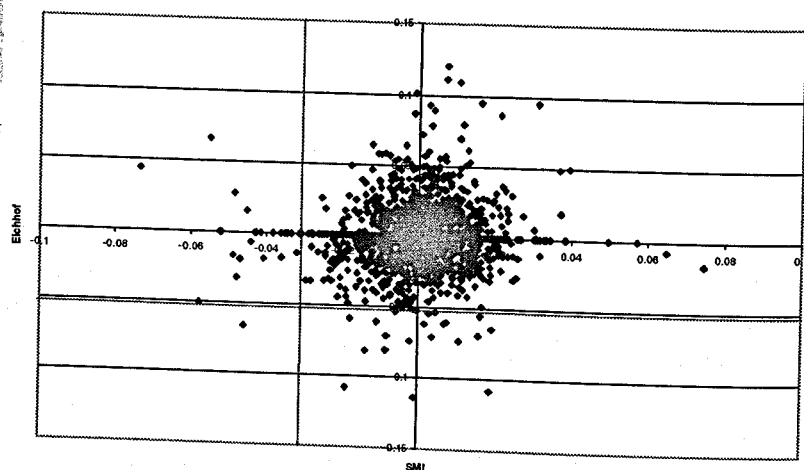


Figure 2.6: Scatterplot of the returns Eichhof vs. SMI: an example of low extremal dependence (one observed joint extreme value).

the extremal linkage measure and column 4 the number of highest order statistics, respectively. Finally, we report the number of observed joint extremes and the p-values for the binomial test in columns 5 and 6.

The hypothesis is that due to international stock markets integration, we would observe (some) dependence in extreme situations. Correlations and conditional crash expectations are calculated both for local currency denominated and for US\$ denominated stock index returns in order to control for the (intuitively) high tail dependence in US\$ exchange rates. All the exceedance probabilities are chosen as 0.01 (i.e. 1%).

Table 2.5 reveals that the average conditional expectation is 1.146 (US\$ denominated returns) and 1.145 (local denominated returns), respectively. Hence, the conditional expectation seems to be invariant under changing the currency of denomination. This goes somehow against intuition, because one would expect the US\$ denominated expectations to be upward biased as a result of the high asymptotic dependence of the US\$ rates. A possible explanation is the fact that stock markets trading at different places evidently exhibit different trading hours. Thus, spillovers may be dampened because a crash in one market may occur when other markets are closed. As Danielsson (2002) points out, this happened for example during the 1987 crisis where the main crash day was spread over two days in Europe and the Far East. A naive analysis would indicate that the US markets experienced larger shocks than other markets, however this is only an artifact of the data. Hence, any cross-country analysis is complicated by market opening hours, and is an additional layer of complexity.

Another observation is that the extremal dependence is not as strong as suspected. A comparison with the single stocks analyzed in the last section (average value of 1.290) confirms this. Straetmans (1998) and Hartmann, Straetmans and De Vries (2000) show similar results for different time periods.³⁴

In order to test for the integration of financial markets, Straetmans (1998) proposes to measure extremal dependence and tests whether asymptotic dependence between stock markets has remained constant over time, estimating the tail dependence over different subsample periods and exploiting the asymptotic normality of the tail dependence estimator in order to test for structural breaks. He found that stock market integration has not risen significantly after the 1987 stock market crash.

³⁴The time series in Straetmans (1998) starts on January 1, 1982 and ends on January 16, 1997. His conditional expectation values are somewhat lower than ours (averages of 1.077 and 1.089, respectively). This may be due to a different choice of the thresholds. Hartmann, Straetmans and De Vries (2000) use weekly data from 1987-1999 and find an average value of 1.160, also with a different threshold.

Index pair	Corr. (\$)	Ce (\$)	<i>k</i>	# observ.	p-value
Canada-US	0.56240	1.1875	110	13	0.000
Canada-UK	0.35185	1.1349	065	07	0.000
Canada-Italy	0.26322	1.1377	144	04	0.001
Canada-Germany	0.34071	1.1249	105	04	0.001
Canada-Japan	0.14231	1.0433	090	01	<i>0.311</i>
Canada-France	0.37358	1.1165	075	07	0.000
US-UK	0.28088	1.0984	080	05	0.000
US-Italy	0.18326	1.0711	128	04	0.001
US-Germany	0.25240	1.1314	132	04	0.001
US-Japan	0.07345	1.0555	103	03	0.006
US-France	0.29170	1.1314	080	07	0.000
UK-Italy	0.43751	1.2186	081	12	0.000
UK-Germany	0.50962	1.2342	096	14	0.000
UK-Japan	0.26703	1.0572	077	03	0.006
UK-France	0.59055	1.2186	081	10	0.000
Italy-Germany	0.54838	1.2923	053	16	0.000
Italy-Japan	0.18671	1.0552	097	04	0.001
Italy-France	0.57491	1.2478	105	17	0.000
Germany-Japan	0.24910	1.0639	082	03	0.006
Germany-France	0.68919	1.3873	103	19	0.000
Japan-France	0.22626	1.0583	110	03	0.006
Average	0.35214	1.1460	095	7.6	0.016

Table 2.5: Bivariate results for G-7 stock index returns: US\$ denominated returns. Correlations, conditional crash expectation, number of highest order statistics and p-values are given in columns 2,3,4 and 6, respectively. Insignificant p-values are shown in italics.

Index pair	Corr. (local)	Ce (local)	<i>k</i>	# observ.	p-value
Canada-US	0.611145	1.2231	096	13	0.000
Canada-UK	0.409583	1.1516	114	09	0.000
Canada-Italy	0.230715	1.0935	075	06	0.000
Canada-Germany	0.374028	1.1244	097	08	0.000
Canada-Japan	0.152408	1.0285	072	00	<i>1.000</i>
Canada-France	0.392942	1.1210	130	07	0.000
US-UK	0.370292	1.1277	122	10	0.000
US-Italy	0.167011	1.0815	078	05	0.000
US-Germany	0.301289	1.1531	053	08	0.000
US-Japan	0.115870	1.0566	065	02	<i>0.054</i>
US-France	0.333830	1.1392	087	09	0.000
UK-Italy	0.384642	1.2047	115	14	0.000
UK-Germany	0.562193	1.2692	105	18	0.000
UK-Japan	0.242047	1.0471	077	04	0.001
UK-France	0.674187	1.2431	109	21	0.000
Italy-Germany	0.468340	1.2607	156	14	0.000
Italy-Japan	0.161109	1.0706	081	04	0.001
Italy-France	0.421747	1.2075	111	14	0.000
Germany-Japan	0.247277	1.0607	093	04	0.001
Germany-France	0.670922	1.3226	101	21	0.000
Japan-France	0.234546	1.0508	107	04	0.001
Average	0.358386	1.1446	097	9.3	0.050

Table 2.6: Bivariate results for G-7 stock index returns: local denominated returns. Correlations, conditional crash expectation, number of highest order statistics and p-values are given in columns 2,3,4 and 6, respectively. Insignificant p-values are shown in italics.

Currency	Max (CHF)	k right (CHF)	Tail right (CHF)	Max (Dmark)	k right (Dmark)	Tail right (Dmark)
Belgian Franc	0.0220	36	3.9509	0.0244	45	2.9205
French Franc	0.0229	50	3.4443	0.0245	21	2.5157
Dutch Guilder	0.0230	48	3.6954	0.0064	44	3.1603
Italian Lira	0.0686	17	3.0799	0.0535	40	2.4898
Irish Punt	0.0674	43	3.7047	0.0703	40	3.0895
British Pound	0.0394	30	5.2849	0.0408	27	4.9268
Danish Krona	0.0269	27	3.5028	0.0286	22	3.1375

Table 2.7: Univariate estimation results for CHF and Dmark denominated currency returns. Max. values, the number of highest order statistics and Hill estimates are given in columns 2/5, 3/6 and 4/7, respectively.

2.4.3 Spillovers between different exchange rate regimes

From a macroeconomic perspective, we are interested in possible spillovers across foreign exchange regimes. The currency pairs under a floating exchange rate regime are CHF currency returns (i.e. currency pairs with the CHF as the numéraire), the pairs under a fixed exchange rate are Dmark currency returns (i.e. currency pairs with the Dmark as the numéraire).³⁵ In the literature on the empirical distribution of exchange rate returns, there is consensus that the tails for regulated currencies exhibit more probability mass (i.e. have fatter tails) than under a free float, see for example Straetmans (1998) or our own results in Table 2.7. Columns 2,3 and 4 show the maximum value, the number of highest order statistics and the tail index estimation (i.e. the Hill estimator), respectively. The average value for the right tail index in CHF is 3.808, whereas the respective value in Dmark is 3.177. This is somewhat contra intuition, as the popular belief is that target zones induce a reduction in extreme volatility. A possible explanation is that incredible exchange rate bands provoke recurrent speculative attacks. The implication of these *univariate* results is that the room for reducing extreme exchange rate volatility is limited in a target zone.

Next, we investigate whether the latter apparent policy ineffectiveness generalizes to a bivariate (multivariate) framework, focusing on the difference between foreign exchange regimes viewed from their bivariate tail. Tables 2.8 (floating exchange rate regime) and 2.9 (fixed exchange rate regime) contain the bivariate estimation results for the currencies,³⁶ respectively. The columns show the

³⁵ The Dmark as the numéraire is chosen as an example of the European Monetary System (EMS), a target zone regime.

³⁶ As mentioned earlier, Straetmans (1998) carried out a similar analysis for a different time

same results as the ones for the international index returns in the last section. Correlations and depreciation expectation (i.e. the conditional "crash" expectation as in 2.30) are calculated for pairs of exchange rates under both floating and fixed exchange rate regimes. In other words, we want to know whether the likelihood of an extremal spillover from one currency market to another is increased under a target zone compared to a free float regime.

Clearly, the time series of nominal exchange rates versus the Swiss Franc are (by far) more highly correlated than EMS currency returns. However, we know that a nonzero correlation may be well in line with a zero conditional crash expectation, depending on the type of the underlying distribution function and therefore calculate the proposed linkage measure. The average values are 1.204 (CHF denominated currency returns) and 1.087 (Dmark denominated currency returns), respectively. Consequently, the risk of spreading speculative attacks or the contagion effect is smaller in a target zone than under a floating exchange rate regime. Straetmans (1998) shows similar results for a different time period.³⁷

The question why tail dependence is influenced by the exchange rate regime remains unanswered so far. A possible explanation goes as follows. If there is an extreme movement of one particular currency within the EMS currency block against the CHF, there is a tendency for the other currencies within the block to do the same. On the other hand, if there is an extreme movement of one particular currency within the EMS currency block against the Dmark, it tends to be of a more specific nature between these two currencies, without affecting the other currencies.

A more concrete explanation uses the fact that in the European Monetary System, countries were forced to keep their exchange rates within prespecified bands. There is empirical evidence that national monetary policies are determined by German monetary policy (the so called German Dominance Hypothesis).

Overall, although exchange rate arrangements such as the EMS are generally considered as being rather ineffective in reducing the univariate occurrence of extremal returns, they reduce the likelihood that these returns occur simultaneously.

period and a different quantile. Furthermore, he compared US\$ currency returns (floating regime) with Dmark currency returns (fixed regime), respectively.

³⁷ The respective time series starts on January 1, 1973 and ends on January 9, 1997. Note that his conditional expectation values are somewhat higher than ours. This is partly due to a different choice of the thresholds. Furthermore, he chose a different floating rate regime, e.g. the US\$, which might further explain the higher values.

Currency pair	Corr. (CHF)	Ce (CHF)	<i>k</i>	# observ.	p-value
Belgium-France	0.773733	1.2947	130	17	0.000
Belgium-Holland	0.781836	1.3117	128	15	0.000
Belgium-Italy	0.556712	1.1033	110	11	0.000
Belgium-Ireland	0.567977	1.1165	105	10	0.000
Belgium-UK	0.404776	1.1020	060	05	0.000
Belgium-Danmark	0.802563	1.4009	105	21	0.000
France-Holland	0.879651	1.5103	105	27	0.000
France-Italy	0.709114	1.1467	120	13	0.000
France-Ireland	0.657233	1.1785	075	12	0.000
France-UK	0.488990	1.1548	138	10	0.000
France-Danmark	0.808080	1.3095	100	20	0.000
Holland-Italy	0.638504	1.1192	110	13	0.000
Holland-Ireland	0.648038	1.1177	098	12	0.000
Holland-UK	0.446572	1.1196	095	09	0.000
Holland-Danmark	0.786877	1.3322	065	20	0.000
Italy-Ireland	0.559303	1.1646	098	14	0.000
Italy-UK	0.481918	1.1480	069	09	0.000
Italy-Danmark	0.610282	1.1018	067	12	0.000
Ireland-UK	0.577806	1.3167	054	13	0.000
Ireland-Danmark	0.603256	1.1049	075	10	0.000
UK-Danmark	0.450576	1.1467	060	08	0.000
Average	0.630180	1.2047	094	13.9	0.000

Table 2.8: Bivariate results for currency returns: floating exchange rate regime. Correlations, conditional crash expectation, number of highest order statistics and p-values are given in columns 2,3,4 and 6, respectively. Insignificant p-values are shown in italics.

Currency pair	Corr. (Dmark)	Ce (Dmark)	<i>k</i>	# observ.	p-value
Belgium-France	0.303150	1.1136	100	05	0.000
Belgium-Holland	0.215051	1.0340	068	03	0.001
Belgium-Italy	0.137511	1.0378	102	02	<i>0.059</i>
Belgium-Ireland	0.138964	1.0302	110	03	0.001
Belgium-UK	0.112817	1.0285	128	02	<i>0.059</i>
Belgium-Danmark	0.502367	1.2240	148	12	0.000
France-Holland	0.189722	1.0528	067	00	<i>1.000</i>
France-Italy	0.413073	1.1940	141	08	0.000
France-Ireland	0.234205	1.1112	132	07	0.000
France-UK	0.235259	1.1068	126	02	<i>0.059</i>
France-Danmark	0.400008	1.1314	136	07	0.000
Holland-Italy	0.137721	1.0583	113	02	<i>0.059</i>
Holland-Ireland	0.108346	1.0557	053	01	<i>0.324</i>
Holland-UK	0.083966	1.0301	059	01	<i>0.324</i>
Holland-Danmark	0.196443	1.0741	120	02	<i>0.059</i>
Italy-Ireland	0.252725	1.1045	107	05	0.000
Italy-UK	0.287013	1.1249	140	03	0.001
Italy-Danmark	0.239113	1.0580	098	03	0.001
Ireland-UK	0.425958	1.1664	080	10	0.000
Ireland-Danmark	0.203502	1.0643	073	07	0.000
UK-Danmark	0.189184	1.0382	149	03	0.001
Average	0.238385	1.0875	107	4.1	0.092

Table 2.9: Bivariate results for currency returns: fixed exchange rate regime. Correlations, conditional crash expectation, number of highest order statistics and p-values are given in columns 2,3,4 and 6, respectively. Insignificant p-values are shown in italics.

2.4.4 Extremal dependence between credit risk and market risk

Finally, we are interested in the extremal dependence between credit risk and market risk, respectively. The former is measured through the credit spread, the latter through market returns. The credit spread is measured as the average spread of corporate bond yields to treasury yields, the average spread of industrial bond yields to treasury yields and the average spread of public utility bond yields to treasury yields, respectively.³⁸ The market risk is measured through the returns of the major (broad) US stock indices, namely the S&P 500 index, the Dow Jones Industrials index and the S&P Industrial index. Table 2.10 shows the main results. Column 1 shows the spreads against the chosen stock indices, which measure market risk. Column 2 shows the correlations, column 3 the calculation of the extremal linkage measure, column 4 the number of highest order statistics, column 5 the number of observed joint extremes and column 6 the p-values for the binomial test, respectively. The long-term bond spreads to treasury are based on seasoned bonds with remaining maturities of at least 20 years,³⁹ the intermediate-term bond spreads to treasury are based on seasoned bonds with remaining maturities averaging 7 years.⁴⁰ We infer from Table 2.10 that the average conditional expectation is 1.186 for the long-term bonds and 1.147 for the intermediate-term bonds, respectively.

2.5 Conclusion

In this chapter we focus on extreme events and characterize the linkage between different returns by their asymptotic tail dependence. We calculate non-parametric estimates for the expected number of exceedances of a high threshold given that there is at least one exceedance, i.e. the asset return is strongly negative, a currency depreciates extremely or the credit spread is very high. The main findings can be summarized as follows.

- On a univariate level, we apply the non-parametric Hill estimator technique and confirm that the normal distribution underestimates the tail

³⁸Note that the average spread of the corporate bonds to treasury is an unweighted average of the corresponding average industrial spreads and the average public utility spreads. See the Appendix for a complete list of the included bonds.

³⁹The daily data for the long-term bonds is from 01.08.1997 to 18.04.2002. The data was kindly provided by Moody's.

⁴⁰The weekly data for the intermediate-term bonds is from 06.08.1997 to 17.04.2002. The data was kindly provided by Moody's.

2.5. CONCLUSION

Spread/Index	Corr.	Ce	k	# obs.	p-value
Longterm Corp – S&P 500 Comp	-0.26606	1.1828	46	2	0.006
Longterm Indust – S&P 500 Comp	-0.25689	1.1909	48	1	<i>0.110</i>
Longterm Publ – S&P 500 Comp	-0.19492	1.1467	30	3	0.000
Average	-0.23929	1.1734	43	2	0.038
Intermed. Corp – Dow Jones Av.	-0.13565	1.1196	19	0	<i>1.000</i>
Intermed. Indust – Dow Jones Av.	-0.12392	1.1785	20	0	<i>1.000</i>
Intermed. Publ – Dow Jones Av.	-0.13452	1.1314	20	0	<i>1.000</i>
Intermed. Corp – S&P Indust.	-0.13869	1.1571	18	0	<i>1.000</i>
Intermed. Indust – S&P Indust.	-0.13336	1.1785	20	0	<i>1.000</i>
Intermed. Publ – S&P Indust.	-0.13605	1.1196	19	0	<i>1.000</i>
Average	-0.13369	1.1474	19	0	<i>1.000</i>

Table 2.10: Bivariate results for credit spread vs. market returns. Correlations, conditional crash expectation, number of highest order statistics and p-values are given in columns 2,3,4 and 6, respectively. Insignificant p-values are shown in italics.

risk for stock returns.

- On a bivariate level, we present empirical evidence for asymptotic dependence between different pairs of assets. Moreover, our results clearly show that there is more extremal dependence in the empirical data than predicted by the bivariate normal distribution.
- We test non-parametrically whether the number of bivariate threshold exceedances suggests tail dependence. Our hypothesis tests for independence are rejected for confidence levels of 5% and particularly of 1% in almost all cases and we conclude that extremal dependence does exist in the data and may be particularly relevant for firm-wide risk management.
- In our extensive survey across different asset classes, estimates for the conditional expectation of joint exceedances range from moderate 1.065 up to 1.29. In other words, the conditional probability of joint exceedances of the 99% quantile takes values from 6.5% up to 29%.
- The first application is between the SMI index and single stocks included in the SMI and SPI, respectively. We measure the following average conditional expectation of joint exceedances of the 99% quantile: 1.29 for the SMI returns, 1.18 for the mid cap firms and 1.065 for the small caps.

- The analysis for the international stock markets is carried out both for US\$ denominated and local denominated returns. Against our intuition, we find the conditional expectation to be invariant under changing the currency of denomination. The estimated average conditional expectations are 1.146 and 1.145, respectively.
- Next, we considered linkages across foreign exchange rate regimes. For the currency pairs under a *floating* exchange rate regime (i.e. CHF denominated currency-pairs) we found an average conditional expectation of 1.20 and for the currency pairs under a *fixed* exchange rate regime (i.e. Dmark denominated currency-pairs) an average value of 1.09. Our results show that the risk of spreading speculative attacks is smaller in a target zone than under a floating exchange rate regime. Univariately, we find somewhat fatter tails for the currency pairs under a fixed exchange rate regime
- Finally, we considered the hypothesis that large downward market movements tend to occur together with (extremely) high credit spreads. For long-term bond spreads to treasury (based on seasoned bonds with remaining maturities of at least 20 years) we find a considerable average conditional expectation of 1.17.

The implications of the results is twofold. First, the simultaneity of crashes is crucial for the stability of the international financial markets and the systemic risks in these markets. The second implication is on a firm level. The more markets crash simultaneously, the more in danger are banks holding substantial diversified trading portfolios, possibly threatening the payment and settlement process.

Straetmans (1998) presents a number of extensions and applications to the above analysis. A common presumption in finance is that markets become increasingly correlated during volatile periods. Replacing correlation by conditional probability and volatility by extremal return, he found that the conditional crash probability is increasing with rising univariate extremes. Moreover, he applies the bivariate extreme value theory results to portfolio theory, using extremal dependencies rather than correlations to identify the potential for risk diversification. Replacing the portfolio variance by bivariate tail risk, he derives the 'mean-tail risk' efficient frontier and calculates minimum tail risk portfolios. His presented model is innovative in that it makes explicit the relationship between extremal dependence and risk diversification.

Poon, Rockinger and Tawn (2001) point out that in the conventional multivariate extreme value analysis, as applied in the present chapter, cross-sectional dependence in returns of risky assets is often modelled making the implicit assumption of asymptotic dependence. However, if the true dependence structure is asymptotically independent, then existing finance models lead to over-estimation of the risk of simultaneous extreme events. They provide techniques for distinguishing between two classes of extreme value dependence, asymptotic dependence and asymptotic independence, and for quantifying the degree of dependence for the appropriate dependence class.⁴¹ Their set of (empirical) dependence measures characterizes all classes of extreme value dependence. In the empirical application, they demonstrate that most of the stock index returns in their data sample⁴² do not exhibit asymptotic dependence, suggesting that much of the extreme value dependence reported in previous studies is likely to be over-estimated. Using an asymmetric version of a GARCH(1,1) filter, they find that much, but not all, of the extreme value dependence is caused by changing stock market volatility.

The analysis of extremal dependence in the present chapter serves as a starting point for the analysis in the following chapters. On a univariate level, there are two alternative ways to implement EVT. The first one has been applied in the present chapter and is built around the Hill estimator. The Hill estimator is called a non-parametric or semi-parametric approach since it does not assume that extremes are drawn *exactly* from the asymptotic distribution. The main drawback of the Hill estimator is that it is designed only for data coming from heavy tailed distributions. The second one, called parametric, is based on the concept of exceedances of high thresholds. It consists of estimating parameters by assuming that the realized returns are drawn exactly from the asymptotic distribution of extremes. Maximum likelihood method is the commonly used technique which provides efficient estimates. Frey and McNeil (2000) show that quantile estimates based on the second concept tend to be more stable than the Hill estimator with respect to the choice of the threshold. We will apply the parametric approach in the next chapter, trying to improve conditional VaR models with the EVT-based quantile estimators.

On a bivariate level we will use the results for the calculation of portfolio VaR and S-VaR in Chapter 4. We will build a risk model which reflects the empirical facts found in the present chapter, e.g. extremal dependence between return series. We will assume different dependence structures (e.g. different

⁴¹The respective measures were first developed by Coles, Heffernan and Tawn (2000), with non-financial applications. See also Coles (1999b).

⁴²Using five stock indices, over a 31-year period from 1968 to 2000.

copulas) and investigate how the choice affects the performance of risk models. Moreover, we are concerned with the question of which dependence structure is most appropriate for risk modelling.

Chapter 3

Conditional and Unconditional EVT-based Risk Measurement

3.1 Introduction

The large increase in the number of traded assets in the portfolio of most financial institutions has made the measurement of market risk, i.e. the risk that a financial institution incurs losses on its trading book due to adverse market movements, a primary concern for regulators and for internal risk control. According to the Capital Adequacy Directive by the Bank of International Settlement (BIS) in Basle, the risk capital of a bank must be sufficient to cover losses on the bank's trading portfolio over a ten-day holding period in 99% of occasions. This value is referred to as Value-at-Risk (VaR).¹ Of course, holding period and confidence level may vary according to application: for purposes of internal risk control most financial firms also use a holding period of one day and a confidence level of 95%.²

Alternative measures of market risk have been proposed. In two seminal papers, Artzner, Delbaen, Eber and Heath (2000, 2001) show that VaR has

¹For a good overview of different VaR models, see Dowd (2002).

²Note that from a mathematical viewpoint VaR is simply a quantile of the Profit-and-Loss (P&L) distribution of a given portfolio over a prescribed holding period.

various theoretical drawbacks as a measure of market risk. They propose the use of the so-called expected shortfall (S-VaR) instead.³ From a statistical point of view the main challenge in implementing these risk-measures is to come up with a good estimate for the tails of the underlying P&L distribution; given an estimate of the latter, both VaR and S-VaR are fairly easy to compute. The basic idealization in the following chapter is that returns follow a stationary time series model with a stochastic volatility structure. The presence of stochastic volatility implies that returns are not necessarily independent over time. The latter fact is captured by considering the conditional return distribution, where the conditioning is on the current volatility.⁴

The starting questions of this chapter can be stated as follows:

- How can we further improve the dynamic (or conditional) measurement of risk with methods of EVT?⁵
- In particular, how can we estimate heteroskedastic financial time series using EVT?
- What are the main advantages and drawbacks, respectively, of conditional and unconditional risk modelling techniques?
- What is the evidence from the Swiss market regarding the performance of different conditional and unconditional risk models?

We aim at considering the stylized facts of financial return series (mainly non-*iid* returns, stochastic volatility, fat tails and asymmetry in return distributions) and getting the most appropriate model to estimate the univariate VaR. In studies of today's volatile financial markets it is common to make the distinction between conditional and unconditional asset return distributions.

Unconditionally, we apply the Historical simulation, simple Variance-covariance model and methods based on EVT. EVT based methods have two features which make them attractive for tail estimation: first, they are based on a sound

³The expected shortfall measures the expected loss given that the loss L exceeds VaR; in mathematical terms it is given by $E[L | L > VaR]$.

⁴With a stationary but dependent process as assumed there are two types of return distributions to be considered - the conditional return distribution where the conditioning is on the current volatility and the marginal or stationary distribution (or unconditional distribution) of the process. In practice, the former is used mainly by the risk manager who is concerned with the market movements over the next day, whereas the latter is used mainly by the risk manager who wishes to perform scenario analysis or worst case scenarios.

⁵In particular, we aim at making the *iid* assumption underlying the use of EVT as accurate as possible.

statistical theory and second, they offer a parametric form for the tail of a distribution. Hence, these methods allow for some extrapolation beyond the range of the data, even if care is required at this point. However, none of the unconditional EVT-based methods for quantile estimation yields VaR-estimates which reflect the current volatility background.

Danielsson, Hartmann and De Vries (1998) show similar results as ours for unconditional models, applying a different EVT-based model (i.e. a model based on the Hill estimator). Gençay and Selçuk (2002) investigate the relative performance of VaR models with daily stock market returns of nine different emerging markets. Their results indicate that EVT-based VaR estimates are more accurate at high quantiles. In addition, they find daily returns to have different moment properties at their right and left tails, respectively.

Conditionally, we apply models of volatility dynamics that assume conditional normality⁶ and conditional t-distributed innovations (i.e. standard GARCH-models). Moreover, we apply the model proposed by Frey and McNeil (2000) which overcomes some of the drawbacks of the methods mentioned above. They use GARCH-modelling and pseudo-maximum likelihood estimation to obtain estimates of the conditional volatility. Statistical tests and exploratory data analysis confirm that the error terms or residuals do form, at least approximately, an *iid* series that exhibits heavy tails. They use historical simulation (for the central part of the distribution) and threshold models from EVT (for the tail estimation) to estimate the distribution of the residuals or innovations. The application of the approach is legitimated by the (approximate) independence of the residuals over time. An estimate of the conditional return distribution is then easily constructed from the estimated distribution of the residuals and estimates of the conditional mean and volatility. The approach reflects two stylized facts exhibited by most financial return series, namely stochastic volatility (using a conditional model) and the fat-tailedness (using an EVT-based model) of conditional return distributions over short time horizons.

Danielsson and De Vries (1997b), Danielsson and Morimoto (2000) and Gençay, Selçuk and Ulugülyağci (2001a) compare conditional and unconditional models and find that GARCH type techniques are less accurate than unconditional EVT-based VaR forecasts. Frey and McNeil (2000) find that their conditional EVT-based model yields better estimates of VaR and S-VaR than both unconditional EVT models and GARCH modelling with normally

⁶Note that the weakness of conditional normality is that it does not seem to hold for real data. As shown, for instance, in Danielsson and De Vries (1997b), models based on conditional normality are therefore not well-suited when estimating *extreme* quantiles of the P&L distribution.

distributed error terms. Byström (2001) applies the same set of models and finds similar results.

We compare the conditional and unconditional models on various return series of titles contained in the Swiss Market Index, over a 15 year time period. In contrast to previous studies, we chose single assets in order to observe more extreme movements compared to the more 'diversified' market index cases. Our backtesting results for the VaR show that the model proposed by Frey and McNeil (2000) outperforms both the conditional GARCH-model with normally distributed error terms as well as the unconditional models. For the expected shortfall, our results show that models with normally distributed conditional return distributions yield very bad estimates.

The chapter is structured as follows. In Section 2, we give an introduction to the conditional and unconditional VaR and S-VaR estimation, respectively. We discuss the conditional models; the conditional normal based model, the conditional t model and the conditional EVT-based model, the latter being described in more detail. Moreover, we discuss the issue of threshold selection. Section 3 presents the data and testing issues, while the empirical results are shown in Section 4. The final section gives some ideas for further research.⁷

3.2 Conditional and Unconditional EVT-based Models for VaR and S-VaR

3.2.1 Estimation of VaR and S-VaR

Suppose that the dynamics of X (i.e. daily observations of the negative log return) are given by

$$X_t = \mu_t + \sigma_t Z_t, \quad (3.1)$$

where the innovations Z are a strict white noise process (i.e. independent, identically distributed) with zero mean, unit variance and marginal distribution function $F_Z(x)$. We are interested in estimating quantiles in the tail of the distribution. For $0 < q < 1$, an *unconditional* quantile is a quantile of the marginal distribution denoted by

⁷The results of this chapter have been presented at the CaRisMa workshop II / 2002 in Zurich. I thank the participants of the respective workshop, especially Dr. Patrick Wegmann and Roger Kaufmann, for helpful comments. The same thanks go to Dr. Jon Danielsson from the London School of Economics and to Prof. Alexander McNeil from the ETH in Zurich.

$$VaR_q(x) = x_q(x) = \inf \{x \in R : F_X(x) \geq q\}, \quad (3.2)$$

and a *conditional* quantile is a quantile of the predictive distribution for the return over the next h days denoted by

$$VaR_q^t(h) = x_q^t(h) = \inf \{x \in R : F_{X_{t+1}+\dots+X_{t+h}|\wp_t}(x) \geq q\}, \quad (3.3)$$

where $F_{X_{t+1}+\dots+X_{t+h}|\wp_t}(x)$ is the predictive distribution of the return over the next h days and $\wp_t$ is the information set about the return process available up to time t . The *unconditional* expected shortfall is then defined to be

$$S - VaR_q = E[X | X > x_q], \quad (3.4)$$

i.e. the expected size of a return exceeding the VaR, and the *conditional* expected shortfall as

$$S - VaR_q^t(h) = \left[E \sum_{j=1}^h X_{t+j} \mid \sum_{j=1}^h X_{t+j} > x_q^t(h), \wp_t \right]. \quad (3.5)$$

Principally, we are interested in quantiles and expected shortfall for the 1-step predictive distribution, which are denoted by VaR_q^t and $S - VaR_q^t$, respectively. Since

$$F_{X_{t+1}|\wp_t}(x) = P \{ \sigma_{t+1} Z_{t+1} + \mu_{t+1} \leq x \mid \wp_t \} \quad (3.6)$$

$$= F_Z((x - \mu_{t+1})/\sigma_{t+1}),$$

the measures simplify to

$$VaR_q^t = \mu_{t+1} + \sigma_{t+1} z_q, \quad (3.7)$$

$$S - VaR_q^t = \mu_{t+1} + \sigma_{t+1} E[Z | Z > z_q], \quad (3.8)$$

where z_q is the upper q th quantile of the marginal distribution of Z_t which by assumption does not depend on t . Note that μ_{t+1} is often set to be zero in practical applications, since the mean is negligible if compared with volatility. In the next subsection we look at different approaches to model the dynamics of the conditional mean and volatility in 3.1.

3.2.2 Conditional models

Building on the realization that returns exhibit volatility clusters, conditional volatility forecasts are important for several applications. In many situations where the investment horizon is short (as in our application with daily VaR and S-VaR forecasts, respectively), conditional volatility models may be preferred for risk forecasting. This is especially the case in intra-day risk forecasting. In such situations both GARCH and stochastic volatility models have been applied successfully.

Although the conditional GARCH models adapt more quickly to a new situation, i.e. to the change to a high risk regime, this necessarily implies more volatile risk forecasts than unconditional models, see Danielsson and Morimoto (2000). High volatility of risk measures can be a serious problem in internal risk management (as for example for the allocation of position limits) because it is very hard to manage individual positions with highly volatile position limits.⁸

Generally, the choice of methodology is likely to depend on the situation and question at hand, and both the conditional and unconditional approaches belong to the toolbox of the risk manager.

The conditional normal model

In the present chapter, we use the GARCH (1,1) process for the volatility and an AR(1) model for the dynamics of the conditional mean. In estimating Var_q^t with GARCH-type models it is commonly assumed that the innovation distribution is standard normal so that a quantile of the the innovation distribution Z_t is simply

$$z_q = \Phi^{-1}(q), \quad (3.9)$$

where $\Phi(z)$ is the standard normal distribution function. A GARCH-type model with normal innovations can be fitted by maximum likelihood (see Tsay 2002) and μ_{t+1} and σ_{t+1} can be estimated using standard 1-step forecasts, so that an estimate of Var_q^t is easily constructed using 3.7. It has been shown by Gençay and Selçuk (2002) that the normal distribution underestimates the extent of large losses and also of the largest gains.

GARCH models with normal innovations have generally proved valuable in forecasting common volatilities, however they perform poorly in predicting

⁸ However, it is largely irrelevant for external risk management, e.g. for regulatory purposes. The reason is that the regulatory VaR is typically an order of magnitude higher than required, see Danielsson and Morimoto (2000).

extreme observations in returns.⁹

The conditional t model

Another standard approach is to assume that the innovations have a leptocurtic distribution, such as the student's-t distribution (scaled to have variance one). If we assume that $Z = \sqrt{(v-2)/v}T$, where T has a t-distribution with $v > 2$ degrees of freedom and distribution function $F_T(t)$, the quantile is $z_q = \sqrt{(v-2)/v} F_T^{-1}(q)$. GARCH models with t-innovations can also be fitted with maximum likelihood and the additional parameter v can be estimated.

The advantage of student t innovations is that they are thick tailed and hence will in general provide better predictive densities than the normal based. The disadvantages of non-normal innovations are that its tails generally have a different shape than the empirical return distribution, that multivariate versions of such models are typically hard to estimate, and that recursive forecasts of multi-step ahead VaR levels are difficult to compute, since the GARCH process is not self additive.

Frey and McNeil (2000) and Danielsson and Morimoto (2000) find that the t-distribution underestimates the losses and overestimates the gains (as compared to the empirical distribution). This illustrates the drawback of using a symmetric distribution with data which are asymmetric in the tail. Frey and McNeil (2000) show that with symmetric data the conditional t-distribution works quite well and can, in fact, be viewed as a special case of the model presented in the following subsection.

While in theory it is straight forward to extend the GARCH model to take into account asymmetry, it necessarily complicates the model. Taking the EGARCH model, for example, the asymmetry parameters are typically non-significant, suggesting that the model is misspecified. As a result, this model does not have any special advantage over the standard GARCH model, however, being more complex.¹⁰ Overall, its limitations imply that the conditional t model should be used with care.

⁹ Note that the while the GARCH-normal model unconditionally has heavy tails, it has thin tails conditionally.

¹⁰ An alternative to consider asymmetry would be to use skewed conditional distributions, this however is not commonly done in practice, due to the difficulty of finding an appropriate skewed distribution and estimating its parameters.

A dynamic EVT-based model

The method proposed by Frey and McNeil (2000)¹¹ makes minimal assumptions about the underlying innovation distribution and concentrates on modelling its tail using EVT. They use a two stage approach which can be summarized as follows.¹²

1. Fit a GARCH-type model to the return data making no assumption about $F_Z(z)$, using a pseudo-maximum-likelihood approach (PML). It can be shown that the PML method yields a consistent and asymptotically normal estimator. Estimate μ_{t+1} and σ_{t+1} using the fitted model and calculate the implied model residuals.

2. Consider the residuals to be a realization of a strict white noise or *iid* process and use EVT to model the tail of $F_Z(z)$. Use this EVT model to estimate z_q for $q > 0.95$.

The following two sections describe these two stages in more detail.

The first stage The first stage consists in fitting a GARCH-type model to the return data. For predictive purposes we fix a constant memory n so that at the end of day t our data consists of the last n (negative) log returns. We consider these to be realizations from an AR(1)-GARCH(1,1) process. Hence the conditional variance of the mean adjusted series $\varepsilon_t = X_t - \mu_t$ is given by

$$\sigma_t^2 = \alpha_0 + \alpha_1 \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (3.10)$$

where $\alpha_0 > 0$, $\alpha_1 > 0$ and $\beta > 0$.¹³ The conditional mean is given by

$$\mu_t = \phi X_{t-1}, \quad (3.11)$$

and the full return process by

$$X_t = \phi X_{t-1} + \sigma_t Z_t, \quad (3.12)$$

This model is fitted using the pseudo-maximum-likelihood (PML) method. This means that the likelihood for a GARCH (1,1) model with normal innovations is maximized to obtain parameter estimates $\hat{\theta} = (\hat{\phi}, \hat{\alpha}_0, \hat{\alpha}_1, \hat{\beta})^T$. Whilst this amounts to fitting a model using a distributional assumption (i.e. normality)

¹¹Note that we call the respective VaR model both conditional EVT-based VaR model and dynamic EVT-based VAR model.

¹²The following sections are heavily based on Frey and McNeil (2000).

¹³For more extensive discussion of GARCH models see for example Tsay (2002), for applications in S-Plus see Zivot and Wang (2002).

we do not necessarily believe, the PML method delivers reasonable parameter estimates. In fact, it can be shown that the PML method yields a consistent and asymptotically normal estimator, even if the standardized residuals are not normally distributed;¹⁴ see Gouriéroux (1997).

Estimates of the conditional mean and standard deviation can be calculated recursively from 3.10 and 3.11 after substitution of sensible starting values. Residuals are calculated both to check the adequacy of the GARCH modelling and to use in stage two of the model. They are calculated as

$$(z_{t-n+1}, \dots, z_t) = \left(\frac{x_{t-n+1} - \hat{\mu}_{t-n+1}}{\hat{\sigma}_{t-n+1}}, \dots, \frac{x_t - \hat{\mu}_t}{\hat{\sigma}_t} \right),$$

and should be *iid* if the fitted model is tenable. Figure 3.1 shows correlograms for the raw data (i.e. the continuously compounded returns for the example of Novartis) and their squared values as well as for the residuals and squared residuals. While for the raw data the assumption of *iid* is clearly not justified, this assumption seems to be tenable for the residuals.

Once satisfied with the fitted model, stage one is finished by calculating estimates of the conditional mean and variance for day $t+1$, which are the one-step forecasts

$$\hat{\mu}_{t+1} = \hat{\phi} x_t \quad (3.13)$$

$$\hat{\sigma}_{t+1}^2 = \hat{\alpha}_0 + \hat{\alpha}_1 \hat{\varepsilon}_t^2 + \hat{\beta} \hat{\sigma}_t^2, \quad (3.14)$$

where $\hat{\varepsilon}_t = X_t - \hat{\mu}_t$.

The second stage The aim of stage two is to estimate z_t (for the VaR) and $E[Z \mid Z > z_q]$ (for the S-VaR) using EVT. In particular, we apply the POT model as introduced in Section 1.2.2. We see from the QQ-Plot of the residuals against the normal distribution in Figure 3.2 that the assumption of conditional normality is unreasonable. This figure legitimates the use of EVT in the conditional model.

We fix a high threshold u and we assume that excess residuals over this threshold have a generalized Pareto distribution (GPD) with distribution function $G_{\xi, \beta}(y)$ as given in Equation 1.8.¹⁵ This particular distributional choice is

¹⁴However, the assumption that the standardized residuals are *iid* is still required, see Manganelli and Engle (2001).

¹⁵Note that we replace the shift parameter σ by β , since σ is used for the estimation of the volatility in stage one.

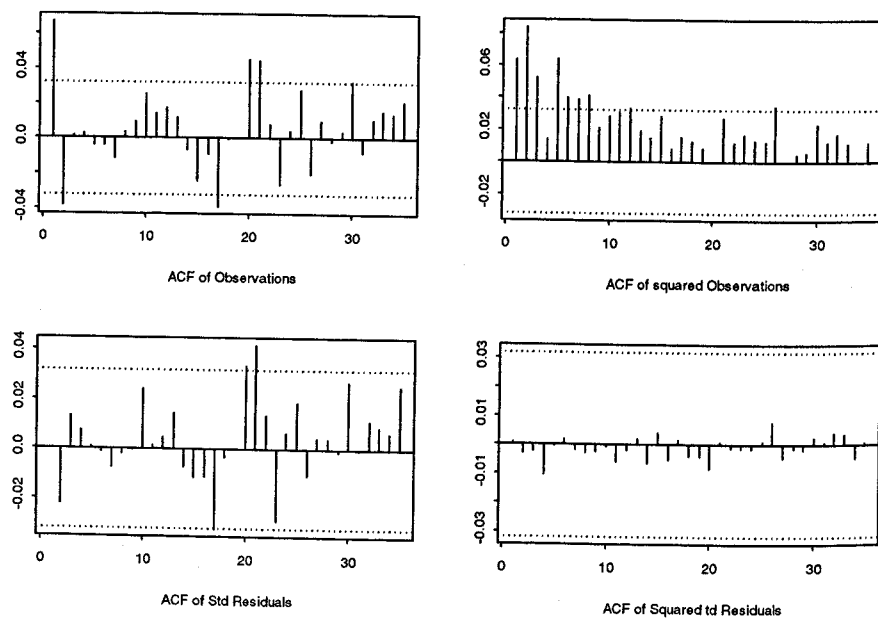


Figure 3.1: Correlograms for the raw data (returns of Novartis) and their squared values as well as for the (standardized) residuals and squared residuals. While the raw data is clearly not *iid*, this assumption may not be tenable for the standardized residuals.

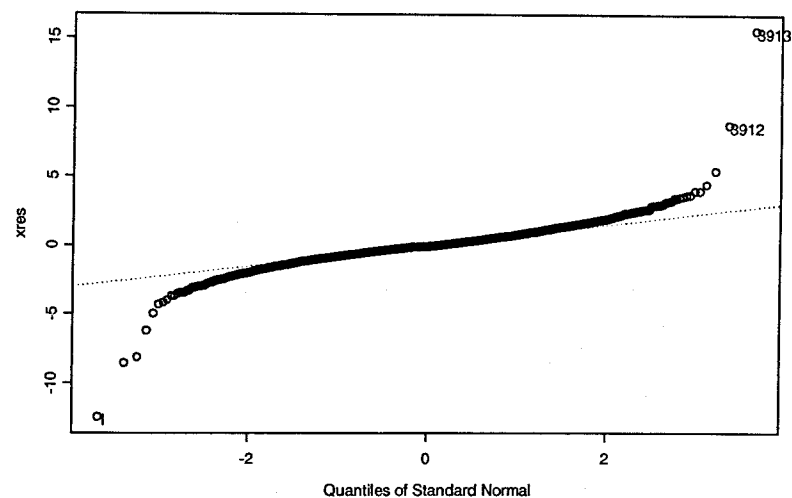


Figure 3.2: The Quantile-quantile plot of residuals against the normal distribution shows that the residuals are leptokurtic. The figure shows the example of Novartis.

motivated by a limit result of EVT. Consider a general distribution function F and the corresponding excess distribution above the threshold u given by

$$F_u(y) = P(X - u \leq y | X > u) = \frac{F(u+y) - F(u)}{1 - F(u)}, \quad (3.15)$$

for $0 \leq y < x_0 - u$, where x_0 is the right endpoint of F . Balkema and De Haan (1974) and Pickands (1975) showed for a large class of distributions F that it is possible to find a positive measurable function $\beta(u)$ such that

$$\lim_{u \rightarrow x_0} \sup_{0 \leq y < x_0 - u} |F_u(y) - G_{\xi, \beta(u)}(y)| = 0. \quad (3.16)$$

We refer again to Embrechts, Klüppelberg and Mikosch (2001) for more details.¹⁶ In the class of distributions for which this result holds are essentially all the common continuous distributions in statistics.¹⁷

We assume that the tail of the underlying distribution begins at the threshold u . From our sample of n points a random number $N = N_u > 0$ will exceed this threshold. If we assume that the N excesses over the threshold are *iid* with exact GPD distribution, the maximum likelihood estimators of the parameters ξ and β are consistent and asymptotically normal.¹⁸ Under the weaker assumption that the excesses are *iid* from $F_u(y)$ which is only approximately GPD, also the estimates of $\hat{\xi}$ and $\hat{\beta}$ are consistent and asymptotically normal.

By letting $u = u_n \rightarrow x_0$ and $N = N_u \rightarrow \infty$ the procedure is asymptotically unbiased provided that $u \rightarrow x_0$ (where x_0 is the right endpoint of F) sufficiently fast. Our best GPD estimator of the excess distribution is obtained by trading off bias against variance. We choose u high to reduce the chance of bias whilst keeping N large (or u low) to control the variance of the parameter values.¹⁹ The choice of u (or N) is the most important implementation issue in EVT and we discuss this issue later on in this chapter.

Consider the following equality for points $x > u$ in the tail of F

$$1 - F(x) = (1 - F(u))(1 - F_u(x - u)). \quad (3.17)$$

¹⁶In particular, theorem 3.4.13, on page 165.

¹⁷More precisely, the class comprises all distributions in the maximum domain of attraction of an extreme value distribution.

¹⁸See Smith (2002).

¹⁹In other words, we choose u high in order that the asymptotic basis for the model is given, and keep u low in order to consider more extreme values in our estimation, which lowers the variance of the estimation.

If we estimate the first term on the right hand side of 3.17 using the random proportion of the data in the tail N/n , and if we estimate the second term by approximating the excess distribution with a GPD fitted by maximum likelihood, we get the tail estimator given in 1.25

$$\hat{F}(x) = 1 - \frac{N}{n} \left(1 + \hat{\xi} \frac{x - u}{\hat{\beta}} \right)^{-1/\hat{\xi}}, \quad (3.18)$$

for $x > u$. In practice, we will fix the number of data in the tail to be $N = k$ where $k \ll n$. This effectively gives us a random threshold at the $(k+1)$ th order statistic. The respective form of the tail estimator $F_Z(z)$ is then

$$\hat{F}_Z(z) = 1 - \frac{k}{n} \left(1 + \hat{\xi} \frac{z - z_{(k+1)}}{\hat{\beta}} \right)^{-1/\hat{\xi}}, \quad (3.19)$$

and for $q > 1 - k/n$ we can invert this tail formula to get the quantile

$$\hat{z}_{q,k} = z_{(k+1)} + \frac{\hat{\beta}}{\hat{\xi}} \left(\left(\frac{1-q}{k/n} \right)^{-\hat{\xi}} - 1 \right), \quad (3.20)$$

as introduced in 1.30, where k is the number of highest order statistics.

To estimate the S-VaR as given in Equation 3.8, we require an estimate for the expected shortfall for the innovation distribution $E[Z | Z > z_q]$. In fact, it can be shown that

$$Z - z_q | Z > z_q \sim G_{\xi, \beta + \xi(z_q - u)}, \quad (3.21)$$

or in words that excesses over the higher threshold z_q also have a GPD distribution with the same shape parameter ξ but a different scaling parameter. We get the following closed form solution for the S-VaR

$$E[Z | Z > z_q] = z_q \left(\frac{1}{1 - \xi} + \frac{\beta - \xi u}{(1 - \xi) z_q} \right). \quad (3.22)$$

This can be estimated analogously to the quantile estimator in 3.20 by replacing all unknown quantities by GPD-based estimates and replacing u by $z_{(k+1)}$. We get the following conditional S-VaR estimate

$$\widehat{S - VaR}_q^t = \hat{\mu}_{t+1} + \hat{\sigma}_{t+1} \hat{z}_q \left(\frac{1}{1 - \hat{\xi}} + \frac{\hat{\beta} - \hat{\xi} z_{(k+1)}}{(1 - \hat{\xi}) \hat{z}_q} \right). \quad (3.23)$$

3.2.3 Unconditional models

The most common model used in practice is the Historical simulation approach.²⁰ Here, the estimated P&L distribution of a portfolio is simply given by the empirical distribution of past gains and losses on this portfolio. The method is therefore easy to implement and avoids ad-hoc assumptions on the form of the P&L distribution.²¹ However, being very sensitive to the particular data window, the method suffers from some serious drawbacks. Extreme quantiles are notoriously difficult to estimate, as extrapolation beyond past observations is impossible and extreme quantile estimates within sample tend to be very inefficient - the estimator is, for extreme quantiles, subject to a high variance.²² If we want to avoid the problems mentioned by considering longer samples, the method is unable to distinguish between periods of high and low volatility.

Another standard approach is the Variance-covariance model, a simple calculation of the standard deviation, making the assumption of normally distributed returns. Its major drawback is that market returns are not normally distributed, having fatter tails than the normal.

The unconditional EVT approach is a simple calculation of the quantile using unconditional EVT (as described in 3.2.2, the second stage). It should be noted that the assumption of independent excesses over threshold is much less satisfactory for the raw return data (as compared to assume *iid* for the residuals as in the dynamic EVT-based model). The asymptotics of the GPD-based tail estimator are therefore much more poorly understood if applied directly to the raw return data. Referring to our starting question, we expect somewhat better results from the conditional EVT-based model than from the unconditional alternative, from this theoretical point of view. After all, pre-whitening of data through fitting of a dynamic model might be a sensible prelude to EVT analysis in practice.

²⁰ See Dowd (2002) for a more extensive discussion of the non-parametric Historical simulation model.

²¹ Another advantage of this approach is that it can incorporate non-linear positions, such as derivative positions, in a natural way.

²² Although the empirical return distribution is very 'dense' and smooth around the mean, it becomes 'discrete' in the tails, hence the high variances. Moreover, at its lower end, the empirical return distribution sharply drops to zero and remains there, which might be considered as imprudent.

3.2.4 Non-parametric vs. parametric estimators

Frey and McNeil (2000) perform a small simulation study to compare the (fully parametric) GPD approach to tail estimation with the non-parametric approach based on the Hill estimator²³ and the non-parametric approach based on the empirical distribution function (i.e. Historical simulation). In their simulation study they generate samples of size $n = 1000$ from Students t-distribution which, as they observe, provides a rough approximation to the observed distribution of model residuals. The sample size corresponds to the window length they use in applications of their proposed two-stage method. They calculate quantile estimates using the Hill estimator and the GPD approach, respectively. Of interest are the mean squared error (MSE) and biases of these estimators, and the dependence of these errors on the choice of k , the number of highest order statistics. For each estimator, they estimate MSE and bias using Monte Carlo estimates based on 1000 independent samples.

Their main findings can be summarized as follows. In finite-samples of the order of 1000 points from typical return distributions EVT quantile estimators (whether maximum-likelihood and GPD-based or Hill-based) are more efficient than the historical simulation method. Furthermore, the GPD-based quantile estimator is more stable (in terms of MSE) with respect to choice of k than the Hill quantile estimator.²⁴ In their application a k value of 100 seems reasonable, but they could have chosen k values of 80 or 150. Regarding the efficiency, they find that for high quantiles with $q \geq 0.99$ the GPD method is at least as efficient as the Hill method. Finally, the GPD method is applicable to light-tailed data or even short-tailed data, whereas the Hill method is designed specifically for the heavy-tailed case.

3.3 Data Sample and Testing Issues

The data set consists of 15 liquid titles of the Swiss Market Index, dating from 22.08.1987 to 19.06.2002, a total of 3913 daily returns for each title.²⁵ In order to compare with less liquid titles, we include nine non-liquid stocks from the

²³ The Hill estimator has been applied in Chapter 2, see Equation 2.21.

²⁴ Although the Hill estimator is generally the most efficient estimator of ξ (it gives the lowest mean squared error (MSE) for sensibly chosen k), it does not provide the most efficient nor the most stable *quantile* estimator.

²⁵ All the data is extracted from Datastream. The titles included are ABB, Adecco, Baloise, Credit Suisse, EMS, Julius Bär, Nestlé, Novartis, Richemont, Roche, Sulzer, SwissRe, UBS and Zurich.

Swiss Performance Index.²⁶ Our choice is based on the idea that with (more volatile) single assets we observe more extreme movements compared to the more 'diversified' market index case. We backtest the proposed models on these assets. In order to backtest the model on the historical series, we calculate the VaR $\widehat{VaR}_q^t$ on each day t using a time window of 1000 days each time. Consequently, we use somewhat less than the last four years of data for each prediction. Hence, a total of 2913 VaR forecasts are calculated for each stock. Regarding the choice of the threshold in the GPD estimation, it is not feasible to examine the fitted model carefully every day and to choose a new threshold for the tail estimator each time; for this reason we fix the threshold at the 90th percentile, estimating the higher percentiles using the (maximum likelihood based) GPD tail estimator. A look at Figure 3.3 shows that there is not much sensitivity of the quantile estimate with respect to the choice of the threshold.²⁷ On each day we determine a new VaR forecast and compare it with the realization on the next day. Whenever the latter exceeds the former, we count a violation of the VaR. Hence, we get a number of violations using our model and compare it with the theoretical expected number of violations.

3.3.1 Backtesting the VaR

In order to test for the success of the VaR forecast models, we apply a binomial test, where we test for significant departure of the realized number of violations from the expected number of violations. Alternatively, we could calculate violation ratios (realized number of violations over expected number of violations) as done e.g. in Danielsson (2002).²⁸

The indicator function for a violation at time $t \in T$ is Bernoulli distributed

$$I_t := 1_{\{x_{t+1} > \widehat{VaR}_q^t\}} = 1_{\{Z_{t+1} > z_q\}} \sim Be(1 - q). \quad (3.24)$$

Therefore, the following sum of the indicator function, i.e. the total number of violations, is binomially distributed under the respective model

$$\sum_{t \in T} I_t \sim B(card(T), 1 - q). \quad (3.25)$$

²⁶These are Agie Charmilles, Conzetta, Eichhof, Huber & Suhner, Jelmoli, Kardex, Nextrom, Oz and Swiss Steel.

²⁷The decision of considering 10% of the data as extreme values is further justified by the simulation study by Frey and McNeil (2000), for a similar data set.

²⁸The advantage of the latter approach is that it directly shows whether the model is under- or overforecasting risk. It is not a statistical test, however.

Under the null hypothesis that the conditional quantiles are correctly estimated by the respective model, the empirical version of the latter statistic is binomially distributed

$$\sum_{t \in T} 1_{\{x_{t+1} > \hat{x}_q^t\}} \sim B(card(T), 1 - q), \quad (3.26)$$

where x_{t+1} is the realization of the VaR on day $t + 1$. We perform a two-sided binomial test of the null hypothesis against the alternative that the model has a systematic estimation error and gives either too few or too many violations. A p-value less than 0.05 (0.01) will be interpreted as evidence against the null hypothesis.

Danielsson (2002) addresses another forecast property; the volatility of the risk forecasts. He uses two measures of fluctuations: the volatility of the VaR and the VaR forecast range, i.e. the maximum and minimum VaR forecasts over the sample period. Although not being well documented in the existed literature, the importance of the issue is real since adjusting positions due to VaR fluctuations results in cost increases. Naturally, if a financial institution has a choice between two risk models both of which forecast equally well, it will chose the one with less volatile forecasts.

Yet another testing issue is addressed by Christofferson (1998) and applied by Sarma (2001); violation clustering or correct conditional average. A correctly specified VaR model should generate the pre-specified failure rate (VaR violations) *conditionally* at every point in time. The basic feature of a 99% VaR is that it should be exceeded in 1% of the cases, and that the probability of the VaR being exceeded at time $t + 1$ remains 1% even after conditioning on all information known at time t . This implies that the VaR should be small in times of low volatility and high in times of high volatility, such that the events where the loss exceeds the forecasted VaR measure are spread over the entire sample period, and do not come in clusters. A model which fails to capture the volatility dynamics of the underlying return distribution will exhibit the symptom of clustering of failures, even if (on average) it may produce the correct unconditional coverage. In order to test for violation clustering, we essentially introduce an indicator function I_t which takes the value of one if there is a violation and the value of zero otherwise. The VaR forecasts are said to have a correct conditional average if the series I_t is *iid* with mean p , where p is the coverage probability (or the expected number of violations).

Hence, in order to completely assess the quality of a risk measure, we should address the following issues: the robustness of the model (with a binomial test

or violation ratios, respectively), the volatility of the model forecasts and the violation clustering.²⁹

3.3.2 Backtesting the S-VaR

To backtest the S-VaR, we are interested in the size of the discrepancy between the realization X_{t+1} and the shortfall estimation $S - VaR_q^t$ in the event of quantile violation. We define residuals as

$$R_{t+1} = \frac{X_{t+1} - (S - VaR_q^t)}{\sigma_{t+1}} = Z_{t+1} - E[Z | Z > z_q].$$

Note that under the model assumptions, these residuals are *iid* and, conditional on a VaR violation, they have expected value of zero. The empirical version of these residuals on days when a violation occurs, i.e. $x_{t+1} > VaR_q^t$, is given by

$$\{r_{t+1} : t \in T, x_{t+1} > \widehat{VaR}_q^t\}, \quad \text{with } r_{t+1} = \frac{x_{t+1} - (S - \widehat{VaR}_q^t)}{\hat{\sigma}_{t+1}}, \quad (3.27)$$

where r_{t+1} stands for the residual, $\hat{\sigma}_{t+1}$ is the volatility forecast on day t for day $t+1$, and $S - \widehat{VaR}_q^t$ is an estimate of the shortfall. Under the null hypothesis that we correctly estimate the VaR and the conditional VaR (S-VaR), these residuals should behave like an *iid* sample with mean zero. A first look at Figures 3.10 and 3.11 shows that for the residuals calculated under the assumption of conditional normality seems doubtful.

To test the hypothesis of mean zero we use a *bootstrap test* that makes no assumption about the underlying distribution of the residuals.³⁰ We conduct a one-sided test against the alternative hypothesis that the residuals have mean greater than zero, or equivalently, that the S-VaR is systematically underestimated, since this is the likely direction of failure. Again, p-values less than 0.01 present some evidence against the null hypothesis of a "correct" model.

²⁹ A good overview of issues regarding the backtesting of market risk models is given in Dowd (2002).

³⁰ For an introduction to the bootstrap test, see Dowd (2002). For further details refer to Frey and McNeil (2000), or Efron and Tibshirani (1993). Furthermore, Shao, Tu and Olkin (1996) show various applications both for the independent and identically distributed (*iid*) case and more complicated cases with non-*iid* data sets. Figure C.1 in the appendix gives an overview of the bootstrap procedure.

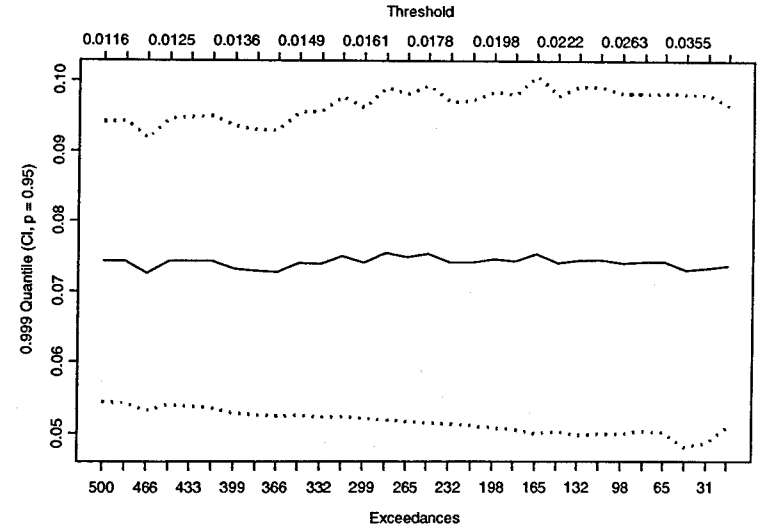


Figure 3.3: Sensitivity of the 99.9% quantile estimate with respect to the choice of the threshold (upper axis) and to the number of exceedances of a certain threshold (lower axis). The figure shows the example of Novartis.

3.4 Empirical Results

3.4.1 Results for the VaR: the unconditional models

In the following section, we present the empirical results.³¹ Tables 3.1 to 3.3 show the results for the unconditional VaR models; the Historical simulation, the Variance-covariance and the unconditional EVT-based model. The first columns show the SMI and SPI titles, respectively. Columns 2, 4, 6 and 8 show the number of VaR violations (expected number of violations in brackets) for different quantiles (i.e. the 95%, 99%, 99.5% and the 99.9% quantile), columns 3, 5, 7 and 9 show the p-values for the binomial test as described in Section 3.3.1, respectively.³² A p-value of 0.05 (0.01) or less indicates that the null hypothesis of a correct model is rejected on a 95% (99%) confidence level. Finally, the last two rows show the percentage of cases where the null hypothesis cannot be rejected (upper row for the 0.05 level, lower row for the 0.01 level).³³ For example, we note that for the Historical simulation, for the 95% quantile, we have 68% of the (25) cases with a p-value of more than 5%, meaning that we accept the null of a correct model³⁴ in 68% percent of the cases under consideration.

The main findings can be summarized as follows. First, the Variance-covariance or normal based model gives the best performance for the 95% quantile. This is according to our intuition that the further we move from the tails or extreme values, the better the estimation by the latter model. Second, for quantiles higher than 95%, the normal based model doesn't give accurate results and is outperformed by both the Historical simulation as well as the EVT-based models. Furthermore, the unconditional EVT-based model is, for every quantile, at least as accurate as the Historical simulation model and it is the best model for high quantiles. For the Historical simulation, we find some evidence that, according to our intuition, extrapolation beyond past observations is not accurate for extreme quantiles.

We note that the results are the same whether we consider the whole set of returns or just the more volatile medium and small capitalized stocks. Further-

3.4. EMPIRICAL RESULTS

	Historical simulation							
	95%	p	99%	p	99.5%	p	99.9%	p
Abb	193(146)	0.00	47(29)	0.00	25(15)	0.01	08(03)	0.01
Adecco	166(146)	0.09	44(29)	0.01	17(15)	0.51	09(03)	0.00
Baloise	147(146)	0.93	39(29)	0.08	22(15)	0.06	10(03)	0.00
CS Group	182(146)	0.00	42(29)	0.02	23(15)	0.05	08(03)	0.01
Ems	124(146)	0.06	25(29)	0.46	13(15)	0.80	05(03)	0.22
Julius Baer	177(146)	0.01	39(29)	0.08	21(15)	0.11	07(03)	0.03
Nestle	164(146)	0.13	44(29)	0.01	22(15)	0.06	06(03)	0.13
Novartis	166(146)	0.09	37(29)	0.16	20(15)	0.19	07(03)	0.03
Richemont	144(129)	0.17	37(26)	0.84	17(13)	0.26	03(03)	0.75
Roche	156(146)	0.40	37(29)	0.16	21(15)	0.11	07(03)	0.03
Sulzer	176(146)	0.01	45(29)	0.01	24(15)	0.02	12(03)	0.00
Swiss Re	154(146)	0.50	28(29)	0.93	17(15)	0.51	07(03)	0.03
Ubs	176(146)	0.01	44(29)	0.01	26(15)	0.01	07(03)	0.03
Unaxis	150(146)	0.73	36(29)	0.22	15(15)	0.89	06(03)	0.13
Zurich	190(146)	0.00	46(29)	0.00	32(15)	0.00	13(03)	0.00
Agie Charm.	152(146)	0.61	31(29)	0.71	17(15)	0.51	07(03)	0.03
Conzetta	155(146)	0.44	35(29)	0.30	16(15)	0.70	05(03)	0.22
Eichhof	167(146)	0.07	42(29)	0.02	22(15)	0.06	06(03)	0.13
Huber+Suhner	206(146)	0.00	56(29)	0.00	33(15)	0.00	11(03)	0.00
Jelmoli	145(146)	1.00	29(29)	1.00	20(15)	0.19	07(03)	0.03
Kardex	156(146)	0.40	33(29)	0.51	15(15)	0.90	04(03)	0.55
Nextrom	235(146)	0.00	66(29)	0.00	38(15)	0.00	06(03)	0.13
OZ Holding	091(091)	1.00	22(18)	0.41	15(09)	0.06	03(02)	0.44
Swiss Steel	140(128)	0.30	30(26)	0.43	17(13)	0.26	07(03)	0.02
SMI	150(132)	0.12	39(26)	0.02	25(13)	0.00	10(03)	0.00
Perc. Signif.	0.68		0.56		0.72		0.36	
	0.80		0.84		0.84		0.76	

Table 3.1: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the historical simulation model. Insignificant p-values for a binomial test in italics.

more, the same is true if considering the stock index only.

Figures 3.4 and 3.5 show Backtesting results for the 99.5% quantile, for the example of Novartis. The figures show log returns (with losses on the positive axes) and VaR forecasts (superimposed), respectively. It is clear from Figure 3.4 that for the Historical simulation, the window length in assessing the probability of spikes is very important. Note how rapidly the VaR prediction changes when new data enter and exit the window. In VaR prediction with Historical simulation, the inclusion or exclusion of one or two days at the beginning of the sample can cause large swings in the VaR estimate.

³¹ All the calculations are performed in the statistical software package S-Plus 2000 for Windows. I thank Claudia Saupper for her support in writing the functions.

³² Note that numbers in italics show p-values higher than 0.01 and numbers in bold italics show p-values higher than 0.05.

³³ Note that p-value less than 0.05 (0.01) are interpreted as evidence (e.g. with significance) against the null hypothesis. Moreover, the higher the p-values the closer the realized number of violations is to the expected number of violations, e.g. the better the model.

³⁴ Or we cannot reject the null hypothesis, correctly.

	VarCov							
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Abb	175(146)	<i>0.02</i>	87(29)	0.00	61(15)	0.00	16(03)	0.00
Adecoo	144(146)	<i>0.90</i>	68(29)	0.00	46(15)	0.00	16(03)	0.00
Baloise	110(146)	0.00	50(29)	0.00	34(15)	0.00	13(03)	0.00
CS Group	156(146)	<i>0.40</i>	63(29)	0.00	41(15)	0.00	13(03)	0.00
Ems	086(146)	0.00	40(29)	<i>0.06</i>	30(15)	0.00	05(03)	<i>0.22</i>
Julius Baer	135(146)	<i>0.37</i>	55(29)	0.00	36(15)	0.00	11(03)	0.00
Nestle	120(146)	<i>0.03</i>	52(29)	0.00	39(15)	0.00	12(03)	0.00
Novartis	122(146)	<i>0.04</i>	41(29)	<i>0.04</i>	34(15)	0.00	06(03)	<i>0.13</i>
Richemont	121(129)	<i>0.53</i>	38(26)	0.00	36(13)	0.00	11(03)	0.00
Roche	127(146)	<i>0.12</i>	46(29)	0.00	39(15)	0.00	09(03)	0.00
Sulzer	158(146)	<i>0.31</i>	68(29)	0.00	50(15)	0.00	14(03)	0.00
Swiss Re	096(146)	0.00	30(29)	<i>0.85</i>	20(15)	<i>0.19</i>	08(03)	<i>0.01</i>
Ubs	131(146)	<i>0.22</i>	65(29)	0.00	39(15)	0.00	14(03)	0.00
Unaxis	126(146)	<i>0.10</i>	50(29)	0.00	36(15)	0.00	07(03)	<i>0.03</i>
Zurich	153(146)	<i>0.55</i>	66(29)	0.00	56(15)	0.00	18(03)	0.00
Agie Charm.	123(146)	<i>0.05</i>	47(29)	0.00	34(15)	0.00	14(03)	0.00
Conzetta	137(146)	<i>0.47</i>	64(29)	0.00	32(15)	0.00	07(03)	<i>0.03</i>
Eichhof	153(146)	<i>0.55</i>	62(29)	0.00	44(15)	0.00	10(03)	0.00
Huber+Suhner	163(146)	<i>0.15</i>	91(29)	0.00	71(15)	0.00	13(03)	0.00
Jelmoli	127(146)	<i>0.12</i>	43(29)	<i>0.02</i>	35(15)	0.00	08(03)	<i>0.01</i>
Kardex	144(146)	<i>0.90</i>	70(29)	0.00	50(15)	0.00	09(03)	0.00
Nextrom	157(146)	<i>0.35</i>	91(29)	0.00	70(15)	0.00	24(03)	0.00
OZ Holding	051(091)	0.00	31(18)	<i>0.01</i>	27(09)	0.00	14(02)	0.00
Swiss Steel	137(128)	<i>0.44</i>	48(26)	0.00	39(13)	0.00	12(03)	0.00
SMI	119(132)	<i>0.25</i>	53(26)	0.00	39(13)	0.00	16(03)	0.00
Perc. Signif.	<i>0.72</i>		<i>0.08</i>		<i>0.04</i>		<i>0.08</i>	
	<i>0.84</i>		<i>0.20</i>		<i>0.04</i>		<i>0.24</i>	

Table 3.2: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the variance-covariance model. Insignificant p-values for a binomial test in italics.

	Gpd							
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Abb	189(146)	0.00	41(29)	<i>0.04</i>	24(15)	<i>0.02</i>	08(03)	<i>0.01</i>
Adecoo	167(146)	<i>0.07</i>	36(29)	<i>0.22</i>	17(15)	<i>0.51</i>	08(03)	<i>0.01</i>
Baloise	148(146)	<i>0.86</i>	37(29)	<i>0.16</i>	17(15)	<i>0.51</i>	06(03)	<i>0.13</i>
CS Group	177(146)	<i>0.01</i>	41(29)	<i>0.04</i>	20(15)	<i>0.19</i>	07(03)	<i>0.03</i>
Ems	119(146)	<i>0.02</i>	25(29)	<i>0.46</i>	13(15)	<i>0.80</i>	02(03)	<i>0.78</i>
Julius Baer	177(146)	<i>0.01</i>	39(29)	<i>0.08</i>	22(15)	<i>0.06</i>	05(03)	<i>0.22</i>
Nestle	158(146)	<i>0.31</i>	38(29)	<i>0.11</i>	26(15)	<i>0.01</i>	02(03)	<i>0.78</i>
Novartis	170(146)	<i>0.05</i>	36(29)	<i>0.22</i>	19(15)	<i>0.29</i>	06(03)	<i>0.13</i>
Richemont	140(129)	<i>0.32</i>	25(26)	<i>0.92</i>	15(13)	<i>0.57</i>	03(03)	<i>0.75</i>
Roche	153(146)	<i>0.40</i>	39(29)	<i>0.16</i>	19(15)	<i>0.11</i>	03(03)	<i>1.00</i>
Sulzer	177(146)	<i>0.01</i>	40(29)	<i>0.06</i>	24(15)	<i>0.02</i>	10(03)	0.00
Swiss Re	157(146)	<i>0.35</i>	23(29)	<i>0.27</i>	14(15)	<i>1.00</i>	06(03)	<i>0.13</i>
Ubs	170(146)	<i>0.05</i>	40(29)	<i>0.06</i>	20(15)	<i>0.19</i>	05(03)	<i>0.22</i>
Unaxis	147(146)	<i>0.93</i>	37(29)	<i>0.16</i>	15(15)	<i>0.89</i>	05(03)	<i>0.22</i>
Zurich	187(146)	0.00	46(29)	0.00	28(15)	0.00	07(03)	<i>0.03</i>
Agie Charm.	153(146)	<i>0.55</i>	31(29)	<i>0.71</i>	17(15)	<i>0.51</i>	07(03)	<i>0.03</i>
Conzetta	157(146)	<i>0.35</i>	25(29)	<i>0.46</i>	11(15)	<i>0.37</i>	05(03)	<i>0.22</i>
Eichhof	169(146)	<i>0.06</i>	34(29)	<i>0.40</i>	18(15)	<i>0.43</i>	04(03)	<i>0.55</i>
Huber+Suhner	198(146)	0.00	52(29)	0.00	29(15)	0.00	08(03)	<i>0.01</i>
Jelmoli	146(146)	<i>0.97</i>	27(29)	<i>0.78</i>	18(15)	<i>0.43</i>	04(03)	<i>0.55</i>
Kardex	154(146)	<i>0.50</i>	30(29)	<i>0.85</i>	11(15)	<i>0.37</i>	04(03)	<i>0.55</i>
Nextrom	241(146)	0.00	66(29)	0.00	34(15)	0.00	07(03)	<i>0.03</i>
OZ Holding	088(091)	<i>0.79</i>	21(18)	<i>0.55</i>	14(09)	<i>0.13</i>	03(02)	<i>0.44</i>
Swiss Steel	141(128)	<i>0.28</i>	27(26)	<i>0.77</i>	15(13)	<i>0.57</i>	05(03)	<i>0.19</i>
SMI	147(132)	<i>0.20</i>	37(26)	<i>0.05</i>	24(13)	<i>0.01</i>	06(03)	<i>0.05</i>
Perc. Signif.	<i>0.68</i>		<i>0.80</i>		<i>0.72</i>		<i>0.68</i>	
	<i>0.84</i>		<i>0.88</i>		<i>0.88</i>		<i>0.96</i>	

Table 3.3: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the unconditional EVT-based model. Insignificant p-values for a binomial test in italics.

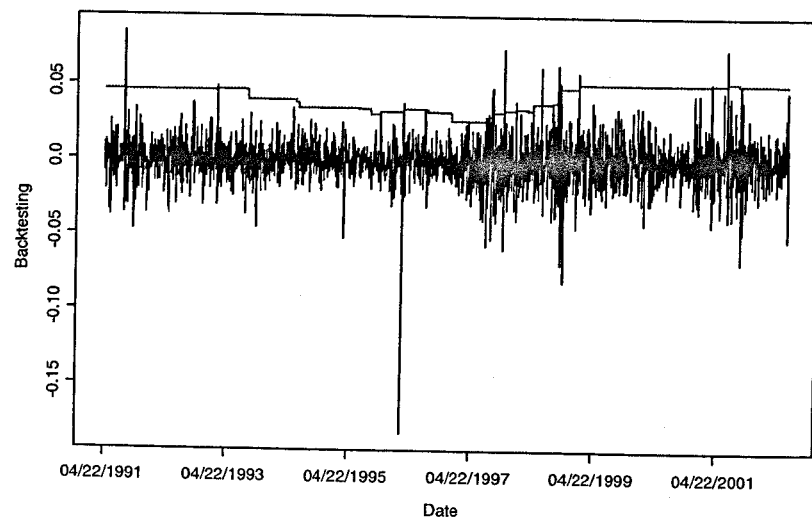


Figure 3.4: Backtesting results for the 99.5% quantile, using the Historical simulation model: log returns (losses on the positive axes) and VaR forecasts (superimposed) for the example of Novartis.

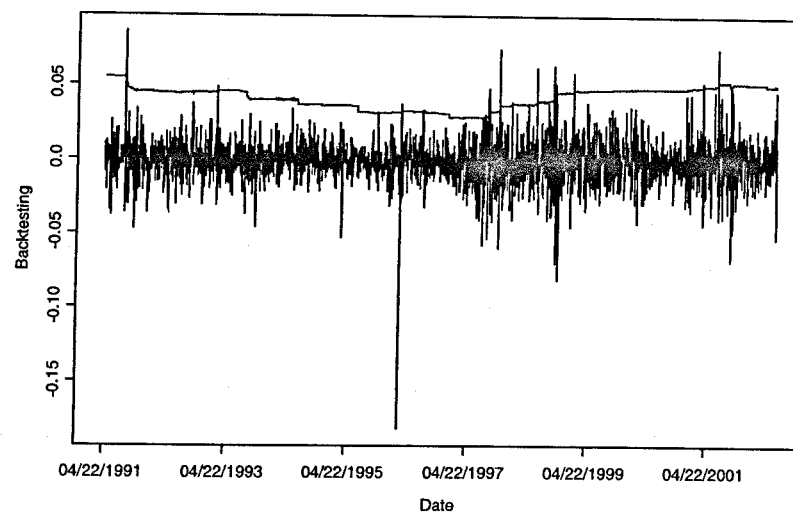


Figure 3.5: Backtesting results for the 99.5% quantile, using the unconditional EVT-based model: log returns (losses on the positive axes) and VaR forecasts (superimposed) for the example of Novartis.

3.4.2 Results for the VaR: the conditional models

Tables 3.4 to 3.6 show the results for the conditional VaR models; the conditional normal, the conditional t and the conditional EVT-based models, respectively. Again, columns 2, 4, 6 and 8 show the number of VaR violations (expected number of violations in brackets) for different quantiles and columns 3, 5, 7 and 9 show the p-values for the binomial test as described in 3.3.1, respectively.³⁵ The main findings are similar as for the unconditional models.

First, the conditional normal based model is accurate only for the 95% quantile, even if it performs worse than the conditional EVT-based model even at this quantile. Second, for quantiles higher than 95%, the normal based model doesn't give accurate results and is outperformed by both the conditional t as well as the conditional EVT-based models. Comparing the latter two models, we note that the conditional EVT-based model has the highest percentage of cases (for every quantile) where the null hypothesis of a correct model cannot be rejected, or more loosely speaking the highest percentage of cases where the null hypothesis is accepted. Note that the conditional EVT-based model is accepted in 99 out of 100 cases (over all titles and quantiles, respectively), on a 1% significance level.

Again the results are unchanged if we consider only the medium and small capitalized stocks. Also, the same is true if considering the stock index only.

If we compare the unconditional with the conditional models, respectively, we note first that the conditional normal based model gives better results than the unconditional normal based model, for the 95%, 99% and 99.5% quantiles. Moreover, the conditional EVT-based model is, for every quantile, better than the unconditional EVT-based model. Reflecting the current volatility background results in somewhat better VaR-estimates.³⁶

Finally, we compare the different models *directly*, as shown in Table 3.7. Direct comparison means that we show the percentage of cases in which one model performs better than an other one, meaning that the realized number of VaR violations is closer to the expected number of VaR violations.³⁷ The comparison of the Variance-covariance with the Historical simulation model, for

³⁵ Again, numbers in italics show p-values higher than 0.01 and numbers in bold italics show p-values higher than 0.05.
³⁶ Note also that in the unconditional model, the *iid* assumption for the raw returns is less accurate than the *iid* assumption for the residuals in the dynamic EVT-based model.
³⁷ Note that in the tables before, we showed the p-values and the percentage of cases where the null hypothesis of a correct model is accepted (or not rejected). Here, we compare two models directly and look which of the two is closer to the mark, meaning closer to the expected number of violations.

	CondNorm							
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Abb	132(146)	<i>0.25</i>	49(29)	0.00	36(15)	0.00	24(03)	0.00
Adecoo	137(146)	<i>0.47</i>	48(29)	0.00	38(15)	0.00	21(03)	0.00
Baloise	108(146)	0.00	40(29)	<i>0.06</i>	26(15)	<i>0.01</i>	11(03)	0.00
CS Group	147(146)	<i>0.93</i>	50(29)	0.00	33(15)	0.00	14(03)	0.00
Ems	105(146)	0.00	43(29)	<i>0.01</i>	29(15)	0.00	13(03)	0.00
Julius Baer	130(146)	<i>0.19</i>	39(29)	<i>0.08</i>	23(15)	<i>0.05</i>	10(03)	0.00
Nestle	166(146)	<i>0.09</i>	37(29)	<i>0.16</i>	20(15)	<i>0.19</i>	07(03)	<i>0.03</i>
Novartis	129(146)	<i>0.16</i>	38(29)	<i>0.11</i>	23(15)	<i>0.05</i>	15(03)	0.00
Richemont	122(129)	<i>0.59</i>	41(26)	<i>0.01</i>	28(13)	0.00	15(03)	0.00
Roche	130(146)	<i>0.19</i>	42(29)	<i>0.02</i>	28(15)	0.00	14(03)	0.00
Sulzer	152(146)	<i>0.61</i>	62(29)	0.00	44(15)	0.00	24(03)	0.00
Swiss Re	118(146)	<i>0.02</i>	29(29)	<i>1.00</i>	22(15)	<i>0.06</i>	10(03)	0.00
Ubs	138(146)	<i>0.52</i>	55(29)	0.00	40(15)	0.00	21(03)	0.00
Unaxis	129(146)	<i>0.16</i>	45(29)	<i>0.01</i>	30(15)	0.00	16(03)	0.00
Zurich	125(146)	<i>0.08</i>	46(29)	0.00	34(15)	0.00	18(03)	0.00
Agie Charm.	132(146)	<i>0.25</i>	46(29)	0.00	33(15)	0.00	21(03)	0.00
Conzetta	131(146)	<i>0.22</i>	57(29)	0.00	41(15)	0.00	26(03)	0.00
Eichhof	144(146)	<i>0.90</i>	61(29)	0.00	46(15)	0.00	26(03)	0.00
Huber+Suhner	142(146)	<i>0.77</i>	70(29)	0.00	54(15)	0.00	34(03)	0.00
Jelmoli	125(146)	<i>0.08</i>	50(29)	0.00	37(15)	0.00	20(03)	0.00
Kardex	136(146)	<i>0.42</i>	66(29)	0.00	51(15)	0.00	36(03)	0.00
Nextrom	130(146)	<i>0.19</i>	63(29)	0.00	49(15)	0.00	39(03)	0.00
OZ Holding	072(091)	<i>0.04</i>	34(18)	0.00	27(09)	0.00	13(02)	0.00
Swiss Steel	133(128)	<i>0.68</i>	51(26)	0.00	40(13)	0.00	23(03)	0.00
SMI	137(132)	<i>0.69</i>	42(26)	<i>0.01</i>	26(13)	0.00	10(03)	0.00
Perc. Signif.	<i>0.84</i>		<i>0.20</i>		<i>0.16</i>		<i>0.00</i>	
	<i>0.92</i>		<i>0.40</i>		<i>0.20</i>		<i>0.04</i>	

Table 3.4: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the conditional normal-based model. Insignificant p-values for a binomial test in italics.

	CondT							
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Abb	150(146)	<i>0.73</i>	37(29)	<i>0.16</i>	23(15)	<i>0.05</i>	07(03)	<i>0.03</i>
Adecco	163(146)	<i>0.15</i>	25(29)	<i>0.46</i>	14(15)	<i>1.00</i>	01(03)	<i>0.38</i>
Baloise	129(146)	<i>0.16</i>	22(29)	<i>0.19</i>	10(15)	<i>0.24</i>	03(03)	<i>1.00</i>
CS Group	158(146)	<i>0.31</i>	24(29)	<i>0.35</i>	09(15)	<i>0.15</i>	01(03)	<i>0.38</i>
Ems	122(146)	<i>0.04</i>	21(29)	<i>0.14</i>	09(15)	<i>0.15</i>	01(03)	<i>0.38</i>
Julius Baer	151(146)	<i>0.67</i>	19(29)	<i>0.05</i>	09(15)	<i>0.15</i>	02(03)	<i>0.78</i>
Nestle	127(146)	<i>0.12</i>	33(29)	<i>0.51</i>	14(15)	<i>1.00</i>	04(03)	<i>0.55</i>
Novartis	151(146)	<i>0.67</i>	25(29)	<i>0.46</i>	15(15)	<i>0.89</i>	04(03)	<i>0.55</i>
Richemont	138(129)	<i>0.39</i>	18(26)	<i>0.14</i>	11(13)	<i>0.68</i>	04(03)	<i>0.33</i>
Roche	148(146)	<i>0.86</i>	28(29)	<i>0.93</i>	16(15)	<i>0.69</i>	04(03)	<i>0.55</i>
Sulzer	168(146)	<i>0.07</i>	32(29)	<i>0.64</i>	15(15)	<i>0.89</i>	03(03)	<i>1.00</i>
Swiss Re	135(146)	<i>0.37</i>	20(29)	<i>0.09</i>	10(15)	<i>0.24</i>	05(03)	<i>0.22</i>
Ubs	156(146)	<i>0.40</i>	37(29)	<i>0.16</i>	14(15)	<i>1.00</i>	02(03)	<i>0.78</i>
Unaxis	153(146)	<i>0.55</i>	25(29)	<i>0.46</i>	10(15)	<i>0.24</i>	03(03)	<i>1.00</i>
Zurich	147(146)	<i>0.93</i>	34(29)	<i>0.40</i>	13(15)	<i>0.79</i>	04(03)	<i>0.55</i>
Agie Charm.	200(146)	0.00	34(29)	<i>0.40</i>	08(15)	<i>0.09</i>	00(03)	<i>0.08</i>
Conzetta	323(146)	0.00	70(29)	0.00	25(15)	<i>0.01</i>	01(03)	<i>0.38</i>
Eichhof	447(146)	0.00	183(29)	0.00	117(15)	0.00	026(03)	0.00
Huber+Suhner	362(146)	0.00	238(29)	0.00	202(15)	0.00	143(03)	0.00
Jelmoli	156(146)	<i>0.40</i>	19(29)	<i>0.05</i>	11(15)	<i>0.37</i>	01(03)	<i>0.38</i>
Kardex	387(146)	0.00	225(29)	0.00	161(15)	0.00	97(03)	0.00
Nextrom	369(146)	0.00	214(29)	0.00	183(15)	0.00	144(03)	0.00
OZ Holding	095(091)	<i>0.67</i>	17(18)	<i>0.82</i>	05(09)	<i>0.19</i>	00(02)	<i>0.27</i>
Swiss Steel	291(137)	0.00	84(26)	0.00	38(13)	0.00	09(03)	0.00
SMI	148(132)	<i>0.17</i>	27(26)	<i>0.92</i>	12(13)	<i>0.79</i>	06(03)	<i>0.05</i>
Perc. Signif.	<i>0.72</i>		<i>0.76</i>		<i>0.76</i>		<i>0.76</i>	
	<i>0.76</i>		<i>0.76</i>		<i>0.80</i>		<i>0.80</i>	

Table 3.5: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the conditional t-based model. Insignificant p-values for a binomial test in italics.

	CondGpd							
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Abb	145(146)	<i>1.00</i>	34(29)	<i>0.40</i>	16(15)	<i>0.69</i>	06(03)	<i>0.13</i>
Adecco	158(146)	<i>0.31</i>	32(29)	<i>0.64</i>	17(15)	<i>0.51</i>	05(03)	<i>0.22</i>
Baloise	132(146)	<i>0.25</i>	26(29)	<i>0.58</i>	11(15)	<i>0.37</i>	05(03)	<i>0.22</i>
CS Group	156(146)	<i>0.40</i>	36(29)	<i>0.22</i>	16(15)	<i>0.69</i>	07(03)	<i>0.03</i>
Ems	122(146)	<i>0.04</i>	24(29)	<i>0.35</i>	11(15)	<i>0.37</i>	03(03)	<i>1.00</i>
Julius Baer	158(146)	<i>0.31</i>	25(29)	<i>0.46</i>	13(15)	<i>0.79</i>	06(03)	<i>0.13</i>
Nestle	128(146)	<i>0.14</i>	30(29)	<i>0.85</i>	13(15)	<i>0.79</i>	03(03)	<i>1.00</i>
Novartis	157(146)	<i>0.40</i>	31(29)	<i>0.71</i>	16(15)	<i>0.69</i>	06(03)	<i>0.13</i>
Richemont	140(129)	<i>0.32</i>	20(26)	<i>0.28</i>	14(13)	<i>0.78</i>	03(03)	<i>0.75</i>
Roche	133(146)	<i>0.29</i>	21(29)	<i>0.14</i>	15(15)	<i>0.89</i>	07(03)	<i>0.03</i>
Sulzer	171(146)	<i>0.04</i>	41(29)	<i>0.04</i>	21(15)	<i>0.11</i>	07(03)	<i>0.03</i>
Swiss Re	134(146)	<i>0.33</i>	19(29)	<i>0.05</i>	11(15)	<i>0.37</i>	07(03)	<i>0.03</i>
Ubs	150(146)	<i>0.73</i>	37(29)	<i>0.16</i>	13(15)	<i>0.79</i>	02(03)	<i>0.78</i>
Unaxis	136(146)	<i>0.42</i>	30(29)	<i>0.85</i>	16(15)	<i>0.69</i>	06(03)	<i>0.13</i>
Zurich	148(146)	<i>0.86</i>	34(29)	<i>0.40</i>	16(15)	<i>0.69</i>	04(03)	<i>0.55</i>
Agie Charm.	149(146)	<i>0.80</i>	29(29)	<i>1.00</i>	15(15)	<i>0.89</i>	05(03)	<i>0.22</i>
Conzetta	140(146)	<i>0.64</i>	33(29)	<i>0.51</i>	15(15)	<i>0.89</i>	06(03)	<i>0.13</i>
Eichhof	160(146)	<i>0.23</i>	35(29)	<i>0.30</i>	22(15)	<i>0.06</i>	06(03)	<i>0.13</i>
Huber+Suhner	162(146)	<i>0.17</i>	42(29)	<i>0.02</i>	19(15)	<i>0.29</i>	06(03)	<i>0.13</i>
Jelmoli	150(146)	<i>0.73</i>	34(29)	<i>0.40</i>	17(15)	<i>0.51</i>	05(03)	<i>0.22</i>
Kardex	144(146)	<i>0.90</i>	32(29)	<i>0.64</i>	10(15)	<i>0.24</i>	05(03)	<i>0.22</i>
Nextrom	195(146)	0.00	35(29)	<i>0.30</i>	23(15)	<i>0.05</i>	04(03)	<i>0.55</i>
OZ Holding	095(091)	<i>0.27</i>	27(18)	<i>0.06</i>	12(09)	<i>0.40</i>	02(02)	<i>1.00</i>
Swiss Steel	145(128)	<i>0.13</i>	27(26)	<i>0.77</i>	13(13)	<i>1.00</i>	05(03)	<i>0.19</i>
SMI	123(132)	<i>0.42</i>	19(26)	<i>0.14</i>	10(13)	<i>0.41</i>	04(03)	<i>0.53</i>
Perc. Signif.	<i>0.88</i>		<i>0.92</i>		<i>1.00</i>		<i>0.84</i>	
	<i>0.96</i>		<i>1.00</i>		<i>1.00</i>		<i>1.00</i>	

Table 3.6: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the conditional EVT-based model. Insignificant p-values for a binomial test in italics.

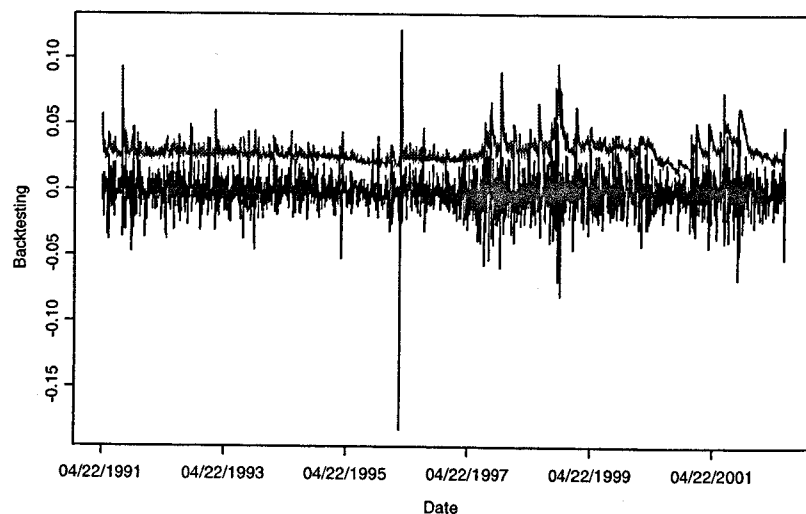


Figure 3.6: Backtesting results for the 99.5% quantile, using the conditional normal-based model: log returns (losses on the positive axes) and VaR forecasts (superimposed) for the example of Novartis.

the 95% quantile, for example, shows that in 60% of the (25) cases, the former is closer to the mark (i.e. closer to the expected number of VaR violations). The results from before are unchanged, however.

Figures 3.6 and 3.7 show Backtesting results for the 99.5% quantile, for the example of Novartis, where we clearly see that the current volatility background is captured in the VaR estimation. Without doing calculations however, we infer from the figures that the conditional model forecasts are (by far) more volatile than the unconditional ones.

3.4.3 The event on September 11, 2001

Figures 3.8 and 3.9 give (graphical) explanations of how some methods fail. The respective figures show enlarged sections of the period corresponding to the

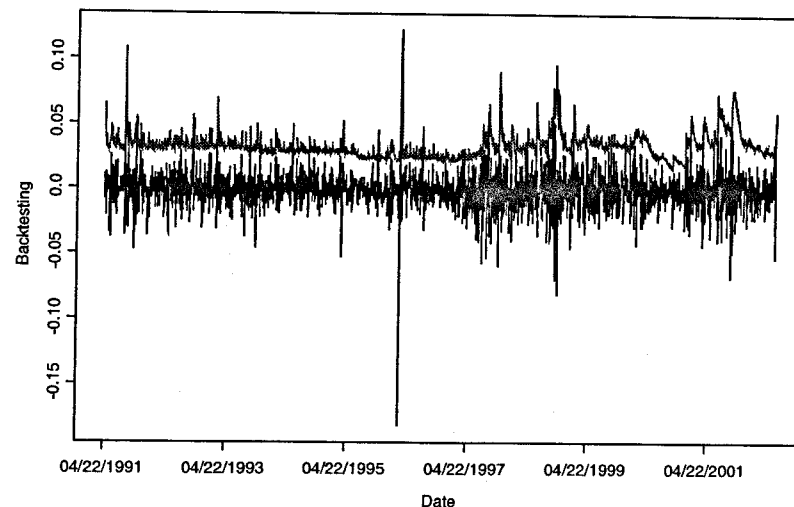


Figure 3.7: Backtesting results for the 99.5% quantile, using the conditional EVT-based model: log returns (losses on the positive axes) and VaR forecasts (superimposed) for the example of Novartis.

	95%		99%		99.5%		99.9%	
VarCov/Hist. sim.	<i>0.60</i>	0.40	0.00	<i>1.00</i>	0.00	<i>1.00</i>	0.04	<i>0.96</i>
VarCov/Gpd	<i>0.60</i>	0.40	0.04	<i>0.96</i>	0.00	<i>1.00</i>	0.00	<i>1.00</i>
Gpd/Hist. sim.	0.46	<i>0.54</i>	<i>0.80</i>	0.20	<i>0.80</i>	0.20	<i>0.95</i>	0.05
CondNorm/VarCov	<i>0.74</i>	0.26	<i>0.80</i>	0.20	<i>0.63</i>	0.37	0.34	<i>0.76</i>
CondGpd/CondNorm	<i>0.64</i>	0.36	<i>0.96</i>	0.04	<i>1.00</i>	0.00	<i>1.00</i>	0.00
CondGpd/Gpd	<i>0.66</i>	0.34	<i>0.63</i>	0.37	<i>0.75</i>	0.25	<i>0.62</i>	0.38
CondGpd/CondT	<i>0.63</i>	0.37	<i>0.74</i>	0.26	<i>0.72</i>	0.28	<i>0.52</i>	0.48

Table 3.7: Direct comparison of two different models: percentage of cases in which one model performs better than an other one, meaning that the realized number of VaR violations is closer to the expected number of VaR violations.

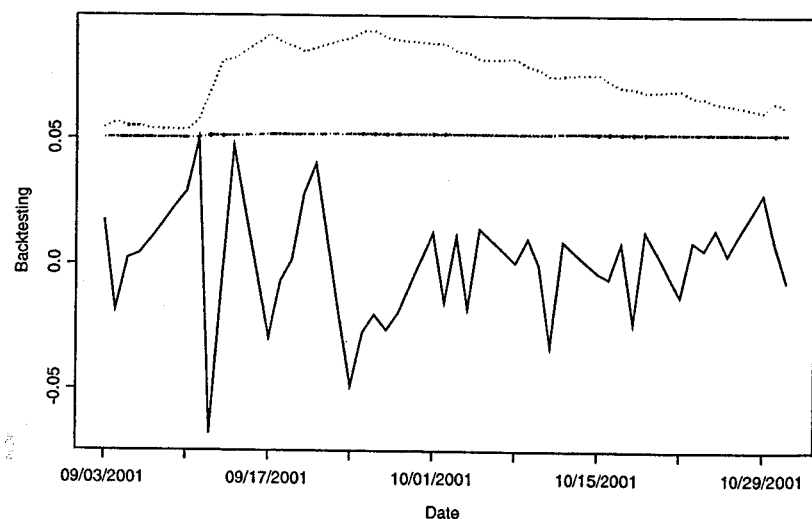


Figure 3.8: Enlarged section corresponding to the event of September 11, 2001, example of novartis. The dotted line shows the path of the conditional EVT-based model, the dashed line shows the path of the unconditional EVT-based model.

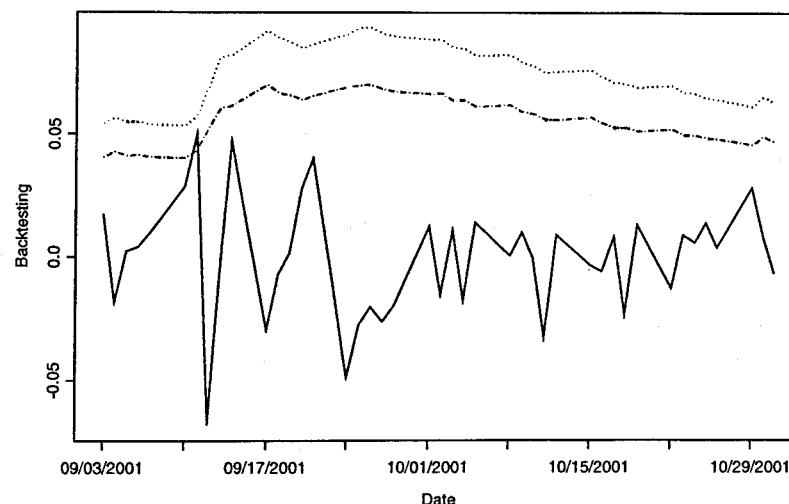


Figure 3.9: Enlarged section corresponding to the event of September 11, 2001, example of novartis. The dotted line shows the path of the conditional EVT-based model, the dashed line shows the path of the conditional normal-based model.

turbulences around the event of September 11, 2001. Figure 3.8 shows that the unconditional EVT-based model cannot respond quickly to changing volatility and tends to be violated several times in a row in stress periods. However, we see also that the GARCH based approach lags behind the returns, but overcompensates by producing large VaR forecasts, leading to violation clustering, whereas the unconditional EVT-based forecast is hardly affected by the event. Figure 3.9 shows that although the conditional normal based model does respond to changing volatility, it does not take into account the leptocurtosis of the residuals. From Figure C.4 in the appendix, we see the similarity between the conditional t and the conditional EVT-based models for our given data set.

3.4.4 Results for the S-VaR

In the following section, we present the empirical results for the S-VaR. Figures 3.10 to 3.13 show the residuals for the Novartis example and quantiles 95% and 99%, respectively. A first look at the figures shows that for the residuals calculated under an assumption of conditional normality the null hypothesis seems doubtful, whereas for the residuals calculated under the conditional EVT-based model the null hypothesis seems to be justified.

Tables 3.8 to 3.10 show the results of the bootstrap test for the unconditional EVT-based, the conditional normal and the conditional EVT-based model, respectively. The first columns show the SMI and SPI titles, respectively. Columns 2, 3, 4 and 5 show the p-values for the bootstrap test as described in subsection 3.3.2, respectively. For example, a p-value of 0.05 (0.01) or less indicates that the null hypothesis of a correct model is rejected, on a 95% (99%) confidence level. Finally, the last two rows show the percentage of cases where the null hypothesis cannot be rejected (upper row for the 0.05 level, lower row for the 0.01 level). For example, we note that for the unconditional EVT-based model, for the 95% quantile, we have 88% of the (25) cases with a p-value of more than 5%, meaning that we accept the null of zero-mean residuals³⁸ in 68% percent of the cases under consideration.

We note that the residuals derived under the assumption of normality always fail the test (on a 5% significance level) with p-values in all cases much less than 0.05. Moreover, the residuals derived under the assumption of the conditional EVT-based model never fail the test (on a 1% significance level), with p-values in all cases more than 0.01. We conclude that an assumption of conditional normality is useless for the purpose of calculating the S-VaR.

3.5 Conclusion

In the present chapter, we were concerned with tail estimation for financial return series, and in particular with the estimation of measures of market risk, e.g. VaR and S-VaR, respectively. We apply different unconditional and conditional models and backtest the models for chosen Swiss stock returns, using daily returns from 22.08.1987 to 19.06.2002. The main findings are summarized as follows.

- Traditional models such as the Variance-covariance model and the Historical simulation model tend to underestimate the risk, leading to more VaR

³⁸ Or we cannot reject the null hypothesis, correctly.

3.5. CONCLUSION

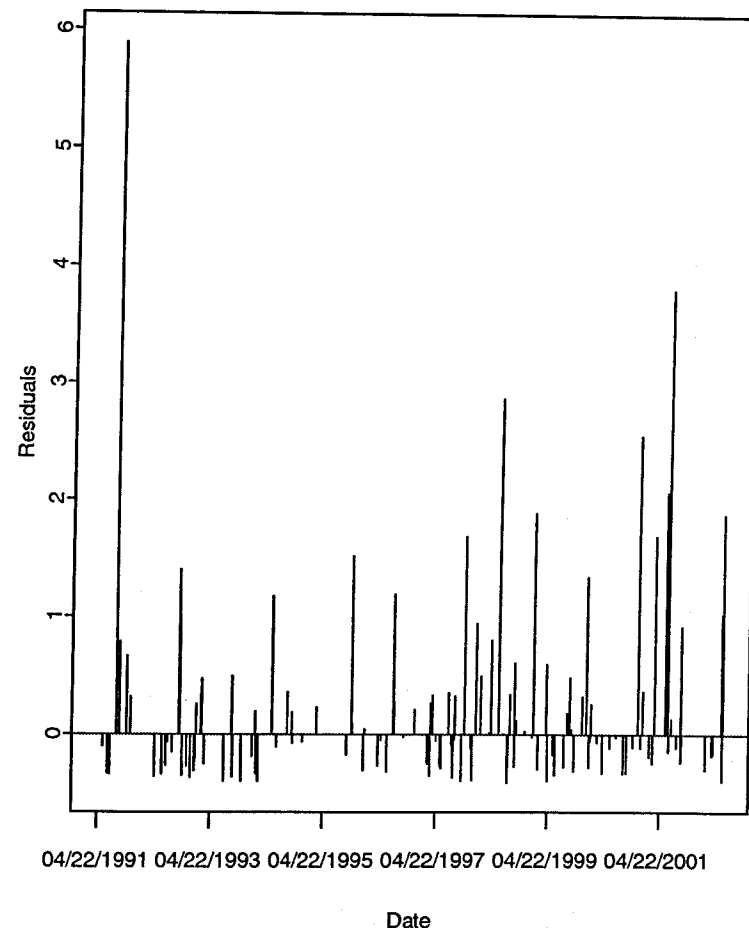


Figure 3.10: Backtesting results: exceedance residuals for the novartis example, 95% quantile, using the conditional normal model. Under the null hypothesis that the model correctly estimates the tail, these residuals should have mean zero.

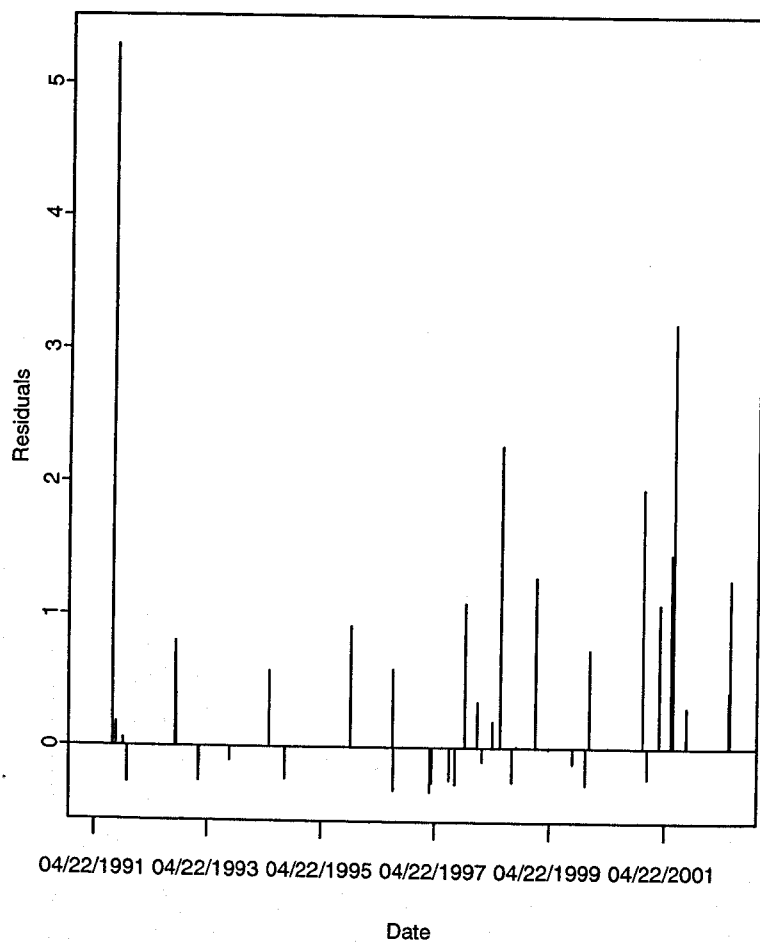


Figure 3.11: Backtesting results: exceedance residuals for the novartis example, 99% quantile, using the conditional normal model. Under the null hypothesis that the model correctly estimates the tail, these residuals should have mean zero.

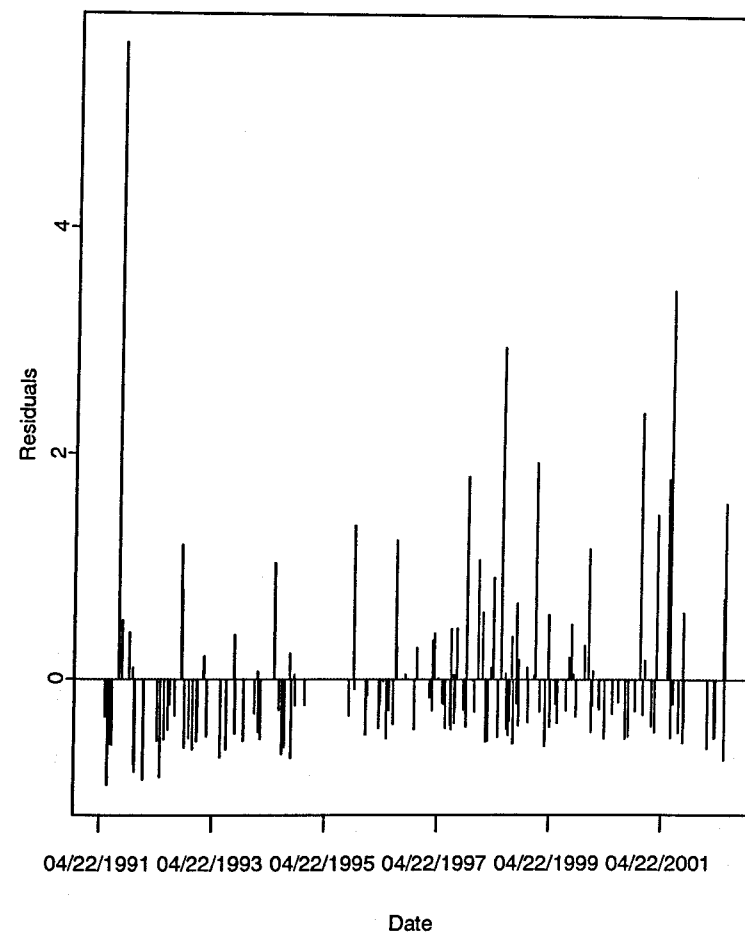


Figure 3.12: Backtesting results: exceedance residuals for the novartis example, 95% quantile, using the conditional EVT-based model. Under the null hypothesis that the model correctly estimates the tail, these residuals should have mean zero.

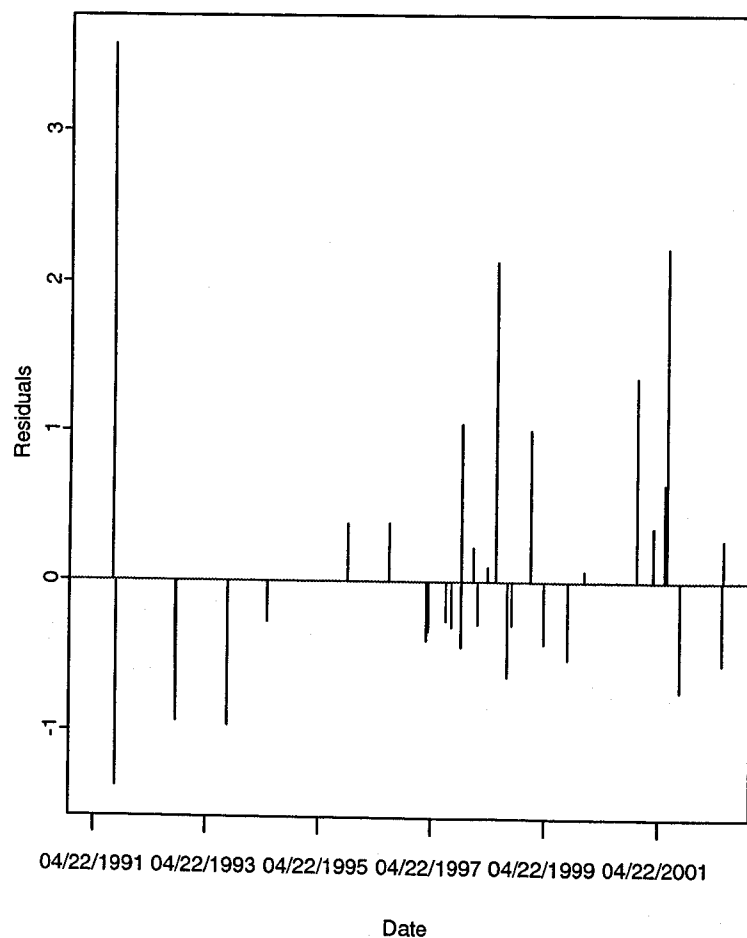


Figure 3.13: Backtesting results: exceedance residuals for the novartis example, 99% quantile, using the conditional EVT-based model. Under the null hypothesis that the model correctly estimates the tail, these residuals should have mean zero.

	Gpd			
	95%	99%	99.5%	99.9%
Abb	<i>0.03</i>	<i>0.16</i>	<i>0.11</i>	<i>0.07</i>
Adeco	<i>0.15</i>	<i>0.30</i>	<i>0.14</i>	<i>0.12</i>
Baloise	<i>0.15</i>	<i>0.20</i>	<i>0.04</i>	<i>0.14</i>
CS Group	<i>0.29</i>	<i>0.24</i>	<i>0.28</i>	<i>0.33</i>
Ems	<i>0.61</i>	<i>0.58</i>	<i>0.53</i>	<i>0.25</i>
Julius Baer	<i>0.20</i>	<i>0.42</i>	<i>0.61</i>	<i>0.51</i>
Nestle	<i>0.25</i>	<i>0.24</i>	<i>1.00</i>	<i>0.23</i>
Novartis	<i>0.48</i>	<i>0.30</i>	<i>0.20</i>	<i>0.11</i>
Richemont	<i>0.47</i>	<i>0.28</i>	<i>0.42</i>	<i>0.12</i>
Roche	<i>0.64</i>	<i>0.94</i>	<i>0.91</i>	<i>0.77</i>
Sulzer	<i>0.02</i>	<i>0.04</i>	<i>0.01</i>	<i>0.03</i>
Swiss Re	<i>0.47</i>	<i>0.01</i>	<i>0.03</i>	<i>0.06</i>
Ubs	<i>0.27</i>	<i>0.47</i>	<i>0.28</i>	<i>0.31</i>
Unaxis	<i>0.24</i>	<i>0.61</i>	<i>0.27</i>	<i>0.25</i>
Zurich	<i>0.04</i>	<i>0.06</i>	<i>0.26</i>	<i>0.04</i>
Agie Charm.	<i>0.43</i>	<i>0.13</i>	<i>0.06</i>	<i>0.60</i>
Conzetta	<i>0.58</i>	<i>0.18</i>	<i>0.07</i>	<i>0.10</i>
Eichhof	<i>0.14</i>	<i>0.37</i>	<i>0.30</i>	<i>0.20</i>
Huber+Suhner	<i>0.06</i>	<i>0.41</i>	<i>0.30</i>	<i>0.05</i>
Jelmoli	<i>0.20</i>	<i>0.05</i>	<i>0.11</i>	<i>0.05</i>
Kardex	<i>0.37</i>	<i>0.32</i>	<i>0.07</i>	<i>0.04</i>
Nextrom	<i>0.06</i>	<i>0.27</i>	<i>0.46</i>	<i>0.44</i>
OZ Holding	<i>0.36</i>	<i>0.43</i>	<i>0.74</i>	<i>0.72</i>
Swiss Steel	<i>0.55</i>	<i>0.10</i>	<i>0.12</i>	<i>0.25</i>
SMI	<i>0.19</i>	<i>0.09</i>	<i>0.60</i>	<i>0.31</i>
Perc. Signif.	<i>0.88</i>	<i>0.92</i>	<i>0.88</i>	<i>0.88</i>
	<i>1.00</i>	<i>1.00</i>	<i>1.00</i>	<i>1.00</i>

Table 3.8: p-values for a one-sided bootstrap test of the hypothesis that the exceedance residuals have mean zero against the alternative that the mean is greater than zero. Insignificant p-values are shown in italics.

	CondNorm			
	95%	99%	99.5%	99.9%
Abb	0.00	0.00	0.00	0.00
Adecco	0.00	0.00	0.00	0.00
Baloise	0.00	0.00	0.00	<i>0.01</i>
CS Group	0.00	0.00	0.00	<i>0.01</i>
Ems	0.00	0.00	0.00	0.00
Julius Baer	0.00	0.00	0.00	<i>0.01</i>
Nestle	0.00	0.00	0.00	0.00
Novartis	0.00	0.00	0.00	0.00
Richemont	0.00	0.00	0.00	0.00
Roche	0.00	0.00	0.00	<i>0.01</i>
Sulzer	0.00	0.00	0.00	0.00
Swiss Re	0.00	0.00	0.00	0.00
Ubs	0.00	0.00	0.00	<i>0.01</i>
Unaxis	0.00	0.00	0.00	0.00
Zurich	0.00	0.00	0.00	0.00
Agie Charm.	0.00	0.00	0.00	0.00
Conzetta	0.00	0.00	0.00	0.00
Eichhof	0.00	0.00	0.00	0.00
Huber+Suhner	0.00	0.00	0.00	0.00
Jelmoli	0.00	0.00	0.00	0.00
Kardex	0.00	0.00	0.00	0.00
Nextrom	0.00	0.00	0.00	0.00
OZ Holding	0.00	0.00	0.00	0.00
Swiss Steel	0.00	0.00	0.00	0.00
SMI	0.00	<i>0.01</i>	0.00	<i>0.01</i>
Perc. Signif.	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>	<i>0.00</i>
	<i>0.00</i>	<i>0.04</i>	<i>0.00</i>	<i>0.24</i>

Table 3.9: p-values for a one-sided bootstrap test of the hypothesis that the exceedance residuals have mean zero against the alternative that the mean is greater than zero. Insignificant p-values are shown in italics.

	CondGpd			
	95%	99%	99.5%	99.9%
Abb	<i>0.30</i>	<i>0.22</i>	<i>0.10</i>	<i>0.90</i>
Adecco	<i>0.69</i>	<i>0.48</i>	<i>0.70</i>	<i>0.85</i>
Baloise	<i>0.29</i>	<i>0.41</i>	<i>0.06</i>	<i>0.76</i>
CS Group	<i>0.33</i>	<i>0.27</i>	<i>0.08</i>	<i>0.06</i>
Ems	<i>0.54</i>	<i>0.49</i>	<i>0.35</i>	<i>0.13</i>
Julius Baer	<i>0.73</i>	<i>0.09</i>	<i>0.04</i>	<i>0.29</i>
Nestle	<i>0.59</i>	<i>0.91</i>	<i>0.81</i>	<i>0.30</i>
Novartis	<i>0.55</i>	<i>0.21</i>	<i>0.13</i>	<i>0.64</i>
Richemont	<i>0.85</i>	<i>0.07</i>	<i>0.42</i>	<i>0.18</i>
Roche	<i>0.57</i>	<i>0.03</i>	<i>0.13</i>	<i>0.26</i>
Sulzer	<i>0.23</i>	<i>0.26</i>	<i>0.21</i>	<i>0.15</i>
Swiss Re	<i>0.39</i>	<i>0.01</i>	<i>0.01</i>	<i>0.05</i>
Ubs	<i>0.47</i>	<i>0.80</i>	<i>0.39</i>	<i>0.24</i>
Unaxis	<i>0.17</i>	<i>0.29</i>	<i>0.16</i>	<i>0.12</i>
Zurich	<i>0.55</i>	<i>0.32</i>	<i>0.14</i>	<i>0.22</i>
Agie Charm.	<i>0.53</i>	<i>0.28</i>	<i>0.30</i>	<i>0.15</i>
Conzetta	<i>0.19</i>	<i>0.21</i>	<i>0.07</i>	<i>0.03</i>
Eichhof	<i>0.25</i>	<i>0.18</i>	<i>0.67</i>	<i>0.68</i>
Huber+Suhner	<i>0.13</i>	<i>0.61</i>	<i>0.36</i>	<i>0.95</i>
Jelmoli	<i>0.22</i>	<i>0.19</i>	<i>0.17</i>	<i>0.12</i>
Kardex	<i>0.29</i>	<i>0.39</i>	<i>0.01</i>	<i>0.16</i>
Nextrom	<i>0.62</i>	<i>0.24</i>	<i>0.27</i>	<i>0.13</i>
OZ Holding	<i>0.37</i>	<i>0.90</i>	<i>0.83</i>	<i>0.73</i>
Swiss Steel	<i>0.60</i>	<i>0.21</i>	<i>0.08</i>	<i>0.16</i>
SMI	<i>0.71</i>	<i>0.25</i>	<i>0.17</i>	<i>0.06</i>
Perc. Signif.	<i>1.00</i>	<i>0.92</i>	<i>0.88</i>	<i>0.96</i>
	<i>1.00</i>	<i>1.00</i>	<i>1.00</i>	<i>1.00</i>

Table 3.10: p-values for a one-sided bootstrap test of the hypothesis that the exceedance residuals have mean zero against the alternative that the mean is greater than zero. Insignificant p-values are shown in italics.

violations than expected.

- For our particular data set, we find the unconditional EVT-based model to be better, for quantiles higher than 95%, than both the Historical simulation as well as the conditional normal based model.
- We find that the conditional approaches which model the conditional distribution of returns are better suited for VaR estimation than unconditional ones.³⁹ The unconditional models cannot respond quickly to changing volatility and tend to be violated several times in a row in stress periods.
- Although the conditional normal based model does respond to changing volatility, it does not take into account the leptokurtosis of the residuals. We find the conditional t or even more the conditional EVT-based models to be preferable.
- For the conditional EVT-based model we accept the Null hypothesis of a correct model in 99% of the cases.⁴⁰ This refers to our starting question; we observe an improvement using a conditional EVT-based model, as compared to the unconditional EVT-based alternative.
- However, although being more robust, the conditional models have the main drawback of producing VaR forecasts which are necessarily more volatile than the unconditional models do.
- The conditional t model works well for our data set with symmetric tails, however, is likely to fail when the tails are asymmetric.⁴¹ The GPD-based model seems preferable from this point of view.
- The comparison of estimates for the S-VaR using the conditional EVT-based model and the standard GARCH-model with normal innovations shows that the innovation distribution should be modelled by a fat-tailed distribution. The assumption of conditional normality is useless for the purpose of calculating the S-VaR.

³⁹The exception is the conditional normal model, which performs worse than the unconditional EVT-based model and the Historical simulation model, respectively, for quantiles higher than 95%.

⁴⁰Note that we have a total, over all titles and quantiles, of 100 backtesting results.

⁴¹Further research will show how the conditional t model performs for financial options with asymmetric payoff patterns.

- Finally, the choice of methodology is likely to depend on the situation and question at hand, on the quantile under consideration, and both the conditional and unconditional approaches belong to the toolbox of the risk manager. While the unconditional distribution is interesting when we look at long-term investment decisions and the occurrence of very rare (stress) events, the conditional distribution is more appropriate when we deal with day-to-day risks and short term risk management, as in the present work. The skilled risk manager is able to specify accurate models for specific situations.

Further research related to the present work could deal with the issue of violation clustering (refer to Section 3.3.1) or with the application of asymmetric data sets, such as financial options.

There are several issues which limit the general applicability of EVT to the financial return series. First, there is the issue of higher dimensions. The dynamic EVT-based model could be extended to bivariate (multivariate) series.⁴² If we are interested in a small number of risk factors and are particularly concerned that joint extreme values may occur, we might use bivariate models to get useful descriptions of the joint tail. For higher dimensions though it seems more appropriate to collapse the problem to a univariate problem by considering a whole portfolio of assets as a single risk.

Second, the derivation of multiperiod VaR estimates. The relaxation of the normality assumption leads to difficulties due to the "square-root-of-time" method, e.g. the practice of obtaining multiperiod volatility predictions by multiplying the one day prediction by the square root of the length of the time horizon. We refer to Danielsson (2002) for some ideas.

Other practical issues in implementing EVT-based measures are the length of the data set for the estimation and the question of how to choose the threshold, or put in different words: from where do extremes matter?⁴³

Danielsson (2002) considers the properties of risk measures in more general, both from internal and external (regulatory) points of view.⁴⁴ In an extensive

⁴²For example, we could estimate a bivariate GARCH model, calculate the residuals and apply bivariate EVT on the residuals. An alternative model for the *bivariate* analysis is presented in the next chapter.

⁴³As mentioned earlier, one solution is given by Danielsson and De Vries (1997a) and Danielsson and De Vries (2002) who propose an automatic double subsample bootstrap procedure for the determination of the optimal threshold. The approach can supplement the usual graphical analysis. It would be interesting, however, to know whether their result can be extended to the GPD approach.

⁴⁴Note that in the present chapter the research question focuses mainly on the possible improvement using a conditional EVT-based risk measure. We will assess risk measures more

survey across data classes⁴⁵ and risk models, he concludes that current risk modelling technology is still in the early stage of development, lacking in the robustness (success) of risk forecasts and/or producing excessively volatile risk forecasts. If not done with great skill and care, the risk forecast may be unreliable. And in the worst case, it may impose significant but unnecessary cost on the financial institution, due to misallocation of capital and excessive portfolio rebalancing.

Chapter 4

Bivariate Risk Measurement based on EVT and Copulas

4.1 Introduction

An important aspect of risk modelling, which is currently receiving increasing attention, is the issue of non-linear dependence, also known as changing correlations. As a stylized fact we know that estimated correlations are much lower when the markets are increasing in value, compared to market conditions when some assets increase and others decrease, respectively. This is especially true when the markets are falling. Indeed, the worse market conditions are, the higher the correlation is, see Ang and Chen (2002) and Longin and Solnik (2000). However, most risk models do not take this into account and produce correlation estimates based on normal market conditions. Consequently, these models will underestimate portfolio risk. More precisely, correlations will underestimate the downside risk. In order to properly measure the dependence structure, we need to use the related concepts of (bivariate) extreme value theory and copulas. In the present chapter, the starting questions are

- How can we build a risk model which reflects the mentioned empirical facts?

generally in our concluding chapter, however.

⁴⁵ Stock indices, stock prices, exchange rates, price indices, gold prices and oil prices are included, respectively.

- In particular, how can we use EVT and related dependence concepts in a Monte Carlo simulation environment?
- How do different assumptions about the dependence structure affect the performance of risk models (Value at Risk (VaR) and the Expected Short-fall¹ (S-VaR), respectively)?
- Moreover, what dependence structure is the most appropriate for risk modelling?

We combine (univariate) extreme value theory and copula theory in a simulation framework. In contrast to chapter 2, we do not necessarily apply concepts of extremal or tail dependence.

Extreme Value Theory, widely used in the actuarial field, is currently experiencing a boom in the field of finance, especially in risk management. Extreme Value Theory is mainly used to address the estimation issues concerning large losses in leptocurtic and potentially asymmetric distributions. Basically, EVT uses limiting distributions for extreme returns, regardless of the actual distribution. Consequently, we do not have to make (any) assumptions about the underlying distribution function of the portfolio returns in order to calculate extreme quantiles and event probabilities. EVT is in this sense unique as a statistical discipline in that it develops techniques and models for describing the unusual rather than the usual. There exists a wide range of textbooks in the area of EVT. Treatments of the mathematical foundations can be found in Leadbetter, Lindgren and Rootzén (1983) or in Resnick (1987). On the statistical side, Reiss and Thomas (2001) and Coles (2001) give thorough overviews of statistical techniques for extreme value modelling. Finally, a detailed treatment of the role of EVT for insurance and financial applications is given by Embrechts, Klüppelberg and Mikosch (2001).

There are several works concerned with univariate EVT-based risk modelling, refer to Chapter 3. Danielsson (2002), Danielsson and Morimoto (2000) and Gençay and Selçuk (2002) compare different conditional and unconditional approaches, respectively. Frey and McNeil (2000) calculate the conditional VaR-measures by filtering return series with a GARCH model and then apply threshold based EVT tools to the *iid* residuals. Byström (2001) extends their approach to the block maxima method. Bali (2002) and Venter and de Jongh (2002) follow similar lines.

¹See Artzner, Delbaen, Eber and Heath (2002) for the definition of the coherent risk measure S-VaR. See Danielsson (2002) for its practical relevance.

The goal of the Multivariate Extreme Value Theory (MEVT) is to construct a methodology to calculate risk measures, i.e. the VaR and the S-VaR, which directly come from the extremal dependence structure between portfolio components. This is achieved by considering their multivariate extreme value probability distributions. However, the extension of univariate to multivariate modelling is far from being obvious. One way of understanding MEVT is as the study of *copulas* (also referred to as dependence function) which arise in the limiting multivariate distribution of componentwise block maxima. Basically, the copula is a function that links univariate margins to the full multivariate distribution. Hence, the risk is splitted into two parts: the individual risks (i.e. the marginal distributions) and the dependence structure between them (i.e. the copula or dependence function).

There are only a few (non-financial) textbooks about dependence concepts, including Joe (1997), Nelsen (1999), Drouot Mari and Kotz (2001). However, there exists a range of overviews and applications of copulas in finance. Bouyé, Durrleman, Nikeghbali, Riboulet and Roncalli (2000) Embrechts, McNeil and Straumann (2002) and Embrechts, Lindskog and McNeil (2003) give general overviews with financial applications. Blum, Dias and Embrechts (2002) show how to handle the measurement of dependence in ART products. Patton (2002b) provides applications of copula theory in financial econometrics. Frey and McNeil (2002, 2003) explore the role of copulas in portfolio credit risk modelling.

In what follows we aim to give an overview of the various empirical applications of copulas in finance. Scaillet (2000b, 2002) and Capéraà, Fougères and Genest (1997) use non-parametric approaches to estimate the copula. The approach of Capéraà, Fougères and Genest (1997) is based on the empirical distribution function and compared to likelihood-based procedures. Their estimator is shown to be uniformly, strongly convergent and asymptotically unbiased. Scaillet (2000b) derives the asymptotic properties of kernel estimators of copulas and applies the method in Scaillet (2002) to european and US stock index returns.

Jondeau and Rockinger (2001a), Ang and Chen (2002), Patton (2001a, 2001b) and Dias and Embrechts (2003) propose conditional dependence models. Jondeau and Rockinger (2001a) use copulas to link marginal distributions, the latter chosen to obey a GARCH-type model with time-varying skewness and kurtosis. They find strong evidence of persistence in dependency for local and US\$ denominated currency return series. Moreover, they show that dependency changes through time and suggest extensions for conditional asset pricing models involving time variation of co-skewness and co-kurtosis. Recent work in empirical finance, see Ang and Chen (2002) and Patton (2002c), has

found evidence that non-linear dependence structures command a premium in the market, meaning that investors demand a higher return on assets which have a "bad" dependence structure.

Bouyé, Gaussel and Salmon (2002) develop copula based dynamic dependency measures and replace the autocorrelation with the so-called auto-concordance. They provide empirical applications using financial returns and transaction based forex data. However, their work relies on the critical assumption that the density of the marginal distributions does not change through time.

Breymann, Dias and Embrechts (2003) analyze bivariate high frequency forex data as a function of the sampling frequency, treating the data as *iid*. Dias and Embrechts (2003) use the same data set, however, they model the dynamics as GARCH-type processes and investigate the stylized facts of the bivariate residuals (i.e. they test for tail-dependence).

There are a few papers concerned with the application of copulas in financial risk measurement. Hürlimann (2002) constructs a parametric family of multivariate copulas and applies the model to the evaluation of the economic risk capital for a portfolio of risks using conditional VaR measures. Bouyé, Durrleman, Nikeghbali, Riboulet and Roncalli (2001) provide simple applications for the practical use of copulas in risk management. Bouyé (2002) investigates a methodology based on the MEVT parametric distributions in two directions: stress scenario and Monte-Carlo based risk measures, and find similar results for the three copulas examined. Embrechts, Hoeing and Juri (2001) develop analytical formula in a non-extremal context, by providing bounds for the risk measures.

In the present chapter, an unconditional simulation model which combines EVT and copula modelling, is suggested. First, we apply EVT to the univariate return series. Next, we transform the returns to uniformly distributed random variables and estimate the parameters for a given copula. We then generate two random variables with uniform marginals and joint cumulative distribution given by the estimated copula from the preceding step. Finally, we calculate the portfolio P/L distribution using the simulated data and estimate the VaR and the S-VaR, respectively. In order to test its quality, we backtest the model (both for VaR and S-VaR, applying a bootstrap test for the latter) and compare it with Historical simulation and Variance-covariance models. Rank (2000) uses a similar approach without applying EVT for the univariate series.

The chapter is structured as follows.² In Section 2, the general theory is presented. The Peak-over-threshold model (POT) is briefly revised, which is used in the empirical part. Furthermore, we introduce the theory of copulas, giving some examples and motivating the choice of some particular copulas. In Section 3, the model itself is presented. Section 4 presents the data and testing issues, while the empirical results are shown in Section 5. The final section gives some ideas for further research.

4.2 General Theory

4.2.1 The POT model again

The POT method, discussed in Section 1.2.2, offers an efficient use of the data set, considering all values exceeding a high pre-specified threshold.

In what follows we apply the fully parametric model based on the generalized Pareto distribution (GPD). Conditional on the event that the random variable X is larger than the threshold u , and denoting these exceedances by y (in other words the probability that a loss exceeds the threshold u by more than an amount y), we are interested in estimating the following conditional probability

$$\Pr(X > u + y \mid X > u) = \frac{1 - F(u + y)}{1 - F(u)}, \quad 0 < y. \quad (4.1)$$

If the parent distribution function F were known, the distribution of threshold exceedances would also be known. Since, in practical applications, this is not the case, approximations that are broadly applicable for high values of the threshold are sought. We know from Section 1.2.2 that the Generalized Pareto distribution (GPD) is the limiting distribution for the distribution of the excesses $(X - u)$. For $u \rightarrow \infty$, the distribution function of $(X - u)$, conditional on $X > u$, is approximately

$$G_{\xi, \sigma}(x) = \begin{cases} 1 - \left(1 + \frac{\xi}{\sigma}x\right)^{-1/\xi} & \text{if } \xi \neq 0 \\ 1 - \exp\left(-\frac{x}{\sigma}\right) & \text{if } \xi = 0 \end{cases} \quad (4.2)$$

²This chapter has been written while the author was studying at the London School of Economics (LSE) in London and will be presented at the CaRisMa workshop I / 2004 in Zurich. I thank Dr. Andrew Patton, Dr. Patrick Wegmann and Prof. Paul Embrechts for helpful comments.

The latter theorem shows that for a sufficiently large u the conditional excess distribution function is well approximated by a GPD. The choice of threshold (see Section 1.3.1) must be high enough for the limit theorem to be valid, but not too high in order to have efficient estimation (i.e. bias variance trade-off). The parameter ξ is the dominant parameter, known as shape parameter.

Having determined a threshold, the parameters of the GPD can be estimated. We use both maximum likelihood and L-moment estimation. Suppose that the values $y_1, \dots, y_k$ are the k excesses of a threshold u . For $\xi \neq 0$ the log likelihood can be derived as

$$\ell(\xi, \sigma) = -k \ln \sigma - \left(\frac{1}{\xi} + 1 \right) \sum_{i=1}^k \ln \left(1 + \frac{\xi}{\sigma} y_i \right), \quad (4.3)$$

provided that $(1 + \sigma^{-1} \xi y_i) > 0$ for $i = 1, \dots, k$; otherwise, $\ell(\xi, \sigma) = -\infty$. In practice, this method works well for the range $-1/2 < \xi < 1/2$, which is often the case. The asymptotic theory of maximum likelihood estimation for EVT is given in Smith (1985). Analytical maximization of the log-likelihood is not possible, so numerical techniques are required, taking care to avoid numerical instabilities when $\xi \approx 0$ in 4.3, and ensuring that the algorithm does not fail due to evaluation outside of the allowable parameter space.

4.2.2 Copula Theory

Introduction

The following subsection gives a short introduction to the copula theory. Suppose we are interested in modelling the stochastic behaviour of two random variables X and Y based on a set of n independent observations of the couple (X, Y) , for example $\{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\}$. Furthermore, suppose we have estimated their respective marginal distributions, $G_1(x)$ and $G_2(y)$, using standard statistical techniques. Hence, we are interested to know how we can test to know whether X and Y are independent, and moreover, how we can describe their dependence structure if they are not independent.

One way to describe the dependence structure between two random variables is to estimate and use their joint distribution function

$$F(x, y) = \Pr \{X \leq x, Y \leq y\}. \quad (4.4)$$

A test of independence between X and Y is simply to test whether $F(x, y)$ is a product of the respective marginal distributions. In order to construct

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a measure of the dependence structure we introduce two uniform distribution random variables $U = G_1(X)$ and $V = G_2(Y)$, i.e. transformations of X and Y , and their joint distribution

$$C(u, v) = \Pr \{G_1(X) \leq u, G_2(Y) \leq v\} = F(G_1^{-1}(u), G_2^{-1}(v)), \quad (4.5)$$

where G_1^{-1} and G_2^{-1} are the quantile functions of X and Y , respectively, i.e.

$$G_1^{-1}(q) = \inf \{x \in \mathbb{R} : G(x) > q\}. \quad (4.6)$$

If we can estimate the function C , we have F , i.e.

$$F(x, y) = C(G_1(x), G_2(y)). \quad (4.7)$$

C is a two-dimensional copula. In words, C is the (joint) cumulative distribution function of two random variables X and Y after transformation to the variables U and V , with uniform $[0, 1]$ marginals. See Nelsen (1999), Joe (1997) and Drouot Mari and Kotz (2001), the recent published textbooks on the subject, respectively. Note that if X and Y are continuous random variables, the function C satisfying the above definition is unique (if X and Y are not continuous random variables, C is uniquely determined on $\text{Range}G_1 \times \text{Range}G_2$, a result known as Sklar's theorem (Sklar (1959), see also Nelsen (1999), p.15).

In order to estimate F , we transform the observations of X and Y , $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, into observations of $U = G_1(X)$ and $V = G_2(Y)$, $(u_1, v_1), (u_2, v_2), \dots, (u_n, v_n)$, where $u_i = G_1(x_i)$ and $v_i = G_2(y_i)$, $i = 1, 2, \dots, n$, and C may then be estimated as the joint distribution of U and V , respectively. Thus, we have an estimate of F . There are a few non-parametric methods developed to estimate the copula (see Deheuvels (1978), Genest and Rivest (1993), Genest, Ghoudi and Rivest (1995), or Capéreau, Fourgères and Genest (1997)). Alternatively, we can also assume that the copula has a particular parametric form and estimate its parameters using, for example, the method of maximum likelihood.

As for the conditions for C to be a copula, C is a two-dimensional copula if and only if it is a function

$$C(u, v) : \mathbf{I}^2 \mapsto \mathbf{I} \quad (\mathbf{I} = [0, 1]), \quad (4.8)$$

that satisfies the following two conditions:

1. For every u and v in $\mathbf{I}$:

$$C(u, 0) = C(0, v) = 0, \quad C(u, 1) = u, \quad C(1, v) = v; \quad (4.9)$$

2. For every $u_1 \leq u_2$, $v_1 \leq v_2$, and $u_1, u_2, v_1, v_2 \in \mathbf{I}$, the following inequality holds:

$$C(u_1, v_1) - C(u_2, v_1) - C(u_1, v_2) + C(u_2, v_2) \geq 0. \quad (4.10)$$

If C is considered to be a distribution function of two random variables U and V , the first condition ensures that U and V have uniform marginals. The second condition, often referred to as the rectangular inequality, simply requires that C is a valid distribution function, i.e. $\Pr\{u_1 \leq U \leq u_2, v_1 \leq V \leq v_2\} \geq 0$.

Parametric Copula Classes and Empirical Copulas

In the following subsection, the most important copula classes are presented, starting with the normal copula.

1. Normal copula

One of the most frequently used copulas (especially for financial modelling) is the copula of a bivariate Gaussian or normal distribution with correlation δ . The copula is defined by

$$\begin{aligned} C(u, v) &= \int_{-\infty}^{\Phi^{-1}(u)} dx \int_{-\infty}^{\Phi^{-1}(v)} dy \frac{1}{2\pi\sqrt{1-\delta^2}} \exp\left\{-\frac{x^2 - 2\delta xy + y^2}{2(1-\delta^2)}\right\} \\ &= \Phi_\delta(\Phi^{-1}(u), \Phi^{-1}(v)), \end{aligned}$$

where Φ^{-1} is the quantile of the standard univariate Gaussian distribution, and Φ_δ is the joint cumulative distribution function of a standard bivariate Gaussian distribution with correlation coefficient δ ($0 < \delta < 1$). Since this copula is a very familiar object to most researchers (especially when used with Gaussian marginals, i.e. the bivariate normal distribution), it has been incorporated in a number of applications simply because it was the only tool available for quite some time (Embrechts, McNeil and Straumann (2000, 2002)). For example, J.P.Morgan's Risk Metrics (see Risk Metrics (1995)) has been using this copula for portfolio risk management by Monte Carlo simulations long before it was related that one were dealing with copulas.

2. The Extreme Value copula class

An important class of copulas is the Extreme Value class (EV copula). A copula is said to be an EV copula if, for all $t > 0$, the scaling property

$$C(u^t, v^t) = (C(u, v))^t, \quad (4.12)$$

holds for all $(u, v) \in \mathbf{I}^2$. EV copulas are max-stable, meaning that, if $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ are iid random pairs from EV copula C and if $M_n = \max\{X_1, X_2, \dots, X_n\}$, $N_n = \max\{Y_1, Y_2, \dots, Y_n\}$, a copula associated with the random pair (M_n, N_n) is also C . It can be shown (see Joe (1997), p. 175) that EV copulas can be represented in the following general form:

$$C(u, v) = \exp\left\{\log(uv) A\left(\frac{\log(u)}{\log(uv)}\right)\right\}, \quad (4.13)$$

where $A: [0, 1] \rightarrow [1/2, 1]$ is a convex function such that $\max(t, 1-t) < A(t) < 1$, for all $t \in [0, 1]$. The function $A(t)$ is called the dependence function. As in the case of univariate random variables, it can be shown that the limiting copula for the sequence $\{(a_n + b_n M_n, c_n + d_n N_n)\}$ is an EV copula, if the sequence converges weakly in distribution for some sequence of numbers a_n, b_n, c_n and d_n , under certain regularity conditions (see Galambos (1987)).

An important example of an EV copula is the well-known Gumbel copula, after Gumbel (1960). It belongs to the EV copula class as well as to the Archimedean copula class (introduced below), and has the following form:

$$C(u, v) = \exp\left\{-[(-\log u)^\delta + (-\log v)^\delta]^{1/\delta}\right\}, \quad (4.14)$$

with $\delta \geq 1$, and the dependence function of the following form:

$$A(t) = (t^\delta + (1-t)^\delta)^{1/\delta}. \quad (4.15)$$

Other copula families belonging to the EV copula class are the Galambos copula (Galambos (1975)), the Hüsler and Reiss copula (Hüsler and Reiss (1989)), the Tawn copula (Tawn (1988)) and the BB5 copula (Joe (1997)).

3. The Archimedean copula class

Another general class of copulas is the Archimedean copula. A copula is called an Archimedean copula, if its distribution function can be written in the following general form:

$$C(u, v) = \phi^{-1} [\phi(u) + \phi(v)], \quad (4.16)$$

with some function $\phi(t) : \mathbf{I} \rightarrow \mathbb{R}^+$, which is continuous, strictly decreasing, convex, and satisfying $\phi(1) = 0$. This function ϕ is called the Archimedean generator (see Nelsen (1999), Chapter 4). $\phi(0)$ is defined as $\lim_{t \rightarrow +0} \phi(t)$, and $\phi^{-1}(z) = 0$ for all $z > \phi(0)$, if $\phi(0) < \infty$. Notice that the Gumbel copula is an EV copula as well as an Archimedean copula because it can be written in the above standard form of an Archimedean copula with a generator function of $\phi(t) = (-\log t)^\delta$. An important example of an Archimedean copula is the Kimeldorf and Sampson copula³ (Kimeldorf and Sampson (1997)), with the following form:

$$C(u, v) = (u^{-\delta} + v^{-\delta} - 1)^{-1/\delta}, \quad (4.17)$$

where $0 < \delta < \infty$, and its generator function is

$$\phi(t) = t^{-\delta} - 1. \quad (4.18)$$

Other copula families belonging to the Archimedean copula class are the Frank copula (Frank (1979)), the Joe copula (Joe (1993)), the BB1 copula (Joe (1997)) and the BB2 copula (Joe (1997)).

4. The Archimax copula class

Capéraà, Fougères and Genest (2000) combined the EV and the Archimedean copula classes into a single class called Archimax copulas. The Archimax copulas are copulas which can be represented in the following form:

$$C(u, v) = \phi^{-1} \left[(\phi(u) + \phi(v)) A \left(\frac{\phi(u)}{\phi(u) + \phi(v)} \right) \right], \quad (4.19)$$

where $A(t)$ is a valid dependence function and ϕ a valid Archimedean generator. Archimax copulas reduce to Archimedean copulas for $A(t) = 1$ and to EV copulas for $\phi(t) = -\log(t)$. Capéraà, Fougères and Genest (2000) proved that it is a valid copula for any combination of valid function $\phi(t)$ and $A(t)$, respectively. The BB4 copula (Joe (1997)) is an example of this class of copula with

$$\phi(t) = t^{-\theta} - 1 \text{ and } A(t) = 1 - (t^{-\delta} + (1-t)^{-\delta})^{-1/\delta}. \quad (4.20)$$

³This copula is also called Clayton copula, see for example in Nelsen (1999). From Figure 4.1 we clearly see that this copula takes negative (extremal) dependence into account.

The respective distribution function is given by

$$C(u, v) = \left(x^{-\theta} + u^{-\theta} - 1 - \left[(u^{-\theta} - 1)^{-\delta} + (v^{-\theta} - 1)^{-\delta} \right]^{-1/\delta} \right)^{-1/\theta}, \quad (4.21)$$

where $\theta \geq 0, \delta > 0$.

5. Empirical copulas

If $u_{(1)} \leq u_{(2)} \leq \dots \leq u_{(n)}$ and $v_{(1)} \leq v_{(2)} \leq \dots \leq v_{(n)}$ are the order statistics of the univariate samples, the empirical copula $\hat{C}_{emp}$ is defined, at the point $(i/n, j/n)$, by the following formula:⁴

$$\hat{C}_{emp} \left(\frac{i}{n}, \frac{j}{n} \right) = \frac{1}{n} \sum_{k=1}^n 1_{\{u_k \leq u_{(i)}, v_k \leq v_{(j)}\}}, \quad i, j = 1, 2, \dots, n. \quad (4.22)$$

Figure 4.1 (and Figure D.3 and D.4 in the appendix, respectively) show copula densities of various distributions, all with standard normal marginal distributions. We note the different extremal dependence patterns for the different copulas. As outlined in the introduction, it is a stylized fact that estimated correlations are much lower when the markets are increasing in value, compared to market conditions when some assets increase and others decrease, respectively. This is especially true when the markets are falling. Consequently, we expect the dependence structure to be captured better using the Kimeldorf Sampson or the BB3 copula, as opposed to the gumbel or the normal copula, respectively. This is also seen in Figure D.5, D.6 and D.7 in the appendix which show the densities of bivariate distributions with normal marginals and different copulas. Moreover, we plot 3D surface (perspective) plots of the densities of different copulas and contour plots of the distribution function of the Normal and the Kimeldorf Sampson copula, see Figures D.8, D.9, D.10, D.11 and D.12 respectively.

4.3 Calculation of Portfolio VaR using EVT and Copulas

The aim of the following section is to present a simulation-based model to calculate the Value-at-Risk (VaR) and the Expected Shortfall (S-VaR) of a two-asset

⁴For a more detailed discussion of the empirical copula, see Nelsen (1999).

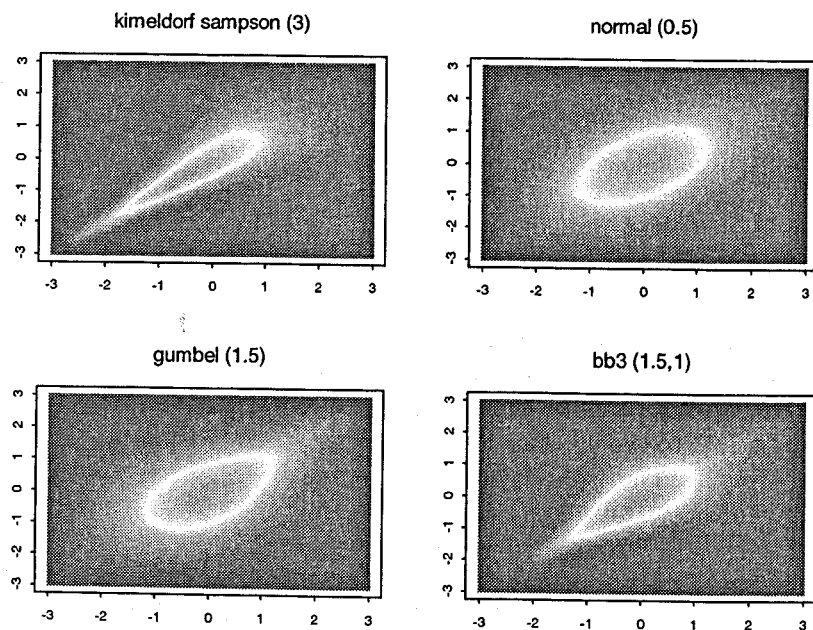


Figure 4.1: Copula densities of various distributions, all with standard normal marginal distributions (parameter values in brackets). Note the different extremal dependencies between the different copulas.

portfolio using copulas. We propose an unconditional model combining EVT and copulas. The general procedure is as follows. First, we apply EVT to the univariate returns series. Next, we transform the returns to uniformly distributed random variables and estimate the parameters for a given copula. We then generate two random variables with uniform marginals and joint cumulative distribution given by the estimated copula from the preceding step. Finally, we calculate the portfolio P/L distribution using the simulated data and estimate the VaR and the S-VaR, respectively. In order to test its quality, we backtest the model (both for VaR and S-VaR, applying a bootstrap test for the latter) and compare it with Historical simulation and Variance-covariance models. The following single steps are implemented:⁵

1. Application of univariate Extreme Value Theory

In a first step, we apply univariate EVT to estimate the extreme value parameters for the upper and for the lower tails, respectively. More precisely, we fit a generalized Pareto (GPD) distribution to excesses over a certain threshold, as described in Section 4.2.1⁶. We calculate L-moment parameter estimates for the GPD distribution:⁷ The L-moments method for parameter estimation consists of equating estimates of sample L-moments to the GPD distribution's L-moment. Alternatively, we use the method of maximum likelihood (ML), as described in Section 4.2.1, which does the same job.⁸ Because it is not feasible to examine the fitted model carefully every day and to choose a new threshold for the tail estimator (as described in Section 1.3.1) each time, we set the threshold (fix) at a level where we use 10% of the data for the tail estimation.

2. Calculation of the cumulative probability of the fitted GPD object from step 1

⁵All the calculations are performed in the statistical software package S-Plus 6.1 for Windows. Furthermore, for some calculations, the S-Plus financial toolbox S-Plus finmetrics is used. The functions to implement the GPD (POT) model were written by Prof. Alexander McNeil, contained in the EVIS library which is available on the internet. The functions to estimate and simulate copulas were written by Prof. René Carmona.

⁶For a more detailed description of the procedure used to fit a GPD distribution to excesses over a certain threshold (e.g. the Peak-over-threshold approach (POT)), we refer to McNeil (1999a). Figure D.1 in the appendix shows an example.

⁷See Hosking and Wallis (1987) for a detailed description of moment estimators for the generalized Pareto distribution. Moreover, Zivot and Wang (2002) describe how to use the function in S-Plus.

⁸See Prescott and Walden (1980) for a description of ML estimators for the generalized extreme-value distribution.

In the second step, we calculate the cumulative probability of the fitted GPD object from step 1, using semi-parametric estimation methods. The result is a vector (of the same length as the sample data set) containing the estimates of the cumulative distribution function from which the sample data was issued. This step, essentially transforming the returns X to uniformly distributed random variables U (as described in 4.2.2), is also referred to as quantile transformation, see Embrechts, McNeil and Straumann (2000, 2002).⁹

3. Estimation of the empirical copula

In this step, we require the input data points to have a uniform (0,1) marginal distribution. This is given from the last step. We calculate the empirical copula from which the sample data was issued.

4. Maximum likelihood parameter estimation for a chosen copula

Next, we compute ML parameter estimates for a chosen parametric copula family¹⁰ (i.e. the Normal and the Kimeldorf Sampson copula, respectively). As an input, we use the estimated empirical copula from the last step, which contains the historical data points. Note that steps 1 to 4 essentially serve to estimate the copula parameters which are used in the following steps to simulate random returns (using the information about the dependence structure given by the copula) and to calculate VaR and S-VaR, respectively.

5. Random Generation from the estimated copula

Given the estimated parameters of the chosen copula from the 4th step, we generate a random sample.¹¹ More precisely, we generate two random variables (i.e. u, v) with uniform marginals and joint cumulative distribution given by the estimated copula from the last step.

6. Calculation of the quantile functions of the generated random variables u, v

Next, we take the parameter estimates from step 1 (i.e. from the fitted GPD distribution) and the generated random variables from step 5 and

⁹For a detailed description see Prescott and Walden (1980).

¹⁰See Rank (2000) or Joe (1997) for a more detailed description of the Maximum likelihood estimation.

¹¹For the empirical analysis, we use either 10'000 and 100'000 simulation runs, respectively, and find no significant differences in the results. See Joe (1997) for a detailed description for the random generation from copulas.

calculate an estimate of the quantile function of the fitted GPD object. In step 2, we transformed the given returns to uniform marginals, in order to estimate the copula parameters. Here we essentially go the other way round and calculate the returns, given the simulated copula, (i.e. two uniformly distributed random variables). The returned values represent simulated returns x and y , respectively.¹²

7. Calculation of portfolio returns

In order to get the continuously compounded portfolio returns we use the following formula

$$r_{pf} = \log(\lambda_1 \exp(x) + \lambda_2 \exp(y)), \quad (4.22)$$

where λ_i are the portfolio weights, and x and y are the asset 1 and asset 2 returns, respectively.

8. Taking the q -th quantile in order to estimate the VaR and the S-VaR, respectively

In order to get the VaR, we calculate the q -quantile of the portfolio return distribution. Finally, we calculate the Expected shortfall, taking the mean of all the cases where VaR is exceeded.

Note that the crucial point in the presented model is the choice of the copula (step 4), which is used to generate random numbers with the respective joint distribution.

4.4 Data, Estimation and Testing

The data set consists of 15 titles of the Swiss Market Index, dating from 22.08.1987 to 19.06.2002, a total of 3913 datapoints for each title.¹³ In order to get ten (arbitrary) portfolios, we chose five highly correlated asset-pairs (highest correlation 0.63) and five low-correlated asset pairs (lowest correlation 0.21), respectively.¹⁴ We backtest the proposed model on these ten portfolios.

¹²For a detailed description see Prescott and Walden (1980).

¹³Source: all the data is extracted from Datastream.

¹⁴We include the following portfolios: Adecco and Baloise, Unaxis and Nestlé, Unaxis and Novartis, Unaxis and Roche, Unaxis and Sulzer, Credit Suisse and UBS, Zurich, Nestlé and Novartis, Swiss Re and Zurich and UBS and Zurich. Figure D.2 shows the scatterplots of the portfolios.

In order to backtest the model on the historical series, we calculate the VaR $\hat{x}_q^t$ on each day t using a time window of 1000 days each time. Consequently, we use somewhat less than the last four years of data for each prediction. Hence, a total of 2913 VaR forecasts are calculated for each portfolio. Regarding the choice of the threshold in the GPD estimation, it is not feasible to examine the fitted model carefully every day and to choose a new threshold for the tail estimator each time; for this reason we fix the threshold at the 90th percentile,¹⁵ estimating the higher percentiles using the GPD tail estimator (POT model). On each day, we go through the eight steps indicated in the last section, determining a new VaR forecast. We compare the VaR forecast on each day with the realization on the next day. Whenever the latter exceeds the former, we count a violation of the VaR. Hence, we get a number of violations using our model and compare it with the theoretical expected number of violations.

In order to test for the success (or robustness) of the VaR estimation models, we apply a binomial test.¹⁶ Under the null hypothesis that the method correctly estimates the conditional quantiles, the following empirical statistic is binomially distributed

$$\sum_{t \in T} 1_{\{x_{t+1} > \hat{x}_q^t\}}, \quad (4.23)$$

where x_{t+1} is the realization of the VaR on day $t+1$. We perform a two-sided binomial test of the null hypothesis against the alternative that the model has a systematic estimation error and gives either too few or too many violations. A p-value less than 0.05 (0.01) will be interpreted as evidence (significance) against the null hypothesis.

To backtest the S-VaR, we are interested in the size of the discrepancy between the realization x_{t+1} and the shortfall estimation $\hat{S}_q^t$. We calculate the following exceedance residuals on days when a VaR violation (i.e. $x_{t+1} > \hat{x}_q^t$) occurs

$$\{r_{t+1} : t \in T, x_{t+1} > \hat{x}_q^t\}, \quad \text{with } r_{t+1} = \frac{x_{t+1} - \hat{S}_q^t}{\hat{\sigma}_{t+1}}, \quad (4.24)$$

¹⁵This decision is justified by a simulation study done by Frey and McNeil (2000) for a similar data set. As mentioned earlier, Danielsson and De Vries (2000) propose a double subsample bootstrap procedure and assess its accuracy with Monte Carlo experiments.

¹⁶See Frey and McNeil (2000) for more details about this test. Alternatively, we could calculate violation ratios (realized number of violations over expected number of violations) as done i.e. in Danielsson (2002).

	Historical simulation							
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Adec-Balois	158(146)	<i>0.31</i>	44(29)	<i>0.01</i>	20(15)	<i>0.19</i>	07(03)	<i>0.03</i>
Unax-Nestle	149(146)	<i>0.80</i>	28(29)	<i>0.93</i>	20(15)	<i>0.19</i>	06(03)	<i>0.13</i>
Unax-Novn	152(146)	<i>0.61</i>	33(29)	<i>0.51</i>	18(15)	<i>0.43</i>	05(03)	<i>0.22</i>
Unax-Roche	145(146)	<i>1.00</i>	39(29)	<i>0.08</i>	21(15)	<i>0.11</i>	04(03)	<i>0.55</i>
Unax-Sulzer	167(146)	<i>0.07</i>	33(29)	<i>0.51</i>	20(15)	<i>0.19</i>	12(03)	0.00
CS-Ubs	169(146)	<i>0.06</i>	43(29)	<i>0.01</i>	25(15)	<i>0.01</i>	09(03)	0.00
CS-Zurich	172(146)	<i>0.03</i>	41(29)	<i>0.04</i>	25(15)	<i>0.01</i>	11(03)	0.00
Nestle-Novn	163(146)	<i>0.15</i>	40(29)	<i>0.06</i>	22(15)	<i>0.06</i>	08(03)	<i>0.01</i>
SwRe-Zurich	157(146)	<i>0.35</i>	39(29)	<i>0.08</i>	24(15)	<i>0.02</i>	11(03)	0.00
Ubs-Zurich	175(146)	<i>0.02</i>	41(29)	<i>0.04</i>	23(15)	<i>0.05</i>	11(03)	0.00
Perc. Signif.	0.80		0.60		0.70		0.30	
	1.00		1.00		1.00		0.20	

Table 4.1: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the historical simulation model. Insignificant p-values for a binomial test in italics.

	VarCov							
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Adec-Balois	129(146)	<i>0.16</i>	64(29)	0.00	44(15)	0.00	26(03)	0.00
Unax-Nestle	118(146)	<i>0.02</i>	39(29)	<i>0.08</i>	28(15)	0.00	16(03)	0.00
Unax-Novn	113(146)	0.00	44(29)	0.00	34(15)	0.00	18(03)	0.00
Unax-Roche	120(146)	<i>0.03</i>	49(29)	0.00	36(15)	0.00	19(03)	0.00
Unax-Sulzer	141(146)	<i>0.70</i>	50(29)	0.00	39(15)	0.00	23(03)	0.00
CS-Ubs	145(146)	<i>1.00</i>	60(29)	0.00	47(15)	0.00	34(03)	0.00
CS-Zurich	143(146)	<i>0.83</i>	65(29)	0.00	49(15)	0.00	33(03)	0.00
Nestle-Novn	134(146)	<i>0.33</i>	54(29)	0.00	35(15)	0.00	23(03)	0.00
SwRe-Zurich	116(146)	<i>0.01</i>	51(29)	0.00	38(15)	0.00	21(03)	0.00
Ubs-Zurich	136(146)	<i>0.42</i>	59(29)	0.00	48(15)	0.00	34(03)	0.00
Perc. Signif.	0.60		0.10		0.00		0.00	
	0.90		0.10		0.00		0.00	

Table 4.2: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the bivariate Variance-covariance model. Insignificant p-values for a binomial test in italics.

BivGpd-normal copula								
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Adec-Balois	154(146)	<i>0.50</i>	41(29)	<i>0.04</i>	31(15)	0.00	10(03)	0.00
Unax-Nestle	141(146)	<i>0.70</i>	30(29)	<i>0.85</i>	23(15)	<i>0.05</i>	07(03)	<i>0.03</i>
Unax-Novn	141(146)	<i>0.70</i>	34(29)	<i>0.40</i>	22(15)	<i>0.06</i>	08(03)	<i>0.01</i>
Unax-Roche	146(146)	<i>0.97</i>	40(29)	<i>0.06</i>	23(15)	<i>0.05</i>	06(03)	<i>0.13</i>
Unax-Sulzer	161(146)	<i>0.20</i>	38(29)	<i>0.11</i>	23(15)	<i>0.05</i>	11(03)	0.00
CS-Ubs	172(146)	<i>0.03</i>	44(29)	<i>0.01</i>	28(15)	0.00	12(03)	0.00
CS-Zurich	168(146)	<i>0.07</i>	49(29)	0.00	30(15)	0.00	16(03)	0.00
Nestle-Novn	165(146)	<i>0.11</i>	39(29)	<i>0.08</i>	27(15)	0.00	05(03)	<i>0.22</i>
SwRe-Zurich	154(146)	<i>0.50</i>	38(29)	<i>0.11</i>	28(15)	0.00	14(03)	0.00
Ubs-Zurich	162(146)	<i>0.17</i>	44(29)	<i>0.01</i>	29(15)	0.00	14(03)	0.00
Perc. Signif.	0.90		0.60		0.40		0.20	
	1.00		0.90		0.40		0.40	

Table 4.3: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the normal copula. Insignificant p-values for a binomial test in italics.

BivGpd-kimeldorf.sampson copula								
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Adec-Balois	152(146)	<i>0.61</i>	36(29)	<i>0.22</i>	20(15)	<i>0.19</i>	05(03)	<i>0.22</i>
Unax-Nestle	139(146)	<i>0.58</i>	25(29)	<i>0.46</i>	16(15)	<i>0.69</i>	05(03)	<i>0.22</i>
Unax-Novn	134(146)	<i>0.33</i>	26(29)	<i>0.58</i>	18(15)	<i>0.43</i>	06(03)	<i>0.13</i>
Unax-Roche	150(146)	<i>0.64</i>	33(29)	<i>0.51</i>	15(15)	<i>0.89</i>	03(03)	<i>1.00</i>
Unax-Sulzer	156(146)	<i>0.40</i>	32(29)	<i>0.64</i>	19(15)	<i>0.29</i>	08(03)	<i>0.01</i>
CS-Ubs	162(146)	<i>0.17</i>	36(29)	<i>0.22</i>	22(15)	<i>0.06</i>	07(03)	<i>0.03</i>
CS-Zurich	159(146)	<i>0.27</i>	38(29)	<i>0.11</i>	23(15)	<i>0.05</i>	08(03)	<i>0.01</i>
Nestle-Novn	156(146)	<i>0.40</i>	33(29)	<i>0.51</i>	20(15)	<i>0.19</i>	02(03)	<i>0.78</i>
SwRe-Zurich	145(146)	<i>1.00</i>	33(29)	<i>0.51</i>	20(15)	<i>0.19</i>	09(03)	0.00
Ubs-Zurich	154(146)	<i>0.50</i>	39(29)	<i>0.08</i>	23(15)	<i>0.05</i>	06(03)	<i>0.13</i>
Perc. Signif.	1.00		1.00		1.00		0.60	
	1.00		1.00		1.00		0.90	

Table 4.4: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the kimeldorf sampson copula. Insignificant p-values for a binomial test in italics.

BivGpd-bb3 copula								
	95%	<i>p</i>	99%	<i>p</i>	99.5%	<i>p</i>	99.9%	<i>p</i>
Adec-Balois	141(146)	<i>0.70</i>	35(29)	<i>0.30</i>	21(15)	<i>0.11</i>	06(03)	<i>0.13</i>
Unax-Nestle	129(146)	<i>0.16</i>	23(29)	<i>0.27</i>	16(15)	<i>0.69</i>	04(03)	<i>0.55</i>
Unax-Novn	129(146)	<i>0.16</i>	27(29)	<i>0.78</i>	15(15)	<i>0.89</i>	05(03)	<i>0.22</i>
Unax-Roche	132(146)	<i>0.25</i>	29(29)	<i>1.00</i>	15(15)	<i>0.89</i>	04(03)	<i>0.55</i>
Unax-Sulzer	146(146)	<i>0.97</i>	30(29)	<i>0.85</i>	18(15)	<i>0.43</i>	06(03)	<i>0.13</i>
CS-Ubs	165(146)	<i>0.11</i>	35(29)	<i>0.30</i>	22(15)	<i>0.06</i>	07(03)	<i>0.03</i>
CS-Zurich	160(146)	<i>0.23</i>	37(29)	<i>0.16</i>	22(15)	<i>0.06</i>	10(03)	0.00
Nestle-Novn	156(146)	<i>0.40</i>	32(29)	<i>0.64</i>	17(15)	<i>0.51</i>	01(03)	<i>0.38</i>
SwRe-Zurich	145(146)	<i>1.00</i>	31(29)	<i>0.71</i>	17(15)	<i>0.51</i>	09(03)	0.00
Ubs-Zurich	155(146)	<i>0.44</i>	37(29)	<i>0.16</i>	20(15)	<i>0.17</i>	07(03)	<i>0.03</i>
Perc. Signif.	1.00		1.00		1.00		0.60	
	1.00		1.00		1.00		0.80	

Table 4.5: Backtesting results: Theoretically expected number of violations (in brackets) and number of violations using the bb3 copula. Insignificant p-values for a binomial test in italics.

	95%		99%		99.5%		99.9%	
VarCov / Hist. sim.	0.50	0.50	0.00	<i>1.00</i>	0.00	<i>1.00</i>	0.00	<i>1.00</i>
VarCov / BivGpd	0.30	<i>0.70</i>	0.00	<i>1.00</i>	0.00	<i>1.00</i>	0.00	<i>1.00</i>
BivGpd / Hist. sim.	0.50	0.50	<i>0.60</i>	0.40	<i>1.00</i>	0.00	<i>1.00</i>	0.00
BivGpd(norm) / BivGpd	0.30	<i>0.70</i>	0.10	<i>0.90</i>	0.00	<i>1.00</i>	0.00	<i>1.00</i>
BivGpd(norm) / VarCov	0.50	0.50	<i>1.00</i>	0.00	<i>1.00</i>	0.00	<i>1.00</i>	0.00
BivGpd / Gpd	<i>0.70</i>	0.30	<i>0.90</i>	0.10	<i>0.90</i>	0.10	<i>0.80</i>	0.20

Table 4.6: Comparison of different models: bivariate Variance-covariance (var-cov), Historical simulation, bivariate Gpd based using the kimeldorf.sampson copula (bivgpd), bivariate Gpd based with a normal copula (bivgpd(norm)) and univariate Gpd based (Gpd). Note that the table shows the percentage of cases in which the respective model performs better.

	VarCov			Kimeldorf.samp copula			Normal copula		
	95%	99%	99.5%	95%	99%	99.5%	95%	99%	99.5%
Adec-Balois	0.00	0.00	0.00	<i>0.03</i>	<i>0.12</i>	<i>0.13</i>	0.00	0.00	<i>0.14</i>
Unax-Nestle	0.00	0.00	0.00	<i>0.70</i>	<i>0.14</i>	<i>0.57</i>	<i>0.14</i>	0.00	<i>0.20</i>
Unax-Novn	0.00	0.00	0.00	<i>0.36</i>	<i>0.10</i>	<i>0.43</i>	<i>0.03</i>	<i>0.03</i>	<i>0.11</i>
Unax-Roche	0.00	0.00	0.00	<i>0.32</i>	<i>0.62</i>	<i>0.23</i>	<i>0.03</i>	<i>0.03</i>	<i>0.13</i>
Unax-Sulzer	0.00	0.00	0.00	<i>0.26</i>	<i>0.06</i>	<i>0.10</i>	<i>0.03</i>	<i>0.01</i>	<i>0.01</i>
CS-Ubs	0.00	0.00	0.00	<i>0.21</i>	<i>0.17</i>	<i>0.17</i>	0.00	<i>0.01</i>	<i>0.01</i>
CS-Zurich	0.00	0.00	0.00	<i>0.03</i>	<i>0.07</i>	<i>0.32</i>	0.00	0.00	0.00
Nestle-Novn	0.00	0.00	0.00	<i>0.52</i>	<i>0.77</i>	<i>0.99</i>	<i>0.01</i>	<i>0.05</i>	<i>0.70</i>
SwRe-Zurich	0.00	0.00	0.00	<i>0.09</i>	<i>0.01</i>	<i>0.01</i>	0.00	0.00	0.00
Ubs-Zurich	0.00	0.00	0.00	<i>0.05</i>	<i>0.24</i>	<i>0.20</i>	0.00	0.00	<i>0.03</i>

Table 4.7: p-values for a one-sided bootstrap test of the hypothesis that the exceedance residuals have mean zero against the alternative that the mean is greater than zero. Insignificant p-values are shown in italics.

where r_{t+1} stands for the residual and $\hat{\sigma}_{t+1}$ is the volatility forecast on day t for day $t+1$. Under the null hypothesis that we correctly estimate the VaR and the conditional VaR (S-VaR), these residuals should behave like an *iid* sample with mean zero. A first look at Figure 4.5 shows that for the residuals calculated under the assumption of normality seems doubtful. To test the hypothesis of mean zero we use a bootstrap test that makes no assumption about the underlying distribution of the residuals.¹⁷ We conduct a one-sided test against the alternative hypothesis that the residuals have mean greater than zero, or equivalently, that the S-VaR is systematically underestimated, since this is the likely direction of failure. Again, p-values less than 0.01 present some evidence against the null hypothesis of a "correct" model.

4.5 Empirical Results

4.5.1 Results for the VaR

In the following section, we present the empirical results. Tables 4.1 to 4.6 show the results for the VaR. From the described copulas, we chose the Normal copula (Table 4.3), the Kimeldorf Sampson copula (Table 4.4) and the BB3 copula (Table 4.5), respectively. The choice of the kimeldorf.sampson copula and the BB3 copula has been made due to the fact that these copulas take into account

¹⁷For further details again refer to Efron and Tibshirani (1993). Furthermore, Shao, Tu and Olkin (1996) show various applications both for the independent and identically distributed (*iid*) case and more complicated cases with non-*iid* data sets.

negative extremal dependence. In addition, we apply an unconditional bivariate Variance-covariance model (Table 4.2) as well as the Historical simulation for comparison (Table 4.1).¹⁸ Note that the model using the normal copula is expected to give somewhat better results than the bivariate Variance-covariance model, because the former uses EVT for the quantile estimation, as described in Section 3. The first columns show the chosen portfolios. The first five portfolios are the low correlation portfolios (with a correlation of about 0.20), the second five portfolios are the high correlation portfolios (with a correlation of about 0.55). Columns 2, 4, 6 and 8 show the number of VaR violations (expected number of violations in brackets) for different quantiles, columns 3, 5, 7 and 9 show the p-values for the binomial test, respectively.¹⁹ For example, a p-value of 0.05 (0.01) or less indicates that the null hypothesis of a correct model is rejected, on a 95% (99%) confidence level. Finally, the last two rows show the percentage of cases where the null hypothesis cannot be rejected (upper row for the 0.05 level, lower row for the 0.01 level). Clearly, the bivariate Variance-covariance model underestimates the portfolio risk on quantiles higher than 95% (note that the null hypothesis is rejected in all but one case for quantiles higher than 95%). For higher quantiles, the simulation based model presented in the last section gives reasonable results, depending on the choice of the copula.²⁰ The improvement with the Kimeldorf Sampson copula as well as with the BB3 copula (which take into account negative dependence, see Figure 4.1) over the Normal copula is apparent. The null hypothesis is rejected only once (on a 1% significance level) for the Kimeldorf Sampson copula. Furthermore, calculations including other copulas (e.g. the gumbel copula with a positive dependence) turned out to give much poorer results than the two copulas shown here, implying that a negative dependence structure is indeed appropriate for the given portfolios and the given sample. The choice of the dependence structure seems crucial in financial risk measurement. For the Kimeldorf Sampson and BB3 copula, loosely speaking, we accept the null hypothesis of a correct model in at least 80% (60%) of the cases, over all quantiles, on a 1% (5%) significance level (see last two rows which show the percentage of significant results).

Table 4.6 shows a direct comparison between the performance of two models.

¹⁸Note that with the bivariate Variance-covariance model we are assuming both normal marginals and a normal copula. For the Historical simulation, we calculate the portfolio returns, sort them in ascending order and read off the VaR as the respective quantiles.

¹⁹Note that numbers in italics show p-values more than 0.01 and numbers in bold italics show p-values more than 0.05.

²⁰Note that already the normal copula model performs better than the bivariate Variance-covariance model. This is due to the fact that we apply EVT in the former model only.

Note that for the given portfolios and the given sample, the proposed model gives better performance than the Historical simulation for quantiles higher than 95%. Furthermore, the bivariate Variance-covariance and the Normal copula models are outperformed for every quantile. Also, the Normal copula gives better results than the bivariate Variance-covariance model. Finally, the univariate GPD based model performs worse than the bivariate counterpart.²¹

Figures 4.2 and 4.3 show Backtesting results for different quantiles, for the example of the Nestlé-Novartis portfolio. Note that the left figures show log returns (with losses on the positive axes) and VaR forecasts (superimposed), respectively. The right figures show differences between realized returns and forecasted VaRs (i.e. the magnitude of under- and overestimation of the risk).

4.5.2 Results for the S-VaR

Table 4.7 gives the results for the (coherent) S-VaR,²² showing the p-values for the bootstrap test.²³ Clearly, the bivariate Variance-covariance model is not accurate, even for the 95% quantile.²⁴ The residuals derived under the assumption of normality always fail the test with p-values in all cases much less than 0.01. Moreover, as in the VaR case, we note a significant improvement of the Kimeldorf Sampson copula compared to the Normal copula approach, although the results have to be interpreted with care, since we have only a few datapoints for the bootstrap simulation.²⁵ The results of the BB3 copula are not shown here, as they are very similar to the Kimeldorf Sampson model, differences (possibly) occurring due to the simulation in the bootstrap test. Figure 4.4 shows the exceedance residuals for the Nestlé-Novartis portfolio (upper figure: 95%, lower figure: 99%), assuming the Kimeldorf Sampson copula. Figure 4.5 shows the respective residuals using the bivariate Variance-covariance model. Under the null hypothesis that the model correctly estimates the tail, these residuals should have mean zero. Clearly, Figure 4.5 shows evidence against the normal

²¹ We did not address the question of how *volatile* the risk measure VaR is. We expect the forecasts to be less volatile than for a conditional model, however.

²² See Artzner, Delbaen, Eber and Heath (2002) for the definition and the relevance of coherent risk measures.

²³ Again numbers in italics show p-values less than 0.01 and numbers in bold italics show p-values less than 0.05.

²⁴ This is in line with similar results by Frey and McNeil (2000) and our own results from Chapter 3, for an univariate conditional model combining EVT and a GARCH(1,1) model.

²⁵ For the 99.5% quantile we only have about 20 datapoints (e.g. VaR exceedances) for the backtesting. Moreover, the algorithm failed for the 99.9% quantile, due to the lack of sufficient data points.

distribution assumption. Moreover, Figure 4.4 shows that the assumption of zero mean residuals seems more reasonable for the model proposed in Section 3, using an appropriate copula. Note that there are only 158 (31) residuals for the 95% (99%) quantile.

4.6 Conclusion

The present dissertation is concerned with tail estimation for financial return series in a bivariate framework, assuming different dependence structures. In particular, the risk measures VaR and S-VaR are calculated and backtested. We use a simulation procedure to combine the theory of extreme values and copulas, respectively. The proposed model is compared to different other methods for tail estimation. Our main findings are summarized as follows.

We find that the dependence structure indeed seems to be a crucial part when modelling portfolio risk. In particular, we find that for the VaR

- for quantiles higher than 95%, the proposed model gives a better performance²⁶ compared with traditional approaches such as the Historical simulation and the simple (unconditional) bivariate Variance-covariance model,
- we get different results when assuming different dependence structures; assuming a copula with strong lower tail dependence behaviour (e.g. the Kimeldorf Sampson or the BB3 copula) gives better results than assuming a Normal copula or a copula with strong upper tail dependence. From the backtesting results we conclude that the former copulas better describe the dependence structure for the given portfolios and sample,
- the proposed unconditional model has the potential advantage over conditional models to result in less volatile VaR-forecasts.
- We also calculate the S-VaR as an alternative risk measure with good theoretical properties and find that we only get reasonable results if we assume a non-normal dependence structure. The assumption of normality seems useless for the purposes of calculating expected shortfall.²⁷

²⁶ Note that better performance means better backtesting results applying the given tests.

²⁷ Note that these results are in line with findings of Frey and McNeil (2000) and our own results from Chapter 3 for univariate conditional models.

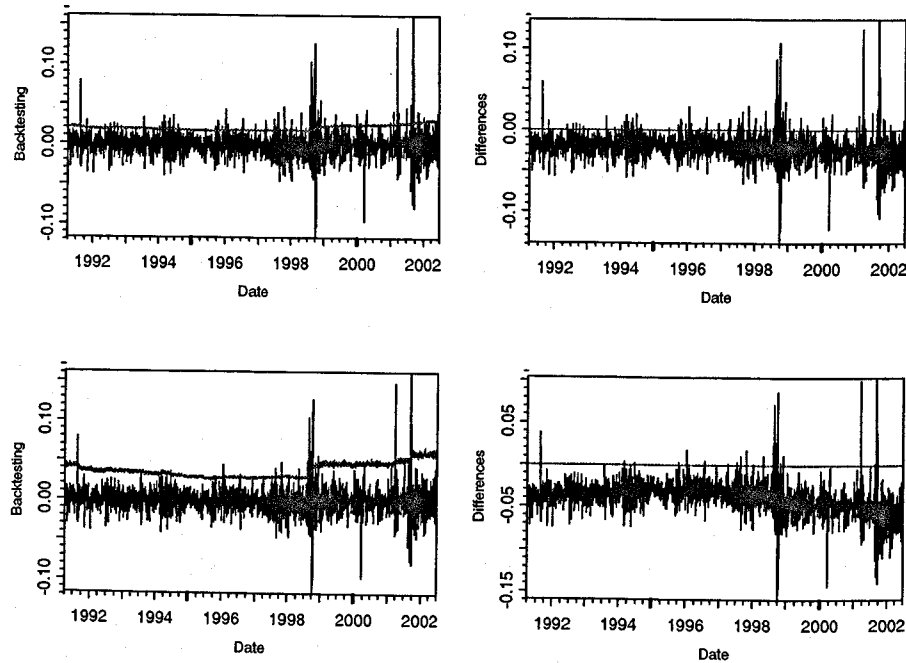


Figure 4.2: Backtesting results for the 95% (upper figures) and for the 99% quantiles (lower figures). Left figures: log returns (losses on the positive axes) and VaR forecasts (superimposed); right figures: differences between realized returns and forecasted VaRs (magnitude of under- and overestimation of the risk)

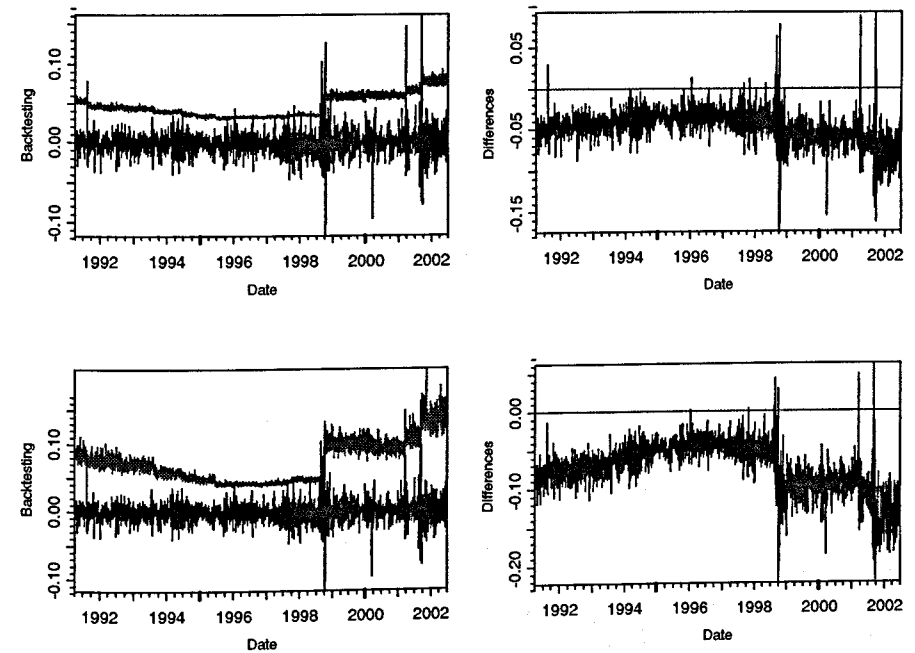


Figure 4.3: Backtesting results for the 99.5% (upper figures) and for the 99.9% quantiles (lower figures). Left figures: log returns (losses on the positive axes) and VaR forecasts (superimposed); right figures: differences between realized returns and forecasted VaRs (magnitude of under- and overestimation of the risk)

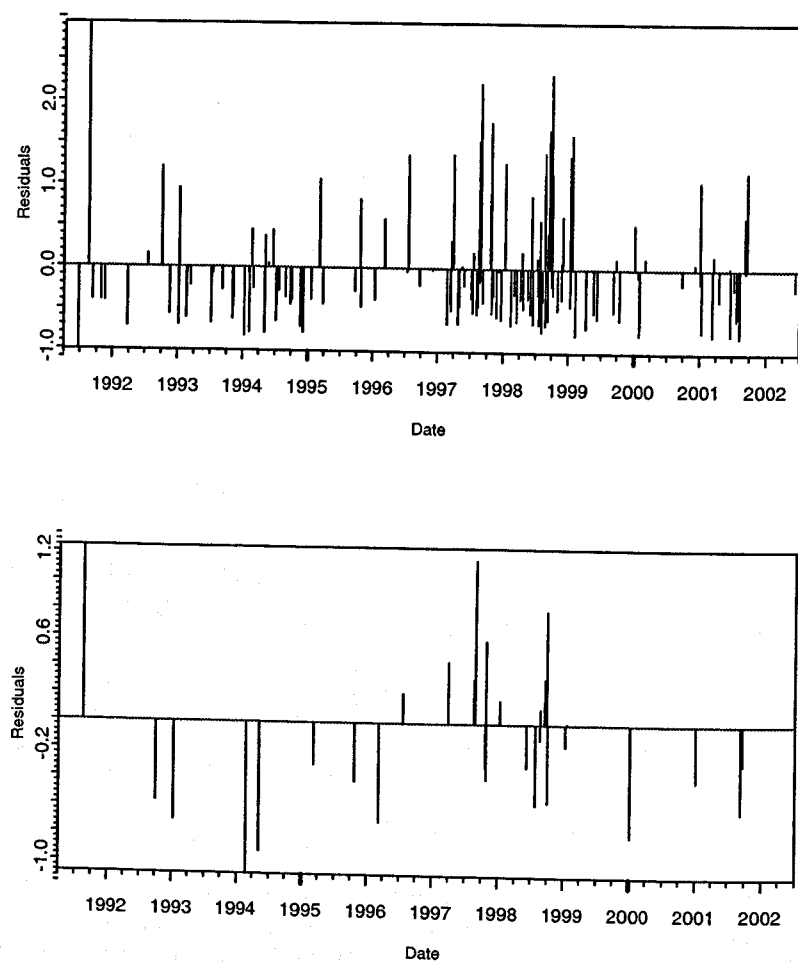


Figure 4.4: Backtesting results: exceedance residuals for the Nestle-Novartis portfolio (upper figure: 95%, lower figure: 99%), assuming the kimeldorf sampson copula. Under the null hypothesis that the model correctly estimates the tail, these residuals should have mean zero. Note that there are only 158 (31) residuals for the 95% (99%) quantile.

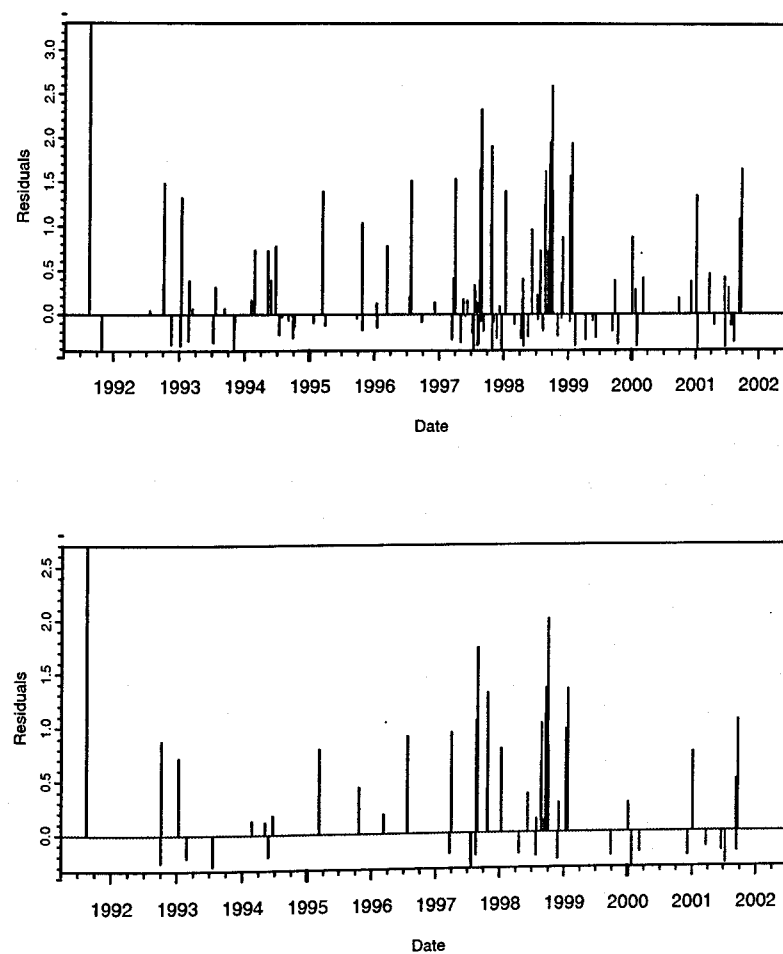


Figure 4.5: Backtesting results: exceedance residuals for the Nestle-Novartis portfolio (upper figure: 95%, lower figure: 99%), assuming the (unconditional) bivariate Variance-covariance model. Under the null hypothesis that the model correctly estimates the tail, these residuals should have mean zero. Note that there are only 158 (31) residuals for the 95% (99%) quantile.

- We conclude that the choice of the dependence structure can be crucial in financial risk measurement and for making decisions in Financial Risk Management.

Some questions remain to be studied, however: *How* to (more systematically) find the copula that best models the dependence structure of the real multivariate data? How to construct specific tests to discriminate the more robust copulas? What do we know or can we infer about the stylized facts of copulas, as opposed to stylized facts of financial return series?²⁸

Furthermore, what about dynamic models? How do models assuming conditional copulas, as proposed for example in Patton (2001a),²⁹ perform compared to standard bivariate GARCH models? Respective results would provide further interesting insights.

And what about higher dimensions? The use of high-dimensional copulas is still in its infancy.³⁰ Realistically, however, parametric approaches involving copulas are only viable in a small number of dimensions. If we are interested in a small number of risk factors and are particularly concerned that joint extreme values may occur, we might use such models to get useful descriptions of the joint tail. For higher dimensions though, there are simply too many parameters to estimate and too many different tails of the multivariate distribution to worry about (curse of dimensionality). In such situations, collapsing the problem to a univariate problem by considering a whole portfolio of assets as a single risk seems to be more appropriate.

²⁸For a first approach in this direction, see Breymann, Dias and Embrechts (2002).

²⁹Patton (2001a) uses marginal distributions which are t-GARCH, each with their own tail fatness parameter, and a Clayton copula (e.g. the kimeldorf sampson copula) which is modeled as evolving in a GARCH-type manner. The model can be thought of as generalizing multivariate GARCH models from conditional second moments to entire conditional densities.

³⁰See i.e. Hürlimann (2002).

Chapter 5

Limits of EVT as a Tool to model Financial Risk

5.1 Introduction

The claim made in the literature is that EVT offers an interesting (and in many ways, the correct) view on the quantitative measurement of risk, especially for skewed P&L distributions. In its more advanced form, EVT describes the behaviour of extremal events for stochastic processes, evolving dynamically in time and space. The main virtue of EVT is that it gives the user a critical view on and a methodological toolkit for issues like skewness, fat tails, rare events and stress scenarios. However, there is always going to be an element of doubt, as one is *extrapolating* into areas one doesn't know about.

The aim of the following chapter is to critically assess the pros and cons of EVT as a tool to model risk, summarizing the main findings of the previous chapters.

5.2 The Potential of EVT in Financial Risk Modelling

The main contribution of EVT can be seen as the improvement of theoretical limitations of existing risk modelling techniques. In particular, we see the following main advantages that EVT offers to Financial Risk Modelling:

- *Fat tails in financial time series.* Existing databases show that the distribution of credit, market and operational risks are heavy-tailed, so that estimates beyond VaR become crucial. Indeed, the main use of EVT is to address the estimation issues concerning large losses in leptokurtic distributions.
- *Asymmetry in financial time series.* The distribution of model residuals is often found to be leptokurtic. For this kind of data, the EVT approach is preferable, because it can deal with asymmetries in the tails. Moreover, the approach is based on a sound theoretical theory.
- *Possibility of out-of-sample extrapolation.* The proposed risk level of the regulators is the 99% VaR. In other words, regulators require banks to hold capital against an event that occurs on average once every hundred days or roughly 2.5 times a year. But presumably, regulatory capital is held to prevent systematic failure. However, systematic failures are extremely rare events and certainly do not occur with an expected frequency of 2.5 times a year. Hence, EVT could be used to produce risk forecasts and confidence intervals beyond the 99% VaR.
- *Distribution of losses above threshold.* A significant drawback of VaR and related risk-measuring methodologies is that it provides us only with a point estimate of the loss distribution. However, when projecting risks or losses presumably one is interested in the distribution of the loss given that a certain extremal threshold has been exceeded, i.e. when calculating the coherent S-VaR. That's exactly what EVT provides us with; the limit distribution above a given threshold. Moreover, EVT estimation provides smoothness and avoids coarseness or bias in the tail estimates.
- *Non-linear dependence.* One problem with conventional risk forecasting models is that they overwhelmingly rely on correlation and covariance matrices as estimates of the dependence structure among assets. However, if the underlying distribution is non-elliptical¹ (so that it is no longer described purely by its first two moments), the dependence structure of assets is no longer captured by linear correlations and we better rely on MEVT and related dependence concepts.
- *EVT as a tool to complement VaR models.* Applying EVT-related concepts necessarily implies a trade-off between accuracy of risk forecasts and

¹Elliptical distributions are densities taking constant value on ellipsoids, i.e. the multivariate normal and t distributions.

ease of computation. Considering its limits, EVT is here to complement VaR models, however not to replace them. EVT alone is not enough.

- *Handling operational risk.* Even given the extremely optimistic view that databases on well-defined operational risk losses will be available in the not-too-distant future, the nature of these rare, high-impact losses renders them very different from the loss data we know from market and credit risk. EVT offers a generic approach towards operational risk.
- *Handling exotic options, emerging-market stocks and exotic currencies.* The VaR approach is, even for market risks, only appropriate if we consider liquid markets where the relevant information is available. For exotic options, emerging-market stocks and exotic currencies we need more reliable tail estimates as provided by EVT.
- *EVT offers the relevant tools.* Another advantage of EVT is that it provides both the theoretical and computational *tools* to model financial risk.

5.3 Pitfalls of EVT in Financial Risk Modelling

There are a number of pitfalls in the use of EVT in financial risk measurement, which will be discussed in the following. Above all, in order to estimate way in the tails, one has to make mathematical assumptions on the tail model. These assumptions are very difficult to verify in practice. Consequently, there is an intrinsic model risk.

5.3.1 Violations of iid assumption

The violation of the iid assumption was discussed in Chapter 1. Generalizations to dependent data have been attempted, see e.g. Leadbetter, Lindgren and Rootzén (1983), but they typically require "anti-clustering" conditions on the extremes, which rule out the sort of volatility clustering often found in financial asset returns.² However, according to Diebold, Schuermann and Stroughair (2000) there are two directions proposed to improve the performance of the tail index estimator in the presence of dependent data.

First, as discussed in Section 1.6, one can use the block maxima method, reducing dependence with the use of per-period maxima. The use of a GEV is justified if the periods are taken to be long enough, and maximum likelihood is

²Recall section 1.6 about how to deal with dependence in a POT framework.

easily implemented. The limits have been mentioned, however; the aggregation (e.g. the use of weekly, monthly or annual maxima) reduces efficiency, and the problem of the appropriate amount of aggregation is introduced.

The second direction is to estimate the tail of the conditional distribution, as done in Chapter 3 of the present thesis. The key is the recognition of the fact that, if the adopted volatility model is a good approximation to the true volatility dynamics, then the standardized residuals will be approximately *iid*.

5.3.2 Violations of the power law assumption

Most of the tail estimation literature, as for example Longin (1996), assumes that returns are in the maximum domain of attraction of a Fréchet distribution,³ so that the tail of the survival function follows a power law. But how to check the latter assumption, and what is the evidence?

Longin (1996) applies both parametric, non-parametric and regression models for tail estimation and finds that the asymptotic distribution of extreme returns is the Fréchet distribution. He performs both statistical (simple tests of $\xi = 0$, corresponding to the Gumbel distribution against $\xi > 0$, corresponding to the Fréchet case) and graphical tests (e.g. plotting the ordered extreme returns against the theoretical values predicted by the Gumbel or the Fréchet distribution, respectively). Moreover, the results are fairly stable over time, the shape of the tails being constant over time. The distribution of extremes is also found to be stable under temporal aggregation, as extremes selected from daily, weekly, and monthly returns follow a Fréchet distribution.

Inoue (1998) develops a more general framework for consistent testing of hypothesis involving conditional distributions of time series.

5.3.3 The size of the tail

The finite-sample properties of the parameter estimators depend crucially on the choice of k . There is, however, no canonical "optimal" choice.

In particular, we have seen that there is an important bias-variance trade-off when varying k for fixed sample size: increasing k and therefore using progressively more data, reduces variance but increases bias. Increasing k reduces variance because more data are used, however it increases bias because the power law is assumed to hold only in the extreme tail. As shown, k is typically chosen

³Recall section 1.2.1.

via ad hoc rules of thumb, sometimes supplemented with plots of the empirical tail function⁴ with the estimated tail superimposed. Diebold, Schuermann and Stroughair (2000) argue that such plots are invaluable - although they are used sometimes, they are not used often enough - and they need to be supplemented with complementary and rigorous rules for suggesting a choice of k , such as proposed by Danielsson and De Vries (1997a) and Danielsson and De Vries (2002). They propose an automatic double subsample bootstrap procedure for the determination of the optimal threshold. The approach can supplement the usual graphical analysis. However, the approach is based on the non-parametric Hill estimator and it would be interesting to know whether their result can be extended to the GPD approach.

Beirlant, Vynckier and Teugels (1996) show that, based on classical ideas on regression diagnostics, algorithms can be constructed searching for an appropriate set of extreme values to use in the estimation procedure.

5.3.4 Availability and relevance of data

If applying EVT we have, by definition, relatively few extreme value observations to work with. This means that any estimate based on EVT will necessarily be very uncertain, relative to any estimates we might make of more central quantiles or probabilities. EVT based estimates therefore tend to have relatively wide confidence intervals attached to them. This uncertainty is not a fault of EVT as such, however an inevitable consequence of the scarcity of data.

Furthmore, there is a trade-off between the accuracy of the statistical estimation and the relevance of the data used.

5.3.5 Finite-sample estimation and inference

Diebold, Schuermann and Stroughair (2000) highlight the contrast between the probability theory and the statistical theory. They claim that the probability theory is often elegant and rigorous, whereas the statistical theory remains primitive and skeletal in many aspects. First, the properties of the existing approaches to estimation of the tail index haven't been adequately explored. Little is known about the finite-sample properties of the tail index estimator under various rules for cutoff selection.

In Financial Risk Management, we are concerned not on the tail index estimate per se, but rather on the extreme quantiles and probabilities. The extreme

⁴Also called empirical survival function, since the survival function is simply one minus the cumulative density or distribution function, $1 - F(x)$.

quantile and probability estimators are highly nonlinear functions of the tail index. Hence, poor sampling properties of the tail index estimator will likely translate into even worse properties of the quantile and probability estimators.

5.3.6 MEVT and computation

Introductions to MEVT are given in Embrechts, De Haan and Huang (2000), Coles (2001), Fougères (2003) and Hürlimann (2002). If we are interested in a small number of risk factors and are particularly concerned that joint extreme values may occur, we might use such models to get useful descriptions of the joint tail. However, as pointed out in previous chapters, handling extremes for higher dimensional portfolios is difficult. There are too many parameters to estimate and too many different tails of the multivariate distribution to worry about. In such situations, collapsing the problem to a univariate problem by considering a whole portfolio of assets as a single risk seems to be more reasonable.

The last important issue is computation. Applications to portfolio risk measurement are likely to require (computationally expensive) Monte Carlo simulations. Furthermore, the convergence of maximum likelihood estimated parameters in univariate models is not guaranteed. Alternative estimators are available and can be considered (such as the method of moments).

5.4 Beyond EVT: Limits to Financial Risk Modelling

There are some limits inherent to risk modelling in general, beyond the use of EVT. Knowledge of these limits to risk modelling itself is a crucial issue and should be considered as the very foundation of modern risk management. The aim of the following section is to discuss the most important respective issues.

5.4.1 Endogeneity of risk

The general imperfection of economic models

Often people forget that economics is, after all, a social science and as such is concerned with the behaviour of human beings, i.e. in complex systems. Hence, no wonder that one day researchers found that some parameters of the system are themselves endogenous, i.e. that market participants, assumed to behave

exogenously, rather react on the outcome of the model. In other words, in economics it is often the case that a theory, once everybody knows about it, cannot be proven, just because everybody is considering it in his or her own actions. Assuming that market participants have rational expectations or a minimal capacity to learn or adapt, there is an inherent instability in economic laws. The latter is particularly true for financial markets, implying a substantial model risk in financial market modelling. Hence, there is a considerable uncertainty in quantitative and statistical models in risk management. Moreover, it is important that the uncertainty or instability is inherent to the system: it is a consequence of the way market participants build expectations and of their capability to learn. This is the main difference to the modelling of risks that are determined by nature.

Implications of the endogeneity of risk

The risk forecasting models based on VaR that are currently employed are based on the assumption that one's own action, based on a volatility forecast, does not affect future volatility itself. However, as argued by Danielsson, Embrechts, Goodhart, Keating, Muennich, Renault and Shin (2001) and Heri and Zimmermann (2001) this reasoning is faulty. Volatility is determined in the market, in large part by the behaviour of all individual market participants - in other words, risk is endogenous rather than exogenous by definition. In times of crisis, this endogeneity may matter enormously if agents become more homogenous as a result.⁵ Using similar risk models, they may pursue similar strategies to mitigate the adverse effects of the on-setting crisis. In such a case, individual actions do not cancel each other out but may in fact reinforce each other. The effect is a pure externality - individual banks do not take it into account when making decisions, yet it affects the stability of the banking system as a whole. However, this externality is not acknowledged by existing forecasting models.

In particular, the use of VaR or similar risk-modelling techniques may no longer be justified in such a situation, mainly for two reasons. First, the process that drives the underlying data will have undergone a structural break and is no longer governed by the behaviour of heterogenous but rather by that of relatively homogenous market players. Hence, a central assumption of VaR-modelling, namely the stationarity of the underlying stochastic process, is violated. Second, data used to estimate forecasting models has also undergone a structural break.

⁵This was the case e.g. during the 1987 crash when portfolio insurance was very much en vogue, see Morris and Shin (1999), or during the crash of the US dollar against the Yen in October 1998 when the US\$ fell from 131 Yen to 112 over two days, see Danielsson (2000).

As a result, data immediately preceding the onset of a crisis becomes useless for the purpose of estimating risk.⁶ Therefore, most risk models are likely to provide little guidance during crisis periods.

5.4.2 Empirical properties of risk models

In an extensive survey across different asset classes, Danielsson (2000) examines the robustness of various commonly used *VaR* methods (including EVT-based methods) of *VaR* estimation. He tests the precision with which volatility is forecasted and finds that existing models provide estimates that can differ by an order of magnitude of five. The Variance-covariance model, for example, comes out as the worst performer while there is no model that *clearly* outperforms all others. Overall, the impression is that *VaR* models are not robust. They do not provide accurate and consistent risk forecasts across different assets, time horizons and risk levels within the same asset class.

Furthermore, estimates show that *VaR* forecasts are almost as volatile as returns themselves. There exists a trade-off between the volatility of the forecasts and the reflection of the actual volatility background,

5.4.3 Subadditivity of risk models and coherent risk measures

An important characteristic that any risk measure needs to satisfy is subadditivity. In the context of *VaR*, subadditivity means that the *VaR* of a portfolio of, say, two assets will be bounded by the sum of the *VaRs* of the two individual assets. The property enables risk models to put an upper bound on the *VaR* of any portfolio by simply adding all individual *VaRs*. Embrechts, McNeil and Straumann (2000) show that *VaR* is a subadditive risk measure only for elliptical distributions (like e.g. the normal distribution). In contrast, for non-elliptical distributions, the *VaR* fails to be subadditive. This implies that the *VaR* of a portfolio may be larger than the sum of *VaRs* of the individual assets - not exactly what our intuition about risk diversification implies. Coherent measures such as *S-VaR* have been introduced and applied to mitigate the problem of subadditivity.

⁶For the impacts of current regulation and the issue of information disclosure, see Danielsson, Embrechts, Goodhart, Keating, Muennich, Renault and Shin (2001).

5.4.4 Extension to arbitrary types of risks

Consider e.g. the credit risk. As the data is scarcely available, there does not exist a data basis in order to adequately address issues involving credit risk.⁷ Moreover, the credit informations of rating agencies often are not representative for bank-specific customer portfolios. There exist no current integrated measures for market and credit risks, respectively; this is not due to the lack of modelling techniques but due to insufficiency of data and knowledge of the relevant empirical dependencies.

5.4.5 Aggregation across different risk classes

Another question concerns the aggregation of risk measures. One often faces the problem that risk measures have been calculated for separate risk classes; how can we estimate the risk measure of a global position? Typical examples are one-period risks withing certain risk classes, but also across different classes. In practice, one is often given only the marginal distribution function, together with some notion of dependence. Most often one does not have full information on the joint distribution function or the dependence structure. The key issue here is not the shape of the marginal distribution functions, but rather the "damage" non-Gaussian dependence structures can cause on risk management systems. Also, the problem of subadditivity is encountered when adding up risk measures across different risk classes.

5.4.6 Multiperiod risk measurement

For the calculation of multiperiod risk, e.g. the 10-day *VaR* as proposed by the regulation, most users follow the recommendation by using the so-called square-root-of-time method, multiplying the one-day *VaR* by the square root of 10 to obtain the 10-day *VaR*. However, this method is legitimate only under extremely strong conditions. Returns need to be *iid* and normally distributed. Clearly, both requirements are violated for most of the financial time series under consideration. In fact, scaling rules are extremely difficult to calculate and the use of any particular scaling rule is arbitrary.

⁷The same is (even more) true for operational risks.

5.4.7 Time inconsistency of risks

Current risk models such as VaR promote a short term orientation. However, an institute that manages its risk with a short term horizon foregoes the usage of intertemporal possibilities to diversify risk. Often, there is the illusion that a small short term risk provides the requirements for a small medium- or even long term risk. However, often there is no time consistency of risk.

5.4.8 Other limits

Lack of consensus

Despite the limits to risk modelling per se, there are some restrictions which may hinder the effective risk modelling. One of these restrictions is the lack of consensus over the choice of the standard model to measure risks and the implementation of the latter. Every institution has its own risk models, giving it its own name. Consequently, empirical results on the quality of the estimation are difficult to evaluate.

Limits due to regulative incentives

Heri and Zimmermann (2001) argue that with every regulative norm, part of the responsibility is passed on to the regulative institution. There is an incentive for firms to reduce their effort to the required minimum. In this context, the crucial question is whether a further engagement of the regulator in particular questions regarding the modelling of risk is adequate to reduce systemic risks. Risk management is, after all, a process within which the applied models need to be improved continuously. A further engagement of the regulator might give wrong incentives with this respect.

5.5 Conclusion

Overall, EVT provides a tailor-made approach to the estimation of extreme probabilities and quantiles, such as VaRs at high quantiles. The EVT approach is intuitive and plausible; and it is relatively easy to apply, at least in its more basic forms.⁸ Moreover, it gives us considerable practical guidance on what we should estimate and how we should do it; and it has good performance results.

⁸For more advanced models, there is a trade-off between the accuracy of the estimation and the intellectual requirement.

EVT is also important in what it tells us not to do, and the most important point is not to use distributions justified by the central limit theory - most particularly, the normal distribution - for extreme value estimation.

However, as it has become clear in the present chapter, we should not lose sight of the limitations of EVT approaches. We should recognize that fitting EVT functions to recent historical data is still fraught with pitfalls. After all, the most powerful statistical techniques cannot make short histories reveal once-in-a-lifetime events. This is why, for all the new techniques behind EVT, (additional) stress testing remains a delicate art form.⁹ Or as Diebold, Schuermann and Stroughair (2000) put it: "When the smoke clears, the contribution of EVT remains basic and useful: It helps draw smooth curves through the extreme tails of empirical survival functions in a way that is guided by powerful theory.... [But] we shouldn't ask more of the theory than it can deliver."

Moreover, there is an increasing body of evidence that inherent limitations in risk modelling technology, coupled with imperfect regulatory design, acts more like a placebo rather than a scientifically proven preventer of crashes it is sometimes claimed to be. It is not possible to create the perfect risk model, and a risk forecaster needs to weigh the advantages and disadvantages of various models and data choices to create what inevitably can only be an imperfect model. More importantly, this imperfection itself should be considered a central feature of modern risk modelling.

As Danielsson (2002) points out, a future risk model, incorporating insights from economic as well as from financial theory, in conjunction with financial data during crisis, has the potential to provide more accurate answers by directly addressing issues such as liquidity dynamics. There is a need for a joint market and liquidity risk model, covering both stable and crisis periods.

⁹See Jorion (2001), Chapter 10, for an introduction to stress testing.

Appendix A

Additional Figures to Chapter 1

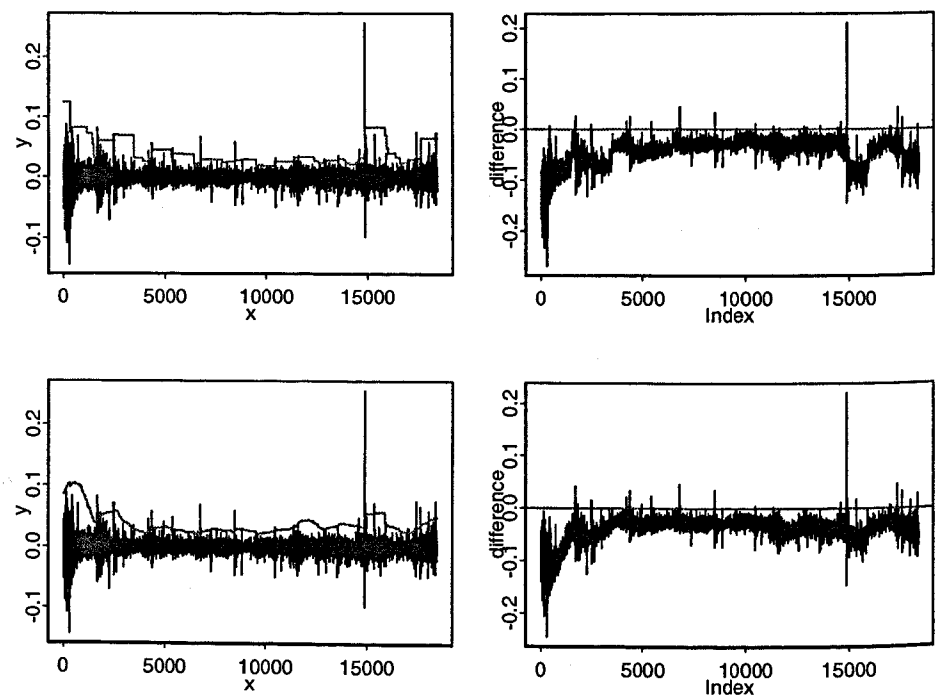


Figure A.1: Backtesting the 99.9% VaR for Dow Jones Index return series: Historical VaR (top figures) and Normal VaR (bottom figures). Returns and VaR estimates on the left figures, differences between realized and predicted VaRs on the right figures, respectively.

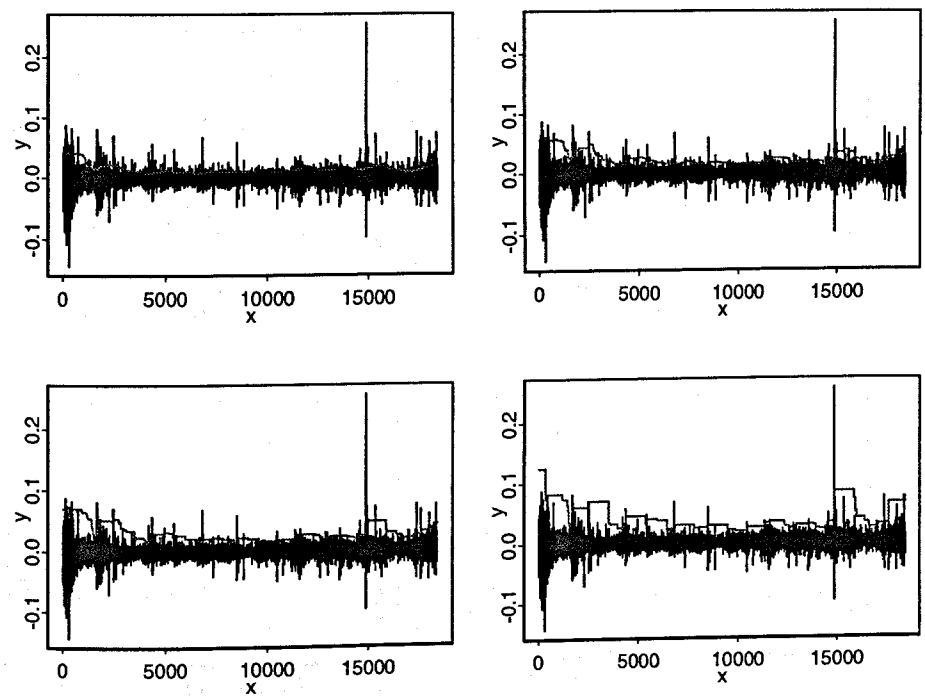


Figure A.2: Backtesting results for different quantiles (top left: 95%, top right: 99%, bottom left: 99.5%, bottom right: 99.9%), using the Historical VaR method. Note that losses are shown on the positive axis, VaR estimates are superimposed.

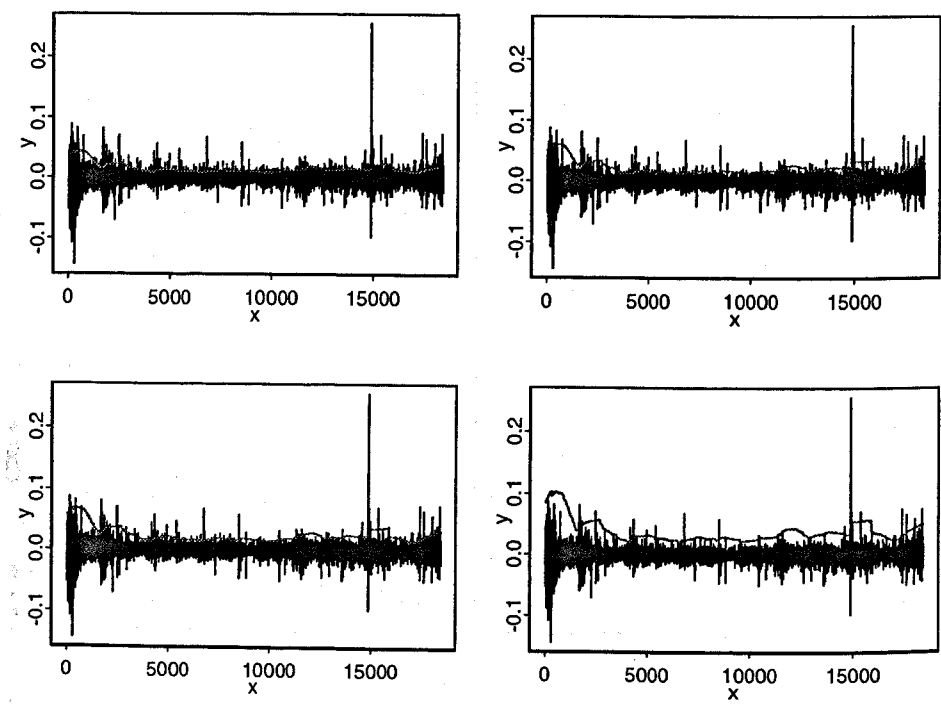


Figure A.3: Backtesting results for different quantiles (top left: 95%, top right: 99%, bottom left: 99.5%, bottom right: 99.9%), using the Normal VaR method. Note that losses are shown on the positive axis, VaR estimates are superimposed.

Appendix B

List of Moodys Credit Indices

APPENDIX B. LIST OF MOODYS CREDIT INDICES

Category	Company	Coupon	Maturity Date	Rating	Date
Aaa-Industrial	BRISTOL MYERS SQUIBB CO	6.8	15.11.2026	AAA	03.12.02
	JOHNSON & JOHNSON	6.73	15.11.2023	AAA	03.12.02
	MERCK & CO INC	6.4	01.03.2028	AAA	03.12.02
Aa-Industrial	MOBIL CORP	7.625	23.02.2033	AAA	03.12.02
	AMERITECH CAP FUNDING CORP	6.55	15.01.2028	AA3	03.12.02
	BELLSOUTH CORP	6.875	15.10.2031	AA3	03.12.02
	COLGATE PALMOLIVE CO MTNS BE	6.45	16.06.2028	AA3	03.12.02
	DU PONT E I DE NEMOURS & CO	7.5	01.03.2033	AA3	03.12.02
	KIMBERLY CLARK CORP	6.375	01.01.2028	AA2	03.12.02
	MCDONALDS CORP	7.05	15.11.2025	AA3	03.12.02
	MINNESOTA MNG & MFG CO	6.375	15.02.2028	AA1	03.12.02
	SCOTT PAPER CO	7	15.08.2023	AA3	03.12.02
	TEXACO CAPITAL INC	6.875	15.08.2023	AA3	03.12.02
Aa-Utility	UNION TEX PETE HLDGS INC	7	15.04.2038	AA1	03.12.02
	CHESAPEAKE & POTOMAC TEL CO MD	7.15	01.05.2023	AA2	03.12.02
	CHESAPEAKE & POTOMAC TEL CO VA	7	15.07.2025	AA2	03.12.02
	DUKE ENERGY CORP	7	01.07.2033	AA3	03.12.02
	FLORIDA PWR & LT CO	7.05	01.12.2026	AA3	03.12.02
	ILLINOIS BELL TEL CO	7.25	15.03.2024	AA2	03.12.02
	MICHIGAN BELL TEL CO	7.5	15.02.2023	AA2	03.12.02
	PACIFIC BELL	7.375	15.06.2025	AA3	03.12.02
	SOUTHWESTERN BELL TEL CO	7.375	15.07.2027	AA2	03.12.02
	WISCONSIN ELEC PWR CO	6.875	01.12.2035	AA3	03.12.02
A-Industrial	APACHE CORP	7.375	15.08.2047	A3	03.12.02
	ARCHER DANIELS MIDLAND CO	6.95	15.12.2097	A1	03.12.02
	BECTON DICKINSON & CO	7	01.08.2027	A2	03.12.02
	CBS INC	7.125	01.11.2023	A3	03.12.02
	CARGILL INC	7.25	01.11.2036	A1	03.12.02
	DAYTON HUDSON CORP	7.65	01.08.2023	A2	03.12.02
	DOW CHEM CO	7.375	01.03.2023	A1	03.12.02
	EATON CORP	7.625	01.04.2024	A2	03.12.02
	KNIGHT RIDDER INC	7.15	01.11.2027	A2	03.12.02
	MCI COMMUNICATIONS CORP	7.125	15.06.2027	A3	03.12.02
A-Utility	MARTIN MARIETTA MATLS INC	7	01.12.2025	A3	03.12.02
	MCDONALDS CORP	7.31	15.09.2027	A1	03.12.02
	MURPHY OIL CORP	7.05	01.05.2029	A3	03.12.02
	PHILIP MORRIS COS INC	7.75	15.01.2027	A2	03.12.02
	PHILLIPS PETE CO	7.125	15.03.2028	A3	03.12.02
	SEARS ROEBUCK ACCEP CORP	7.5	15.10.2027	A3	03.12.02
	CONSOLIDATED EDISON CO N Y INC	7.75	01.06.2026	A1	03.12.02
	EQUITABLE RES INC	7.75	15.07.2026	A2	03.12.02
	NEW YORK TEL CO	7	01.12.2033	A1	03.12.02
	OKLAHOMA GAS & ELEC CO	7.3	15.10.2025	A1	03.12.02
Baa-Industrial	ONEOK INC NEW	6.875	30.09.2028	A2	03.12.02
	ANADARKO PETE CORP	7.25	15.11.2096	BAA1	03.12.02
	CHAMPION INTL CORP	7.15	15.12.2027	BAA2	03.12.02
	CLEAR CHANNEL COMMUNICATIONS	7.25	15.10.2027	BAA3	03.12.02
	CONSOLIDATED RAIL CORP	7.875	15.05.2043	BAA2	03.12.02
	HARCOURT GEN INC	7.2	01.08.2027	BAA1	03.12.02
	INTL PAPER CO	7.625	01.03.2023	BAA2	03.12.02
	KN ENERGY INC	7.25	01.03.2028	BAA2	03.12.02
	KERR MCGEE CORP	7.875	15.09.2031	BAA2	03.12.02
	LITTON INDS INC	7.75	15.03.2026	BAA3	03.12.02
Baa-Utility	LOCKHEED CORP DEL	7.875	15.03.2023	BAA2	03.12.02
	LOUISIANA LD & EXPL CO	7.65	01.12.2023	BAA2	03.12.02
	MASCO CORP	7.75	01.08.2029	BAA1	03.12.02
	MEAD CORP	7.55	01.03.2047	BAA2	03.12.02
	NEWS AMER HLDGS INC	7.7	30.10.2025	BAA3	03.12.02
	NOBLE AFFILIATES INC	8	01.04.2027	BAA2	03.12.02
	NORFOLK SOUTHERN CORP	7.9	15.05.2097	BAA1	03.12.02
	TCI COMMUNICATIONS INC	7.875	15.02.2026	BAA2	03.12.02
	TRW INC	7.75	01.06.2029	BAA2	03.12.02
	TIME WARNER INC	7.57	01.02.2024	BAA1	03.12.02

Figure B.1: Moodys Long Term Bond Index Components.

Category	Company	Coupon	Maturity Date	Rating	Date
Aaa-Industrial	JOHNSON & JOHNSON	6.625	01.09.2009	AAA	03.13.02
	MERCK & CO INC	5.1	22.02.2011	AAA	03.13.02
	WARNER LAMBERT CO	6	15.01.2008	AAA	03.13.02
Aa-Industrial	ABBOTT LABS	5.4	15.09.2008	AA3	03.13.02
	AMERITECH CAP FUNDING CORP	6.15	15.01.2008	AA3	03.13.02
	ATLANTIC RICHFIELD CO	5.9	15.04.2009	AA1	03.13.02
	ILLINOIS TOOL WKS INC	5.75	01.03.2009	AA3	03.13.02
	MCDONALDS CORP MED TERM NT BE	5.35	15.09.2008	AA3	03.13.02
	TEXACO CAPITAL INC	5.7	01.12.2008	AA3	03.13.02
	VASTAR RES INC	6.5	01.04.2009	AA1	03.13.02
	WASHINGTON POST CO	5.5	15.02.2009	AA3	03.13.02
	GTE CALIF INC	5.5	15.01.2009	AA3	03.13.02
	GTE NORTH INC	6.9	01.11.2008	AA3	03.13.02
Aa-Utility	NEW ENGLAND TEL & TEL	5.875	15.04.2009	AA2	03.13.02
	PACIFIC BELL	6.625	01.11.2009	AA3	03.13.02
	SOUTHWESTERN BELL TEL CO	6.625	15.07.2007	AA2	03.13.02
	WPS RESOURCES CORP	7	01.11.2009	AA3	03.13.02
	WISCONSIN PWR & LT CO	7	15.06.2007	AA3	03.13.02
	AIR PRODS & CHEM INC MTNS BE	6.6	11.10.2007	A2	03.13.02
	ALLIEDSIGNAL INC	6.2	01.02.2008	A2	03.13.02
	CAMPBELL SOUP CO	6.75	15.02.2011	A3	03.13.02
	CARGILL INC	6.3	15.04.2009	A1	03.13.02
	DANAHER CORP DEL	6	15.10.2008	A2	03.13.02
A-Industrial	DAYTON HUDSON CORP	5.875	01.11.2008	A2	03.13.02
	DEERE JOHN CAPITAL CORP	6	15.02.2009	A3	03.13.02
	DISNEY WALT CO MTNS BE	5.62	01.12.2008	A3	03.13.02
	GTE CORP	6.46	15.04.2008	A2	03.13.02
	GENERAL MTRS CORP	6.375	01.05.2008	A3	03.13.02
	HEINZ H J CO	6	15.03.2008	A3	03.13.02
	JOHNSON CTLS INC	6.3	01.02.2008	A2	03.13.02
	KNIGHT RIDDER INC	6.625	01.11.2007	A2	03.13.02
	MARTIN MARIETTA MATLS INC	6.9	15.08.2007	A3	03.13.02
	MAY DEPT STORES CO	5.95	01.11.2008	A1	03.13.02
A-Utility	MCCORMICK & CO INC MTN BE	6.8	01.02.2008	A2	03.13.02
	PRAXAIR INC	6.5	01.03.2008	A3	03.13.02
	SARA LEE CORP MTN BE	6.15	19.06.2008	A3	03.13.02
	SEARS ROEBUCK ACCEP CORP	7	15.06.2007	A3	03.13.02
	SHERWIN WILLIAMS CO	6.85	01.02.2007	A2	03.13.02
	TEXTRON INC	6.625	15.11.2007	A3	03.13.02
	VULCAN MATLS CO	6	01.04.2009	A1	03.13.02
	ALABAMA PWR CO	5.375	01.10.2008	A2	03.13.02
	CONSOLIDATED EDISON CO N Y INC	6.45	01.12.2007	A1	03.13.02
	IES UTILS INC	6.625	01.08.2009	A2	03.13.02
Baa-Industrial	LG&E CAP CORP MTN BE 144A	6.46	15.01.2008	A3	03.13.02
	NATIONAL FUEL GAS CO MTNS B E	6.303	27.05.2008	A2	03.13.02
	NEW YORK TEL CO	6.125	15.01.2010	A1	03.13.02
	NORTHN STS PWR CO MINN	6.875	01.08.2009	A1	03.13.02
	ONEOK INC NEW	8	01.02.2009	A2	03.13.02
	PACIFICORP SECD MTN-BOOK ENTRY	6.375	15.05.2008	A3	03.13.02
	PUB SVC COLO SCD 1ST COLTR MTN	7.11	05.03.2007	A3	03.13.02
	SOUTHWESTERN PUB SVC CO	6.2	01.03.2009	A3	03.13.02
	VIRGINIA ELEC & PWR CO MTNS BE	5.73	25.11.2008	A3	03.13.02
	XCEL ENERGY INC MINN	7	01.12.2010	A3	03.13.02
Baa-Utility	BELO CORP	7.125	01.06.2007	BAA3	03.13.02
	BORG WARNER INC	6.5	15.02.2009	BAA2	03.13.02
	COASTAL CORP	7.625	01.09.2008	BAA2	03.13.02
	COLTEC INDS INC	7.5	15.04.2008	BAA2	03.13.02
	COX ENTERPRISES INC	7.875	15.09.2010	BAA1	03.13.02
	DIAL CORP NEW	6.5	15.09.2008	BAA3	03.13.02
	DUKE ENERGY FIELD SVCS LLC	6.875	01.02.2011	BAA2	03.13.02
	ENERGEN CORP MTNS BE	7.625	15.12.2010	BAA1	03.13.02
	FMC CORP MTN BE	7	15.05.2008	BAA3	03.13.02
	FORD MTR CO DEL	7.25	01.10.2008	BAA1	03.13.02

Figure B.2: Moodys Intermediate Term Corporate Bond Index Components.

Appendix C

Additional Figures to
Chapter 3

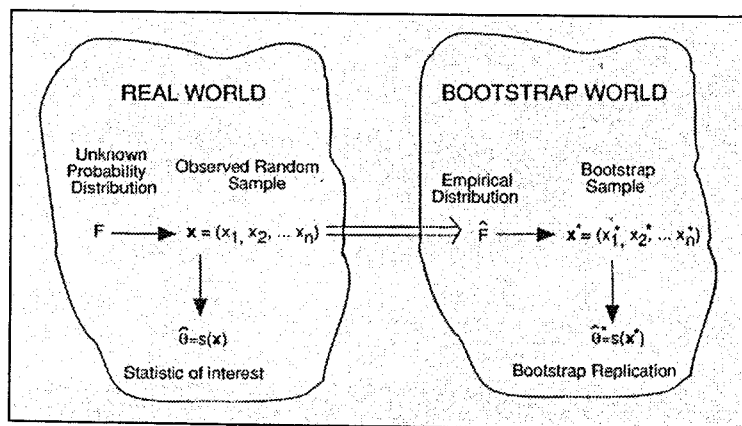


Figure C.1: Backtesting the Expected Shortfall: the procedure for a Bootstrap test as used in Chapter 3 and 4, respectively. Source: Efron and Tibshirani (1993).

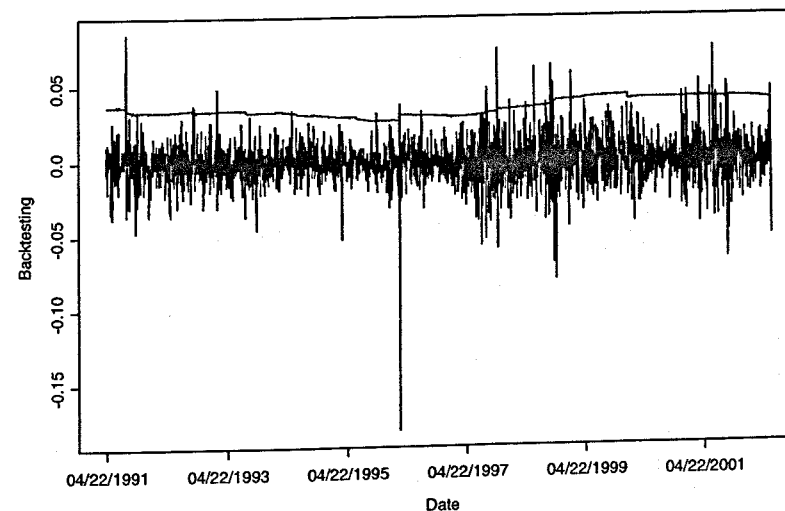


Figure C.2: Backtesting results for the 99.5% quantile, using the Variance-covariance model: log returns (losses on the positive axes) and VaR forecasts (superimposed) for the example of Novartis.

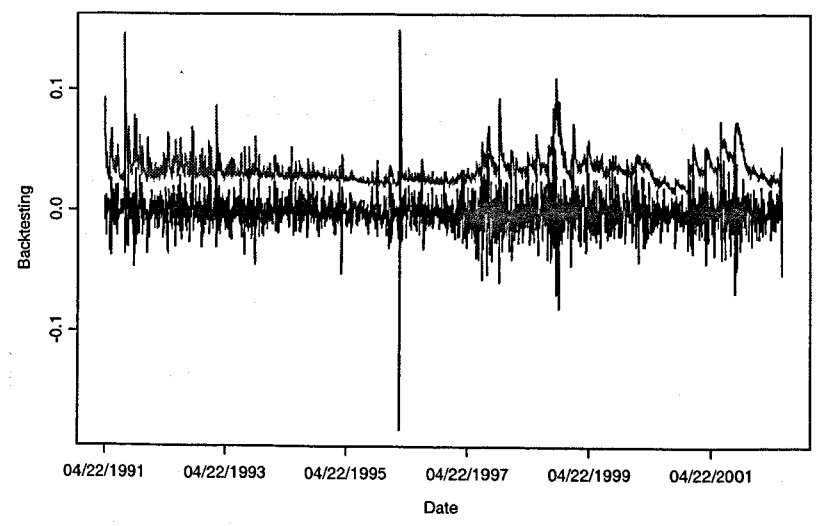


Figure C.3: Backtesting results for the 99.5% quantile, using the conditional t-based model: log returns (losses on the positive axes) and VaR forecasts (superimposed) for the example of Novartis.

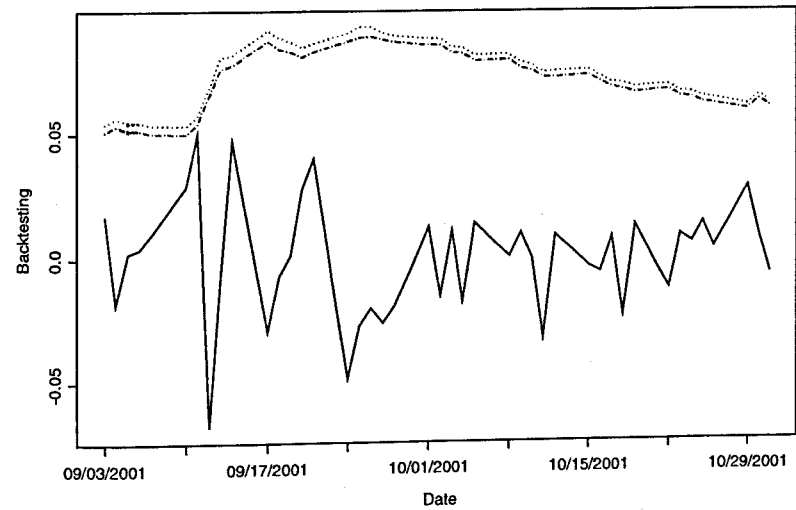


Figure C.4: Enlarged section corresponding to the event of September 11, 2001, example of novartis. The dotted line shows the path of the conditional EVT-based model, the dashed line shows the path of the conditional t-based model.

Appendix D

Additional Figures to Chapter 4

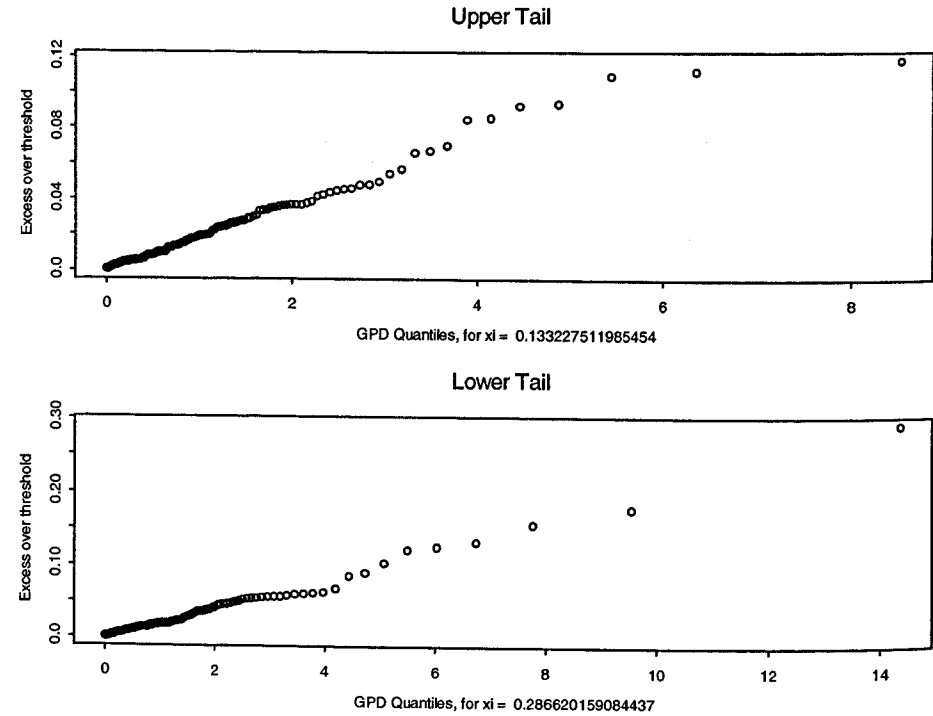


Figure D.1: Result of fitting a generalized Pareto (GPD) distribution to excesses over certain (high) thresholds on the upper and the lower tails, respectively (e.g. a QQ-plot of the excesses over threshold versus quantiles of the estimated GPD).

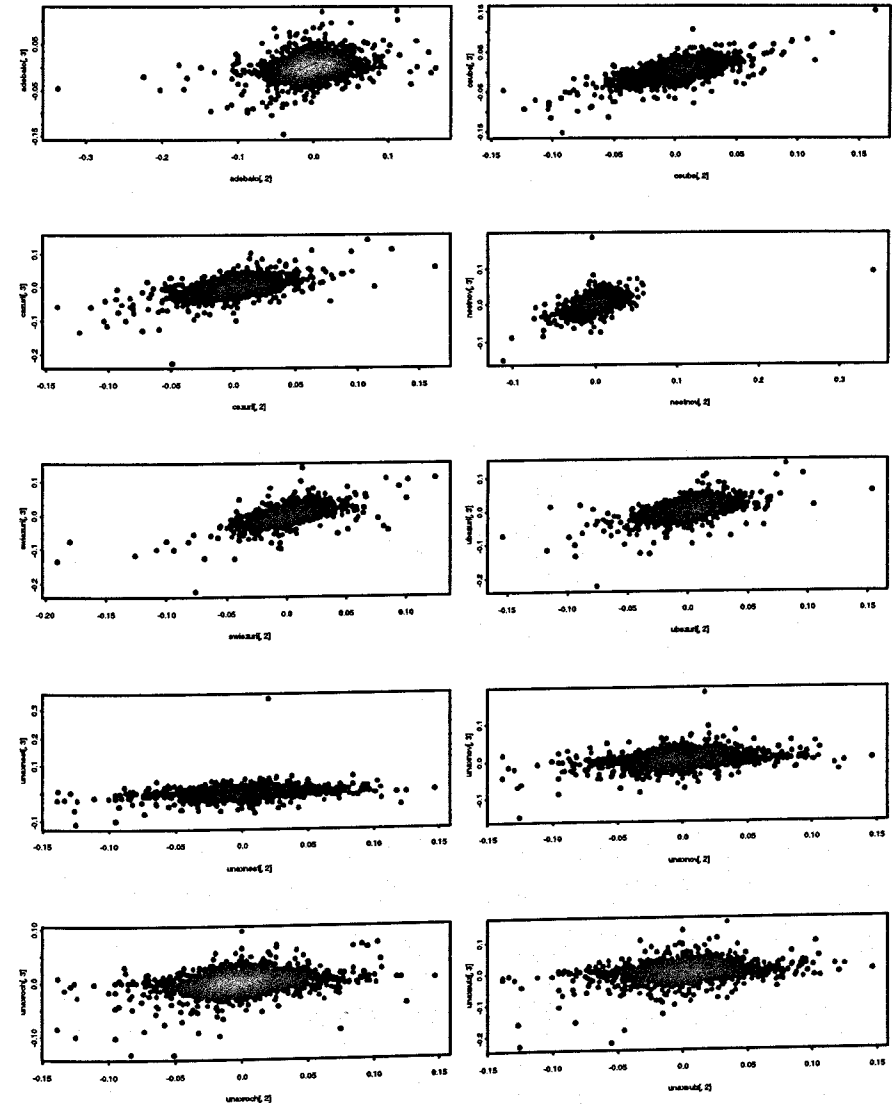


Figure D.2: Scatterplots of the arbitrarily chosen portfolios.

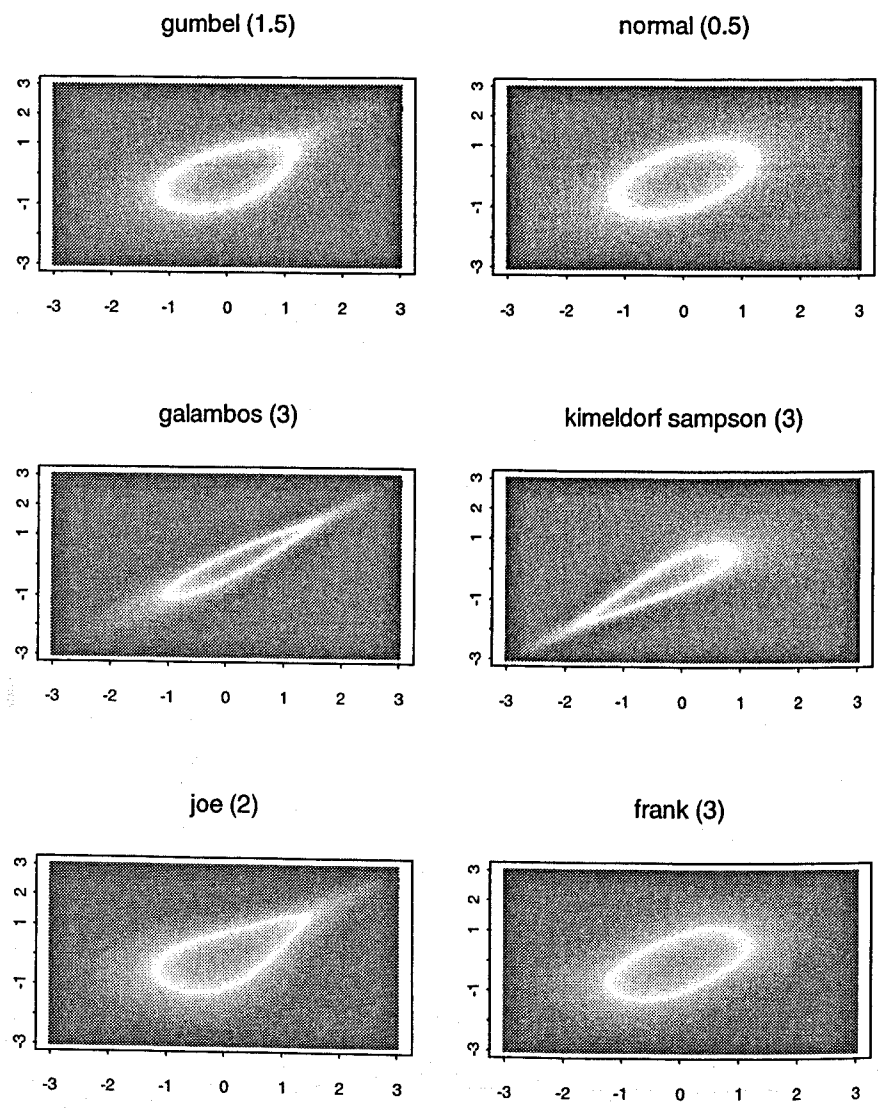


Figure D.3: Copula densities of various distributions, all with standard normal marginal distributions (parameter values in brackets).

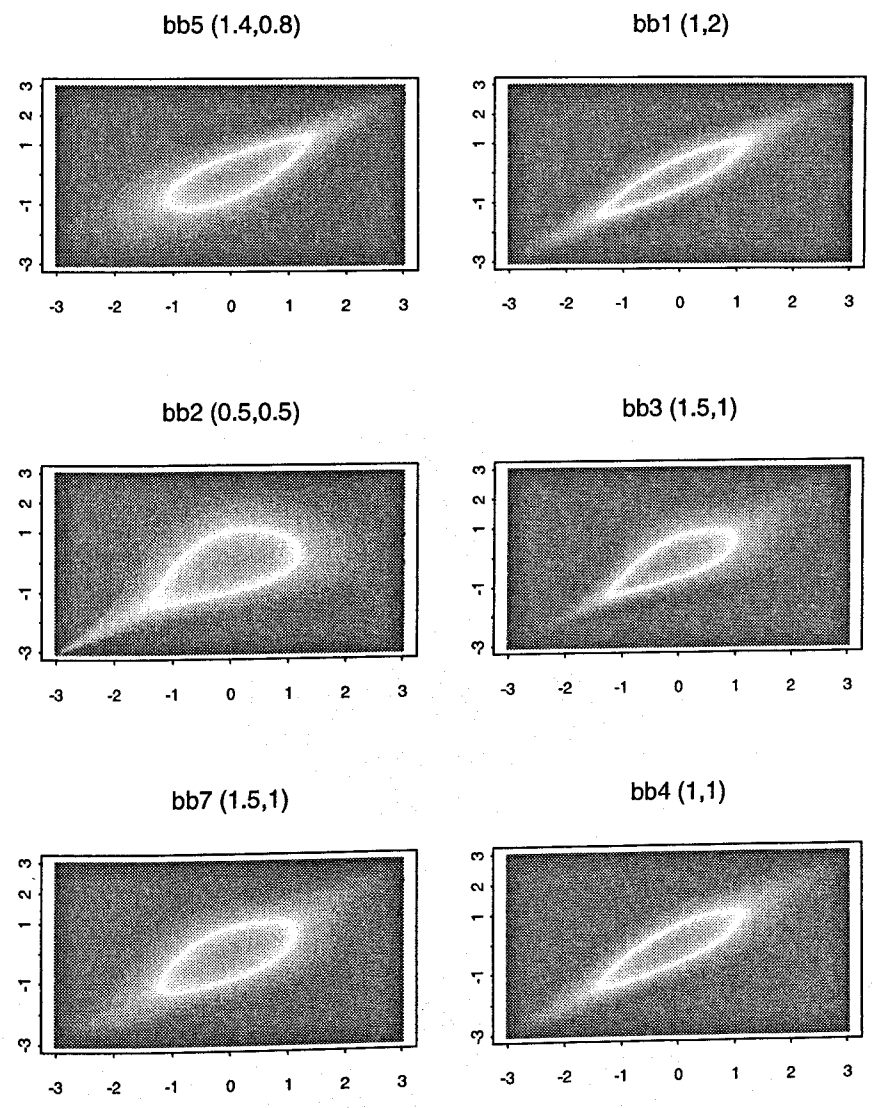


Figure D.4: Copula densities of various distributions, all with standard normal marginal distributions (parameter values in brackets).

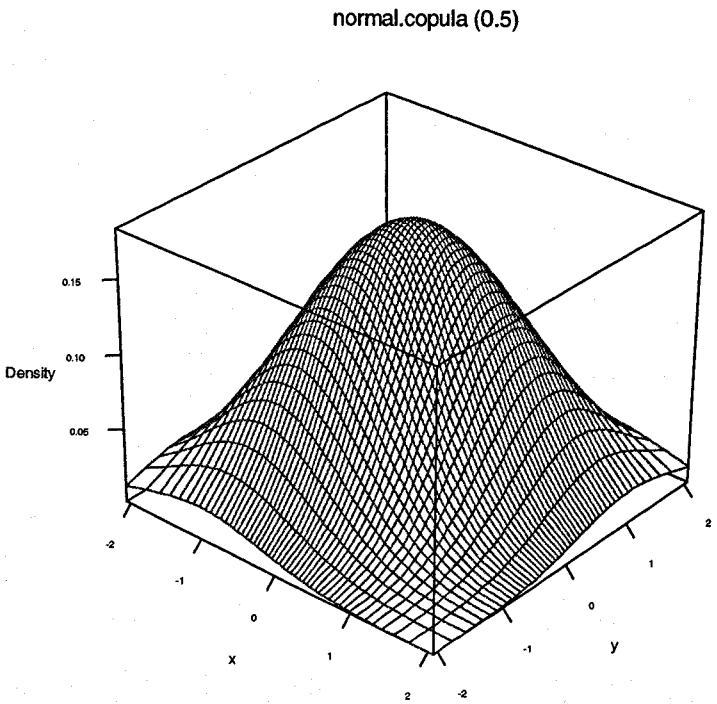


Figure D.5: Density of a bivariate distribution with normal marginals and a normal copula, with parameter value 0.5.

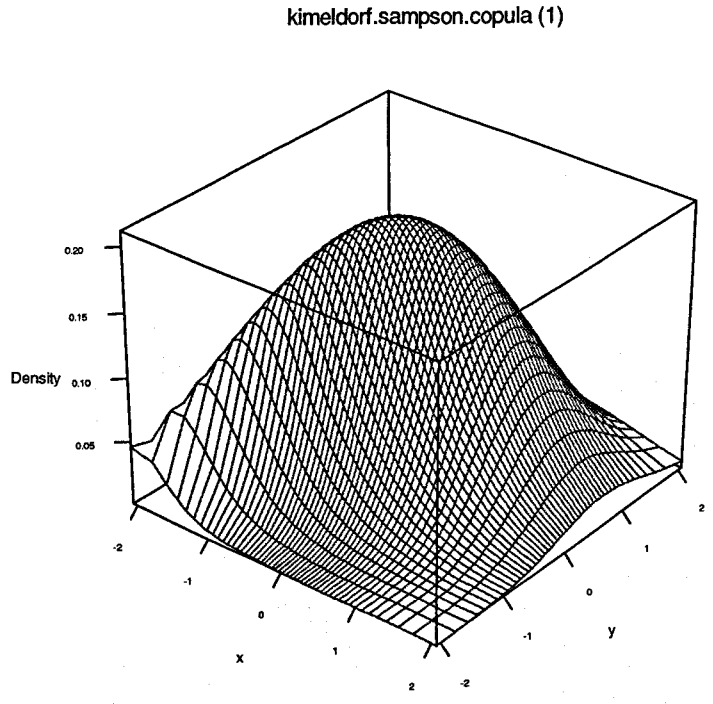


Figure D.6: Density of a bivariate distribution with normal marginals and a kimeldorf sampson copula, with parameter value 1.

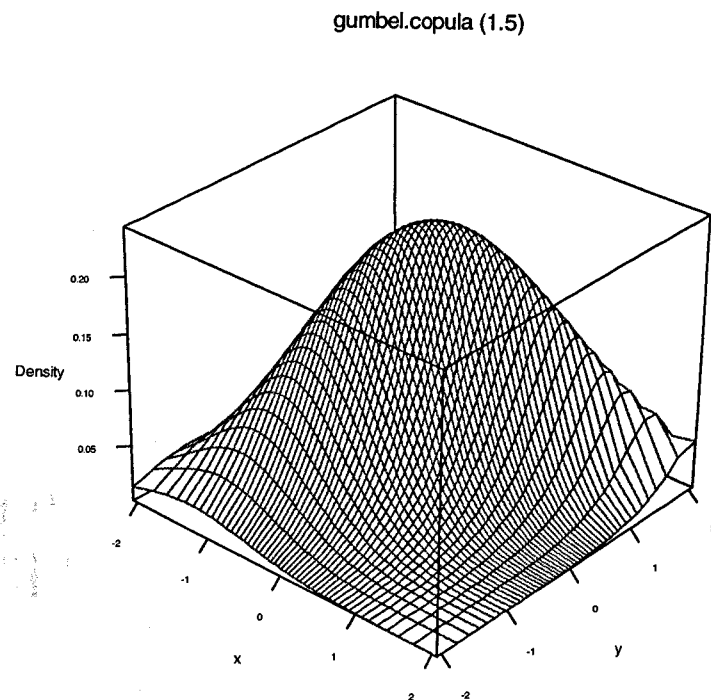


Figure D.7: Density of a bivariate distribution with normal marginals and a gumbel copula, with parameter value 1.5.

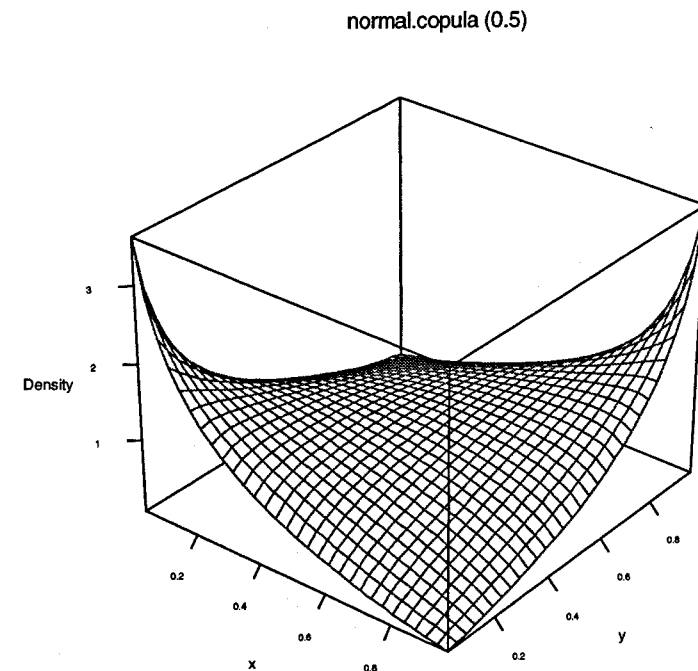


Figure D.8: 3D surface (perspective) plot of the density of a normal copula with a parameter value of 0.5.

kimeldorf.sampson.copula (1)

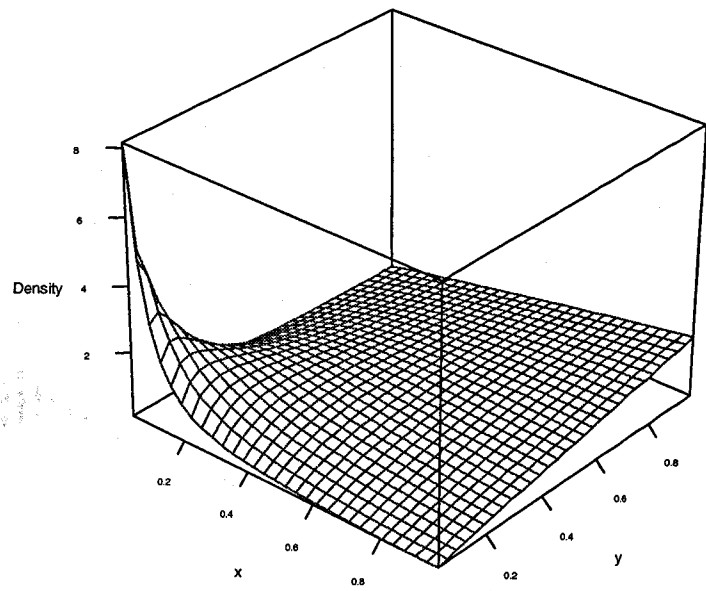


Figure D.9: 3D surface (perspective) plot of the density of a kimeldorf sampson copula with a parameter value of 1.

gumbel.copula (1.5)

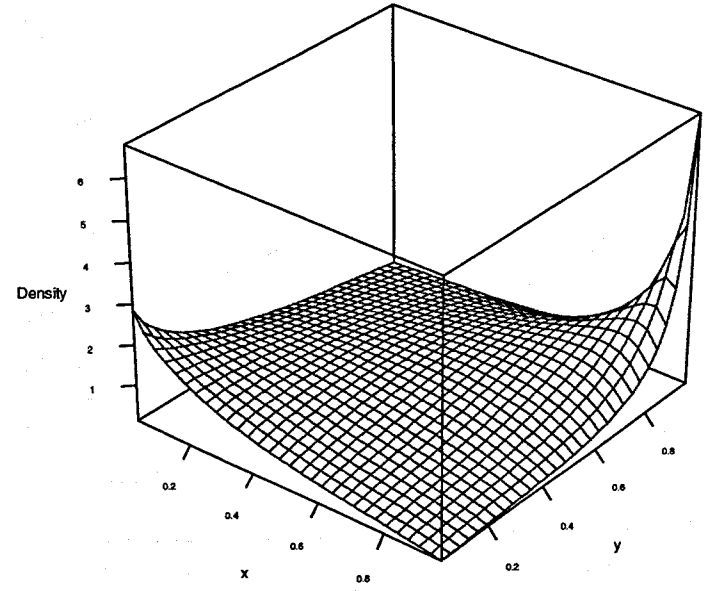


Figure D.10: 3D surface (perspective) plot of the density of a gumbel copula with a parameter value of 1.5.

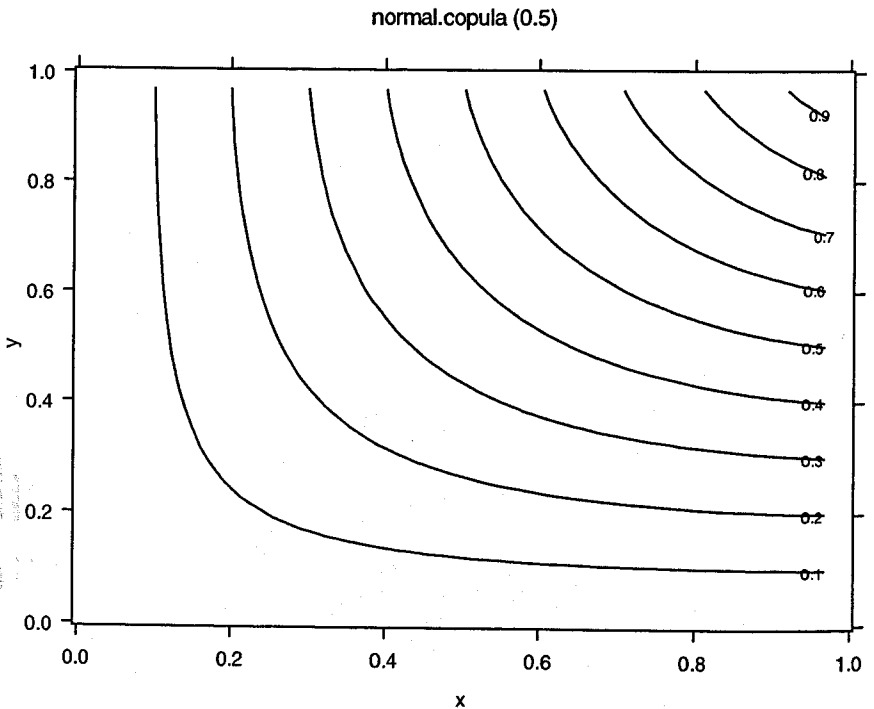


Figure D.11: Contour plot of the distribution function of a normal copula with parameter value 0.5.

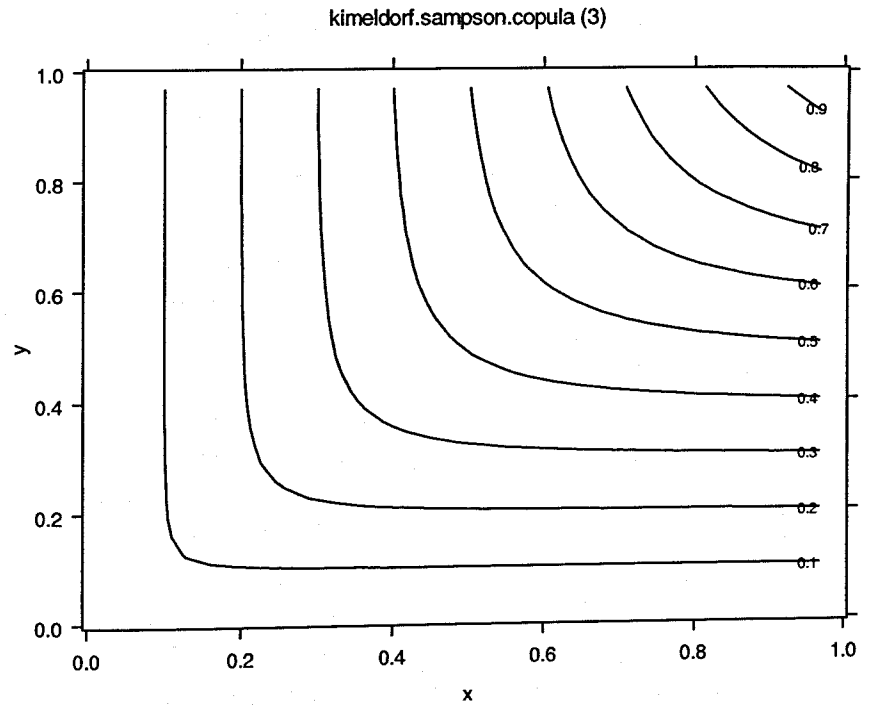


Figure D.12: Contour plot of the distribution function of a kimeldorf sampson copula with parameter value 3.

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CURRICULUM VITAE

EDUCATION

- 10/ 2000 – 11/2003 University of St. Gallen HSG: doctoral studies in Finance, doctoral thesis "Applications of Extreme Value Theory in Financial Risk Modelling"
▪ Doctoral exams (*Dr. oec. HSG*, 29.03.04)
▪ Written examinations (1st year)
- 09/ 2002 - 07/ 2003 London School of Economics LSE: Master of Science in Finance and Economics, (Optional courses: Quantitative Risk Management)
▪ Graduation (*MSc Finance and Economics*)
- 05/ 2002 International Center for Financial Asset Management and Engineering (FAME): Geneva Executive Courses in Finance - "Extreme Value Theory and Dependence Modeling: Risk Management beyond VaR"
- 02/ 2002 - 07/ 2002 Swiss Federal Institute of Technology Zurich (ETH): lecture in Extreme Value Theory
- 05/ 1995 - 10/ 2000 University of St. Gallen HSG: Licentiate in Economics / Business Administration, major in Economics with cross-faculty specialization in Finance and Capital Markets
▪ Graduation (*lic. oec. HSG*)
▪ Pre-diploma examinations (1st and 2nd year)

WORKING EXPERIENCE

- 02/ 2004 - Risk Management Specialist, Ernst & Young AG, Zurich
- 09/ 2001 - 08/ 2002 Project assistant of Prof. Dr. Heinz Zimmermann at the University of Basel
Member of CaRisMa (research association for Capital Markets and Risk Management)
- 10/ 2000 - 06/ 2001 Project assistant of Prof. Dr. Heinz Zimmermann at the University of St. Gallen HSG:
Member of the Risk Management Team (research association for Risk Management)
- 02/ 1997 - 04/ 1997 Internship at Graubündner Kantonalbank Chur (swiss regional bank):
Departments of accounting and controlling
- 02/ 1996 - 04/ 1996 Internship at Graubündner Kantonalbank Chur (swiss regional bank):
Department of law

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